

SPICE Flight Safety Data Package, Phase III

Smoke Point in Co-flow Experiment Flight Project (SPICE)

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ZIN Technologies

Prepared For:
**National Aeronautics and Space Administration
John H. Glenn Research Center at Lewis Field
Exploration Systems Division
Cleveland, Ohio 44135**

Prepared By:
**ZIN Technologies, Incorporated
2001 Aerospace Parkway
Brook Park, Ohio 44142
Under Contract NAS3-99155, 60107-DO-086**

SPICE-SDP-003

SPICE FLIGHT SAFETY DATA PACKAGE PHASE III

Prepared by: _____ Date: _____
Debra Lyden
System Safety Engineer
Zin Technologies, Inc.

Approved by: _____ Date: _____
Tibor Lorik
SPICE
Task Lead
Zin Technologies, Inc.

Accepted by: _____ Date: _____
Robert Hawersaat
SPICE Project Manager
NASA GRC

Concurred by: _____ Date: _____
William Schoren
Lead Safety Engineer
NASA GRC

_____ Date: _____
Gary Kelm
Chief, System Safety and Reliability
NASA GRC

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1 INTRODUCTION

The Smoke Point in Co-flow Experiment (SPICE) Investigation will be operated onboard the International Space Station (ISS) in the Microgravity Science Glovebox (MSG). A previous version of the SPICE design was called the Enclosed Laminar Flames (ELF) experiment, which flew in the Middeck Glovebox on the Space Shuttle STS-87. The Enclosed Laminar Flames (ELF) experiment objective was to study fundamental phenomena related to the mechanisms controlling the stability and extinction of jet diffusion flames. The design modifications made as a result of a carrier change (Middeck GBX to ISS MSG) are considered significant enough that the SPICE Hardware is not considered Series Flight Hardware of the original ELF design.

This Phase III Flight Safety Data Package (SDP) documents the hazards assessed for SPICE. This SDP documents compliance to NSTS/ISS 1700.7B and is prepared in accordance with NSTS/ISS 13830C. A brief overview of the experiment hardware and MSG interfaces is presented first, followed by a brief description of crew procedures. Finally, a safety assessment of SPICE Hazards is presented. Hazards that have not been eliminated by design have been addressed and documented in the Standard Hazard Form (JSC 1230 Form) or in a Unique Hazard Report (JSC 542B Form). Refer to Section 4 of this document for the SPICE Safety Assessment. Refer to Attachment B for the SPICE Standard Hazard Reporting Form, (JSC 1230 Form). Refer to Attachment C for the SPICE Unique Hazard Reporting Forms, (JSC 542B) that support the SPICE safety assessment.

1.1 APPLICABLE DOCUMENTS

(Latest versions available on the PSRP webpage or the PALS webpage)

NSTS/ISS 13830 - Payload Safety Review and Data Submittal Requirements, REV C

NSTS 1700.7 - Safety Policy and Requirements for Payloads Using the STS

NSTS 1700.7 Addendum - Safety Policy Requirements for Payloads Using the International Space Station (ISS Addendum)

NSTS/ISS 18798 - Interpretations of NSTS/ISS Payload Safety Requirements, REV B

NASA-STD-3000 - Man-Systems Integration Standards

NASA-STD-5003 – Fracture Control Requirements for Payloads Using the Space Shuttle

NASA-STD-6001 - Flammability, Odor, Off-gassing, and Compatibility Requirements and Test Procedures for Materials in Environments that Support Combustion (same as NHB 8060.1)

NASA-STD-7001 – Payload Vibroacoustic Test Criteria

NSTS 22648 - Flammability Configuration Analysis for Spacecraft Applications

SSP 30233 – Space Station Requirements for Materials and Processes

SSP 30237 - Space Station Electromagnetic Emission and Susceptibility Requirements

SSP 30240 – Space Station Grounding Requirements

SSP 30245 – Space Station Electrical Bonding Requirements

SSP 50005 - International Space Station Flight Crew Integration Standard (NASA-STD-3000/T)

SSP 52005 - International Space Station Payload Flight Equipment Requirements and
Guidelines for Safety Critical Structures

SSP 57000 - Pressurized Payloads Interface Requirements

MSFC-RQMT-2888 - Microgravity Science Glovebox (MSG) Investigation Interface
Requirements Document

MSFC-ICD-3270 – Interface Control Document for SPICE

MSFC-SPEC-527 / JSC 09604 - Materials Selection List for Space Hardware Systems

1.2 ACRONYMS AND ABBREVIATIONS

AI	Action Item
COTS	Commercial Off-The-Shelf
CM	Combustion Module (CM-1 and CM-2)
DC	Direct Current
EA	Experiment Assembly
ECLSS	Environmental Control & Life Support System
ELF	Enclosed Laminar Flames
EMI	Electromagnetic Interference
FDS	Fire Detection and Suppression
FSDC-3	Fiber Supported Droplet Combustion
SPICE	Smoke Point in Co-flow Experiment
GRC	Glenn Research Center
HR	Hazard Report
Hz	Hertz
ID	Inner diameter
I/O	Input/Output
I/F	Interface
ISS	International Space Station
IVA	Internal Vehicular Activity
JSC	Johnson Space Center
KSC	Kennedy Space Center
LED	Light Emitting Diode

SPICE PHASE III FLIGHT SAFETY DATA PACKAGE
SPICE-SDP-003

LFL	Lower Flammability Limit
MFC	Mass Flow Controller
M&P	Materials & Processes
MIUL	Material Identification and Usage List
MSFC	Marshall Space Flight Center
MSG	Microgravity Science Glovebox
MUA	Material Usage Agreement
NASA	National Aeronautics and Space Administration
NCR	Non-Conformance Report
NSTS	National Space Transportation System
SMAD	Safety and Mission Assurance Directorate
PSRP	Payload Safety Review Panel
QA	Quality Assurance
QC (QD)	Quick Connect. A form of QD.
SDP	Safety Data Package
SSP	Space Station Program
STD	Standard
STS	Space Transportation System
SVTL	Safety Verification Tracking Log
UFL	Upper Flammability Limit
WV	Working Volume

1.3 EXPERIMENT OVERVIEW

The predecessor investigation to Smoke Point in Co-flow Experiment (SPICE) was called Enclosed Laminar Flames (ELF) and was developed by NASA/GRC for flight on the Shuttle Spacelab during the Fourth United States Microgravity Payload (USMP-4 / STS-87) mission in November 1997.

SPICE continues to study fundamental phenomena related to understanding the mechanisms controlling the stability and extinction of jet diffusion flames. SPICE will stabilize an enclosed laminar flame, measure the overall flame shape to validate the theoretical and numerical predictions, measure the flame stabilization heights, and measure the temperature field to verify flame structure predictions.

The differences in ELF and SPICE hardware are the following:

The Gas Bottle Assembly size increased from 75 ml to 150 ml, and maximum pressure was reduced from 350 psi to 230 psi. ELF bottles were open on both ends with a plug placed in one end and the valve in the other. For SPICE only one end of the bottle has an opening.

The original fuel was 50/50 % methane. Propane, Ethylene, Propylene have been added as fuels. The increase in bottle size and decrease in pressure cause the worst case SPICE bottle to have three times the energy of an ELF bottle due to fuel. Also the number of bottles increased from 6 to 12.

For the Experiment Assembly (EA) the Mass Flow Controller model was changed.

There is a new Electronics power video box power box, a new video camera, a new still camera and there is an interface with MSG video drawer.

1.4 PAYLOAD AND MISSION OVERVIEW

The SPICE payload is currently scheduled to fly on the International Space Station (ISS) in the Microgravity Science Glovebox (MSG). The ascent flight on the NSTS is tentatively scheduled to be ULF2. All SPICE hardware will be returned to earth at the same time on a future shuttle flight.

1.4.1 SPICE Payload Hardware

SPICE payload hardware is all new hardware that has been developed from the ELF experiment hardware design.

SPICE consists of an Experiment Assembly (EA), 12 Gas Bottle Assemblies, a Nikon camera, a Video Camera, associated structural support for the cameras, Video Power Unit, and a number of stowage cases for the hardware and harnessing which includes a Control Box. SPICE is providing two bottle stowage cases, a component stowage case, and 2 tape bags. The rest of the hardware will be packed by JSC/MSFC. See figures 1 and 2 for hardware and figure 9b for a case diagram respectively.

1.5 MICROGRAVITY SCIENCE GLOVEBOX (MSG) INTERFACES

SPICE will interface with the MSG for structural support, containment and filtering, conditioned power and science data retrieval / storage / downlink.

1.5.1 Structural Interfaces with MSG

All SPICE hardware (see Figure 1) except the Control Box (see Figures 2 & 2A) will be mounted inside the MSG Working Volume. All components are stowed for launch. The SPICE components mounted in the MSG are the Power-Video Unit, the Experiment Assembly, the Nikon Digital Still Camera & Mount and the Video Camera. All of the brackets are anodized aluminum. The SPICE Experiment Assembly (see section 2.1 for components) is mounted to the floor of the MSG. The SPICE Experiment Assembly has three aluminum angle brackets, each having one captive, M6 thumbscrew. One bracket attaches the 70mm grid on the MSG Airlock Lid and one to the 70mm grid on the MSG Cold Plate. The third bracket will be attached to the mid-section side of the Still Camera Mount.

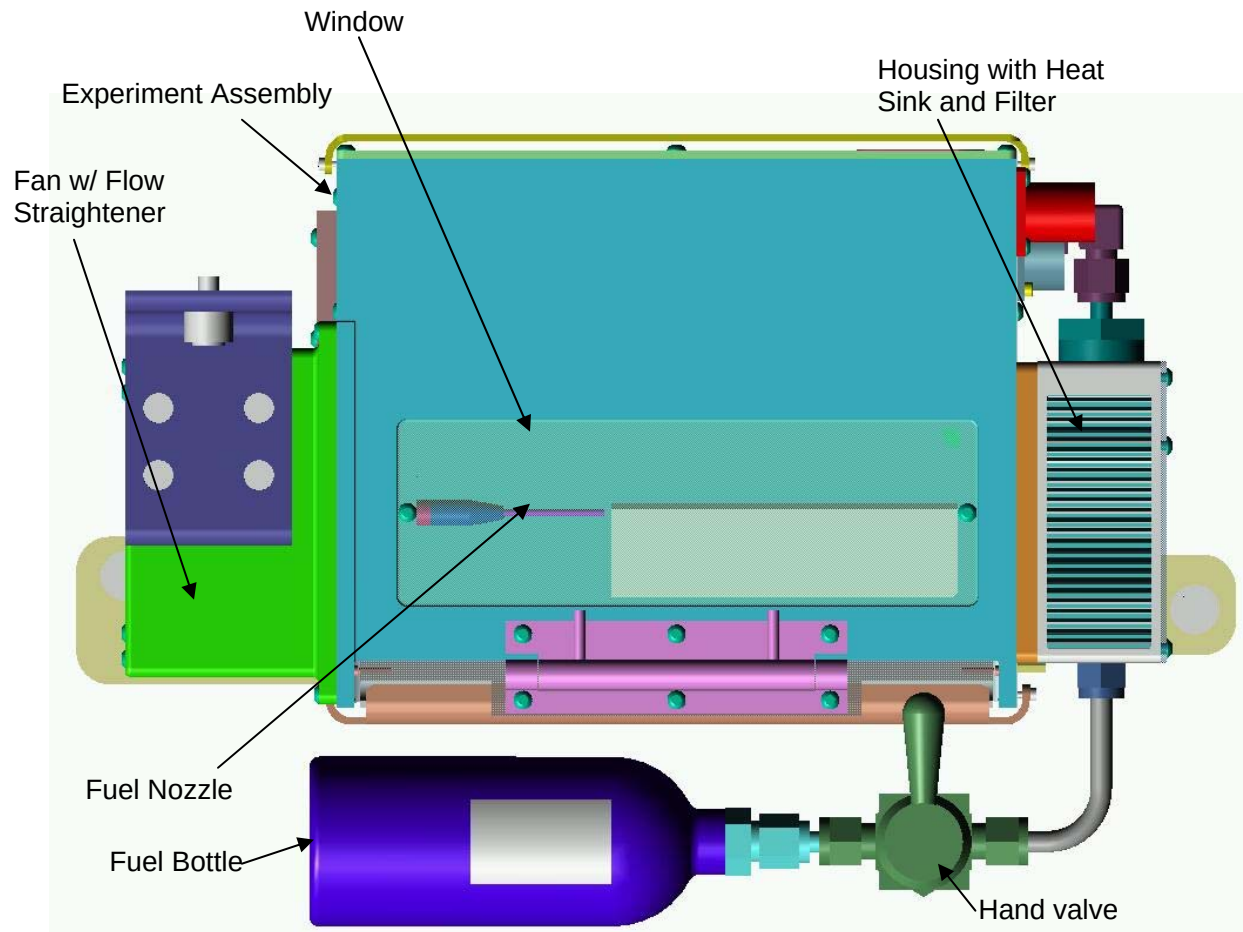
The Still Camera Mount base is attached to the 70mm grid on the MSG Cold Plate by an aluminum angle bracket that has two captive, M6 thumbscrews. The mid-section of the Still Camera Mount is then attached to the third bracket on the Experiment Assembly mentioned above.

The Video Camera Mount Base is attached to the 70mm grid on the MSG Cold Plate by three captive, M6 thumbscrews.

The SPICE Power-Video Unit comprised of a Power Conditioning Box Assembly and Video Titler Box Assembly that are mounted to an Adapter Plate which is mounted to the 70mm grid on the rear wall of MSG using two captive, M6 thumbscrews.

A drawing of the SPICE engineering model inside the MSG is shown in Figure 2.

Figure 1 – Layout of SPICE Hardware in the MSG Top View



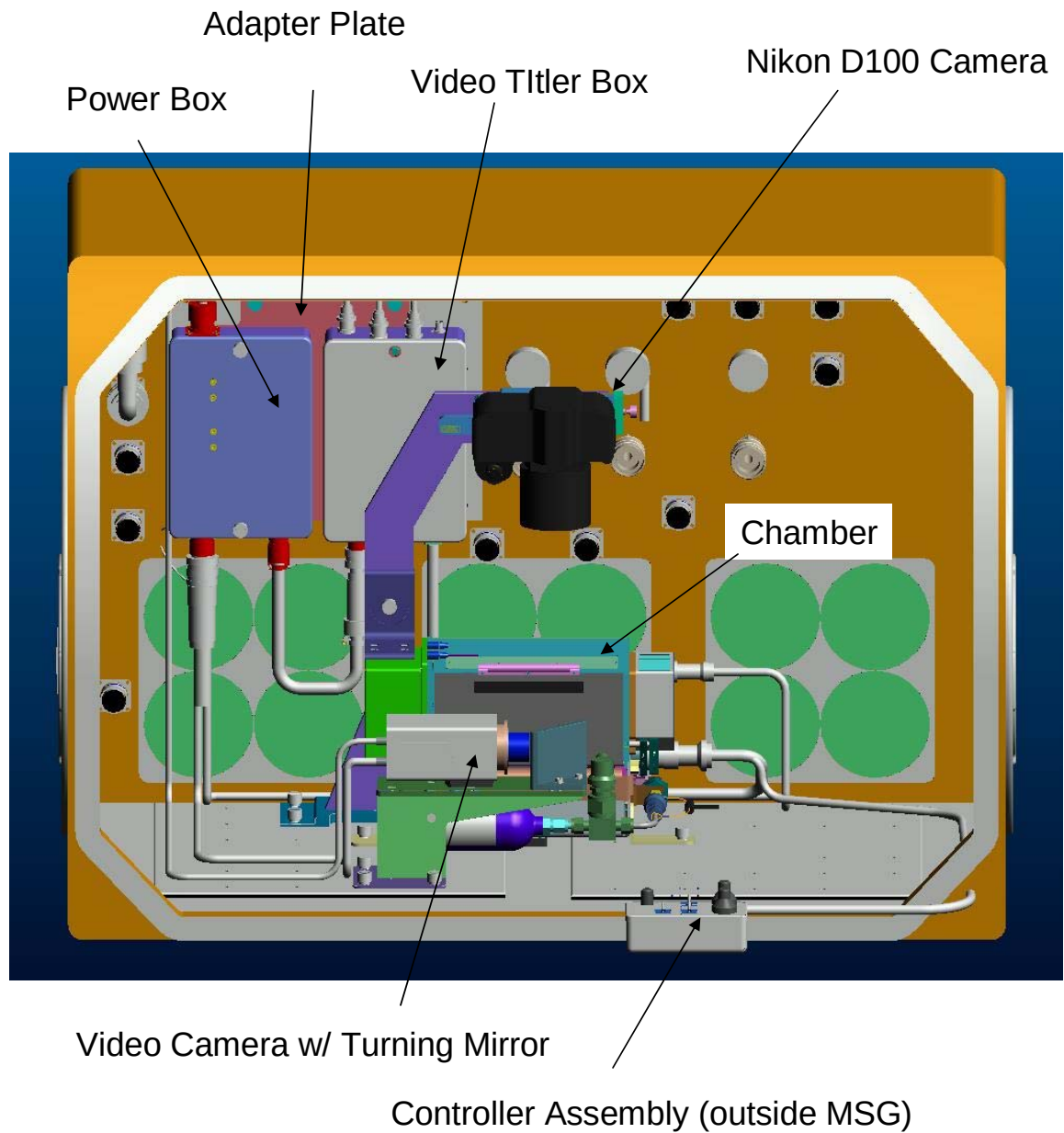
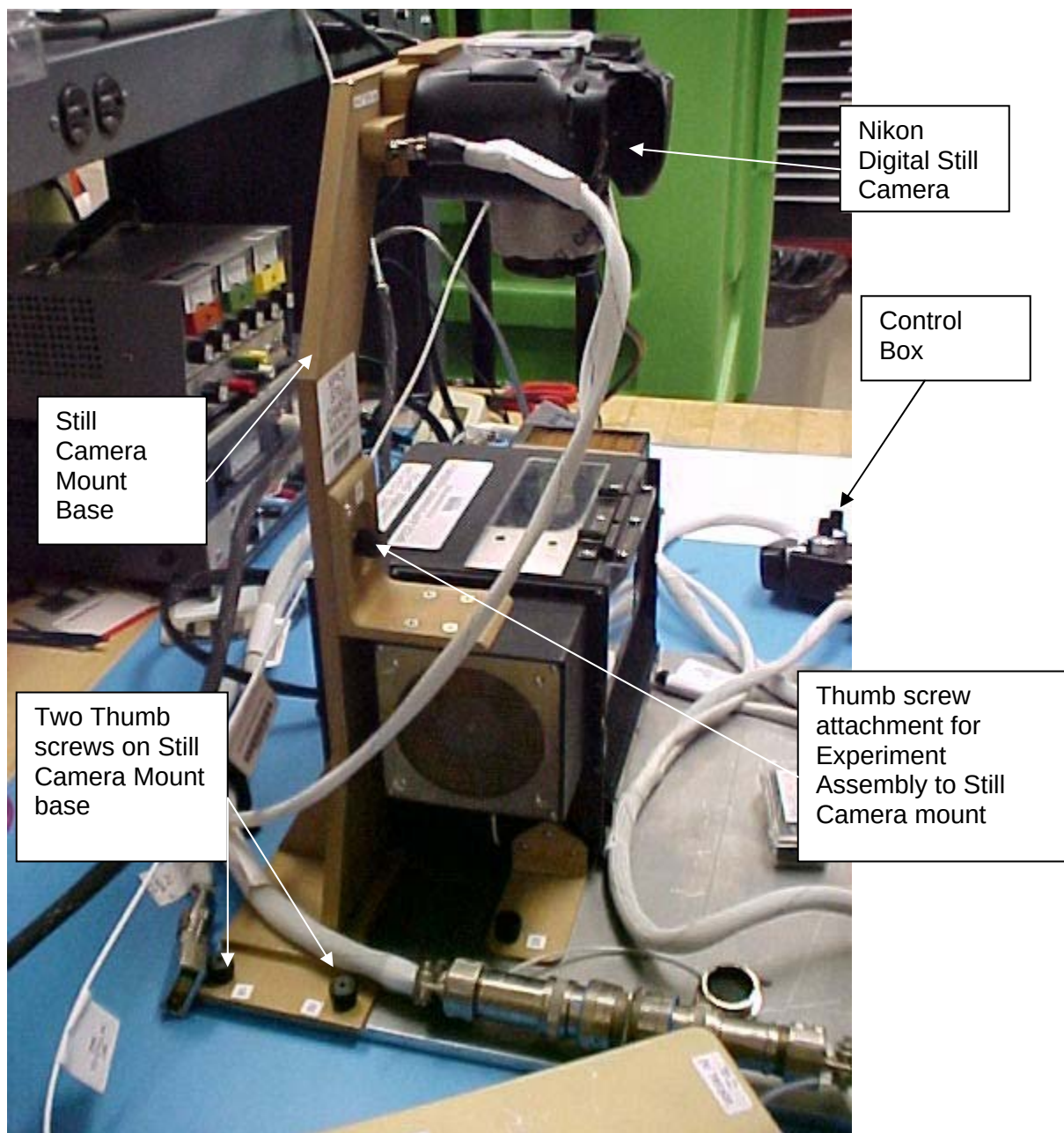


Figure 2 SPICE inside the MSG

Figure 2A Mounting hardware



1.5.2 Power Interfaces to MSG

SPICE to MSG Electrical Interface Diagram is shown in Figure 3.

SPICE will receive conditioned power of +5, +/-12, and + 28 VDC from MSG. MSG conditioned power source will provide circuit protection to SPICE. SPICE will also provide additional circuit protection in the Power Conditioning Box (5 ampere fast-acting cartridge fuses) on the +28 VDC and +5 VDC power lines, See Figure 6.

Nominal power consumption is 27 W with maximum power intake from the MSG of about 42 W during ignition (< 10 seconds typical).

Figure 3 – SPICE Electrical Interfaces with MSG

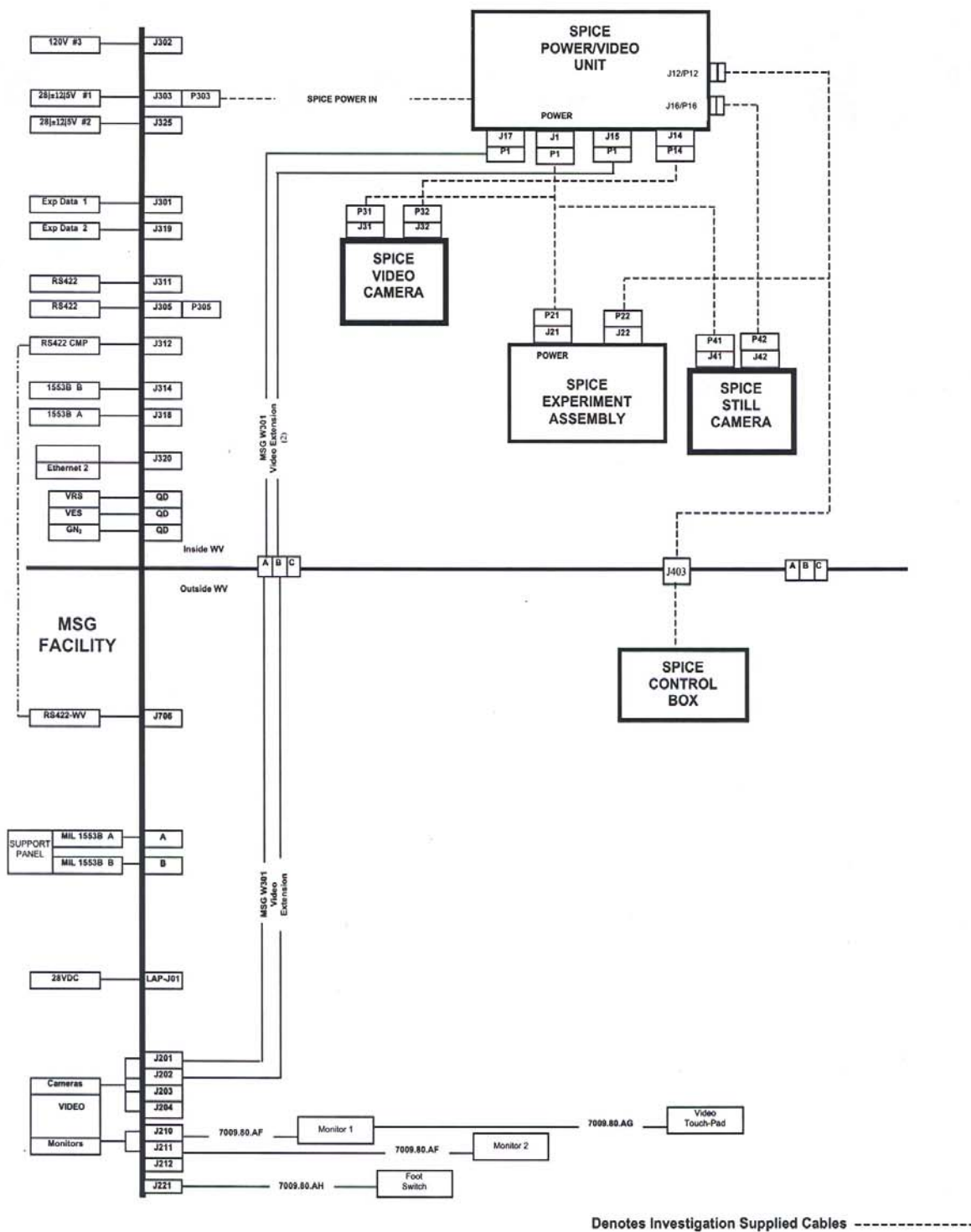




Figure 3a Control Box

1.5.3 Fire Detection and Suppression

SPICE will receive fire detection and suppression capability from MSG and Crew during experiment operations. This is discussed further in Section 4.3.4.

1.5.4 Communications Interfaces with MSG

SPICE will use the MSG Video Drawer containing two video recorders and two monitors to record science data. SPICE Experiment video and digital photos are required to be down linked as soon as possible either during or immediately after each flame test run. The interface with MSG is through the MSG W301 Video Extension. Requirements for SPICE Data & Video handling are defined in the MSG ICD for SPICE Document, MSFC-ICD-3270 Rev A, para. 3.3.

SPICE requests that MSG Working Volume Ambient Temperature be down linked during experiment operations.

SPICE does not interface to the MSG Command and Data Handling (C&DH) System. SPICE does interface to the MSG Video Drawer.

SPICE does not require uplink.

The SPICE experiment uses the MSG J403 connector to interface the SPICE Control Box located external to the MSG Working Volume and the SPICE hardware inside the Working Volume (see figure 3a)

1.5.5 MSG Thermal Interfaces

Although the SPICE Experiment Assembly is mounted to the MSG Cold Plate. Cooling service provided by the Cold Plate is not required for SPICE operations. The heat dissipated by the burning process will raise the temperature of the air flowing through the Experiment Assembly only a few degrees. The heat will be then dissipated to the surroundings.

1.5.6 MSG Working Volume and Air Filtration System

The SPICE experiment combustion process is run with the MSG Working Volume configured in the normal mode (MSG Fans on lowest setting) with the side loading ports and glove ports installed. This mode provides one level of containment for combustion by-products (all by-products are tox level 0). At the end of the combustion process the MSG fans maybe changed to a higher speed and run for a 5-minute period in order to re-circulate the Working Volume air and clean-up combustion byproducts using the Working Volume Air Filtration System.

1.5.7 Crew Interfaces

The crew interfaces with SPICE consist of set up, nozzle change out, video tape change out, IBM micro drive change out, fuel bottle change out, ignition, smoke point setting, still camera shots, and tear down. Section 3.0 describes the operations in more detail.

2 SPICE HARDWARE & SOFTWARE DESCRIPTION

The SPICE experiment will use the Microgravity Science Glove Box (MSG) facility in the International Space Station. The SPICE payload consists of the following hardware:

- One (1) Experiment Assembly, which is comprised of:
 - Interchangeable nozzles (0.4, 0.8 & 1.6 mm) - radiometer
 - Fan for Co-flow - anemometer
 - Exchangeable hot-wire Igniter - quick-connect fuel inlet
 - heat sink/hand guard - mass flow controller
 - Pressure regulator
 - Tattletale Model 8 data logger (anemometer, fan voltage, fuel flow rate, radiometer)
- One (1) Power-Video Unit, which is comprised of:
 - Power Conditioning Box
 - Video Titler Box
 - Adapter Plate
- One (1) Control Box, which is comprised of:
 - Igniter enable switch - fuel enable switch
 - Data enable switch - radiometer gain switch
 - Fuel flow pot - fan velocity pot
- Twelve (12) Fuel Bottle Assemblies comprised of:
 - 150 cc DOT bottle with hand valve and quick connect for attachment to Experiment Assembly.
 - Gas Bottle fuels are propane, ethylene, methane, propylene (100%), and nitrogen diluted propylene (75%, and 50%).
- Four (4) Cables:
 - Power Cable
 - Still Camera Cable

- Control Box Cable

- Video Cable

- Data Acquisition and Video System:
 - Panasonic WV-CP 654 video camera (side view of flame)
 - Nikon D100 35 mm Camera (top view of flame)
 - 6 IBM Micro-drives used to store digital images taken by the Nikon D100 Camera.
 - Horita SCT-50 Video-Titrer (overlay data displays on video) and two Horita VDA-50 Video- Amplifiers.
 - 30 Digital Video Tapes that are stowed in two bags each bag containing 15 tapes.
- Stowage Cases
 - Two (2) bottle cases
 - Component Case

SPICE utilizes the MSG power supply; two color monitors, two video recorders, and the MSG's Working Volume containment capability since the SPICE experiment combustion process is run with the MSG Working Volume configured in the normal mode.

SPICE is powered off MSG Secondary Power (+28, +12, -12 and +5 VDC). The total power consumption is 27 W nominal with a peak of 42 W for less than 10 seconds (typical) during ignition.

The following sections provide the description of the various SPICE components.

2.1 EXPERIMENT ASSEMBLY

The Experiment Assembly (see Figure 4) is composed of an aluminum rectangular chamber 34.6cm X 17.0cm X 26.2cm with a ¼ inch polycarbonate Lexan 9034 window cover and the associated stainless steel plumbing system that regulates the flow of gas. A commercial three (3) inch diameter fan at the inlet to the chamber provides the air Co-flow. The fans are Panaflo FBA08A12HIA axial fans with a nominal speed of 2950 RPM with plastic housing and blades. The fan is placed in an aluminum housing and metal screens are located at the inlet and exit of the fan, and at the exhaust end of the chamber. A ceramic (Cordierite) flow straightener is also located in the fan housing. It is contained in a housing with a 100 mesh screen over it. A heat sink/hand guard is also placed at the exhaust end of the Experiment Assembly. The heat sink is a piece of copper with holes in it. It is contained in an aluminum box with corrugated bronze mesh over the opening. The experiment Assembly is also equipped with a data logger and PCMCIA (data card type of digital storage) storage. The Experiment Assembly Cover opens to facilitate Nozzle and Igniter hardware installation and change-out operations. The thermal mass of the Experiment Assembly precludes the potential to exceed touch temperature limits.

The Experiment Assembly also includes the Fuel Delivery System (See Figure 4A), which consists of stainless steel assemblies and component parts such as the Fuel Bottle Assembly (1-7), a male half Quick Disconnect (8), Filter (10), Pressure Regulator (13), Mass Flow Controller (15), and applicable stainless steel tubing and adapter fittings (9,11,12,14,16). See table 1 attached to Hazard Report SPICE-U2 for the Maximum Design Pressures (MDP) and other information concerning these components. .

Figure 4 – SPICE Experiment Assembly

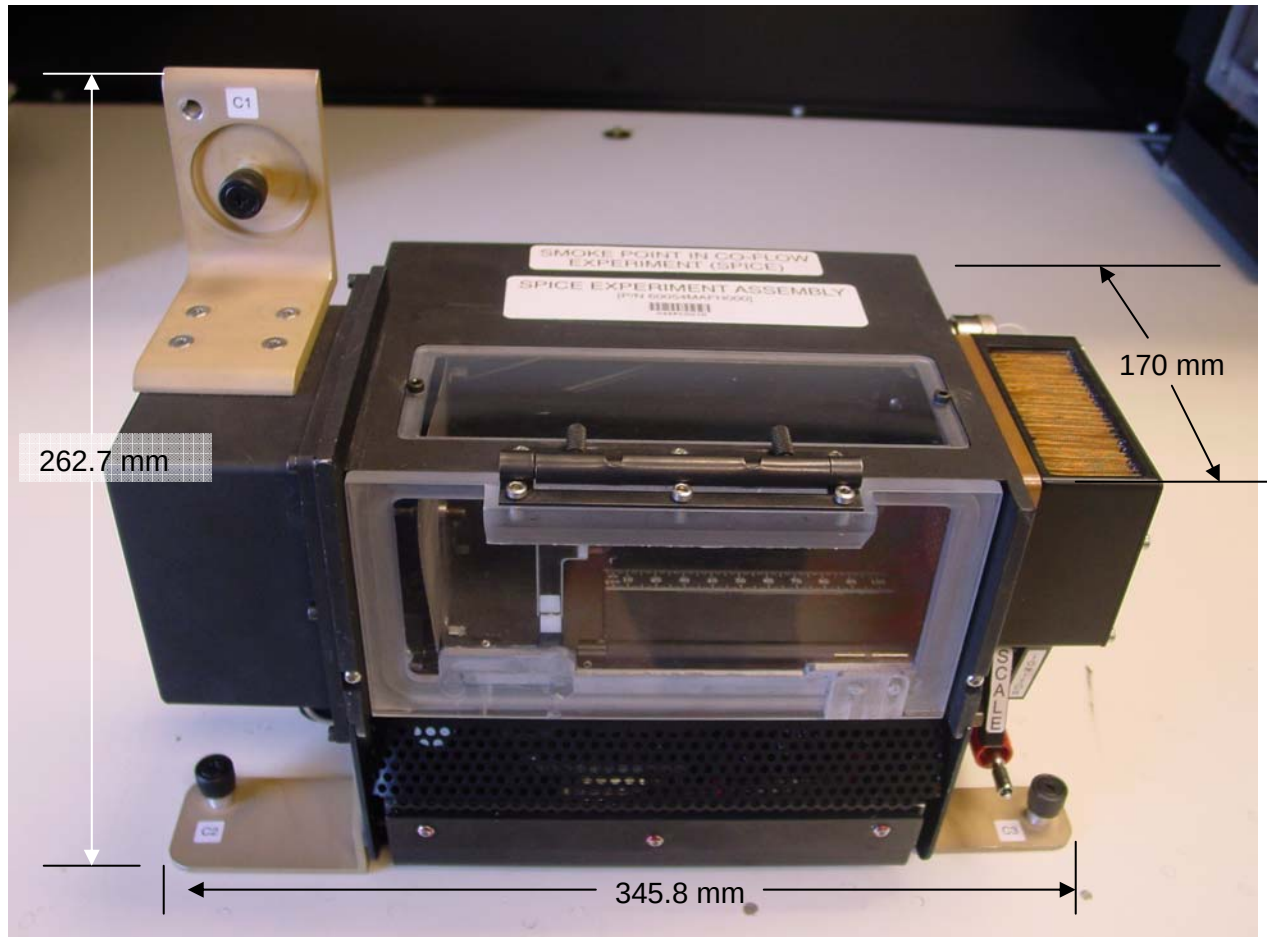


Figure 4A – SPICE Plumbing Diagram

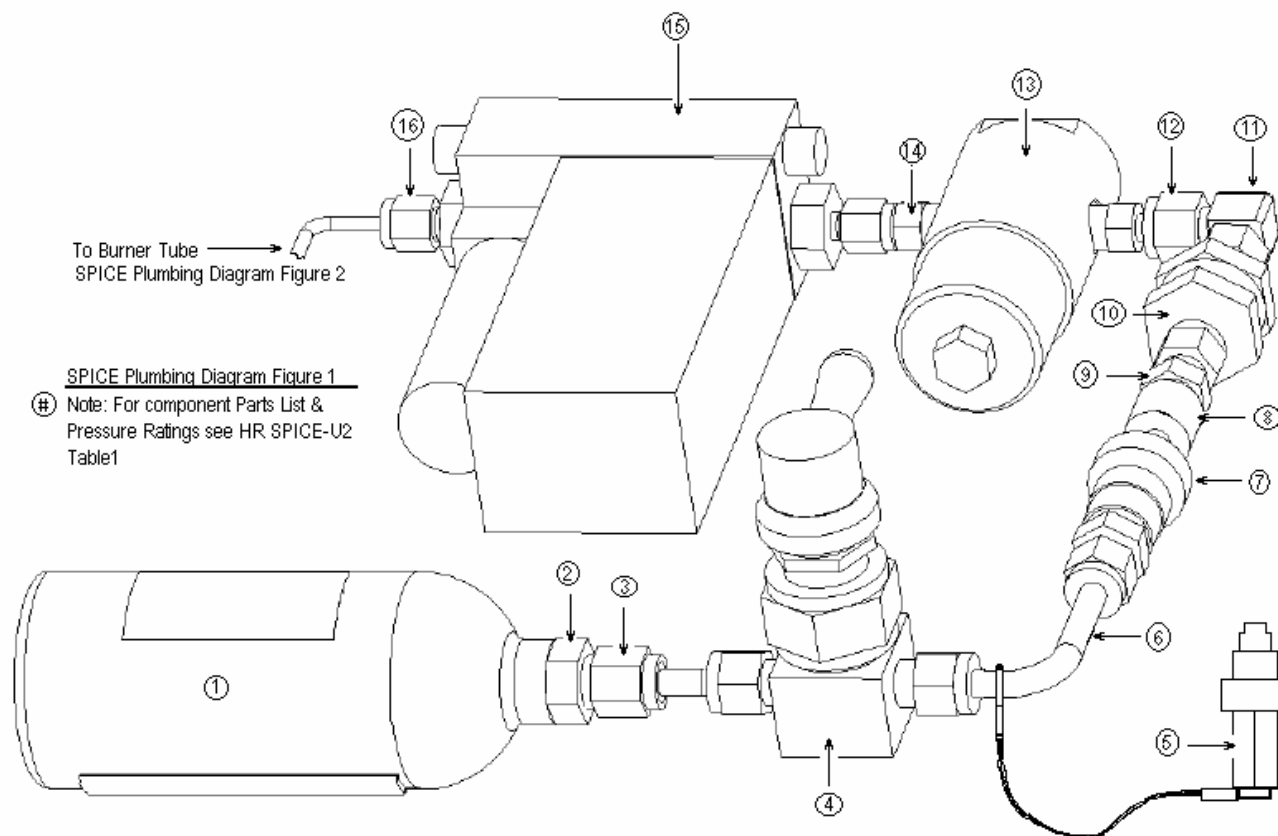


Figure 4A Parts List

1	SAMPLE CYLINDER, 150 CC SINGLE END SWAGELOK No. 304L-05SMS6-150	9	MALE ADAPTER, 1/4 NPT TO 1/4 TUBE SWAGELOK NO. SS-4-TA-1-4
2	SAE/MS MALE CONNECTOR 1/4 TUBE TO 9/16-18 SAE/MS STRAIGHT THREAD, SWAGELOK No. SS-400-1-6ST	10	FILTER, 7 MICRON SWAGELOK NO. SS-4FW-TA18-7
3	PORT CONNECTOR, 1/4 TUBE SWAGELOK No. SS-400-1-PC	11	UNION ELBOW, 1/4" TUBE SWAGELOK NO. SS-400-9
4	DIAPHRAGM VALVE, 1/4 SWAGELOK NUPRO NO. SS-DLS4	12	MALE ADAPTER, 1/4" TUBE TO 1/8" NPT SWAGELOK NO. SS-4-TA-1-2
5	QD PROTECTOR PLUG SWAGELOK No. SS-QC4-BP	13	PRESSURE REGULATOR, 0-50 PSIG GO INC NO. CPR1-1B01A3E178
6	1/4" S.S. TUBING .035 INCH WALL THICKNESS	14	MALE ADAPTER, 1/8" TUBE TO 1/8" NPT SWAGELOK NO. SS-2-TA-1-2
7	QUICK CONNECT BODY SWAGELOK NO. SS-QC4-B-400	15	MASS FLOW CONTROLLER MKS 1179A52CS1BV
8	QUICK CONNECT, DESO STEM, 1/4" SWGLOK SWAGELOK NO. SS-QC4-D-400	16	MALE CONNECTOR 9/16" STRAIGHT THD TO 1/8" TUBE SWAGELOK NO SS-200-1-6ST

2.2 VIDEO TITLER BOX

The Video Titler Box is mounted on an aluminum adapter plate along with the Power Conditioning Box (see Section 2.3). The complete assembly is called the Power-Video Unit (See Figure 5).

This Video Titler Box receives camera video and superimposes SPICE text messages on it. Experiment data is presented on the video for tape recording and for display to crewmembers.

The Video Titler Box is an aluminum enclosure approximately 8 in. x 5.5 in. x 3 in. The Video Titler Box has six (6) connections: Power from Power Conditioning Box, Control (RS-232), two (2) video inputs, one each from the SPICE Video camera and D100 Digital camera and two (2) video outputs which are connected to the MSG Video Drawer via a MSG Working Volume front feed-through connector.

Internal to the Video Titler Box are three (3) conformal coated printed circuit boards and associated wiring. The circuit receives the two (2) video signals from the SPICE Video camera and Digital Camera adds text and outputs the appropriate video format to the MSG Video Drawer. The MSG Video Drawer contains the video monitoring equipment that will recorder the SPICE video signals received onto SPICE provided digital video tapes.

Video Titler Box power is provided from the SPICE Power Conditioning Box (see section 2.3). Control of the Video Titler Box is via (RS-232) link to the SPICE Experiment Assembly.

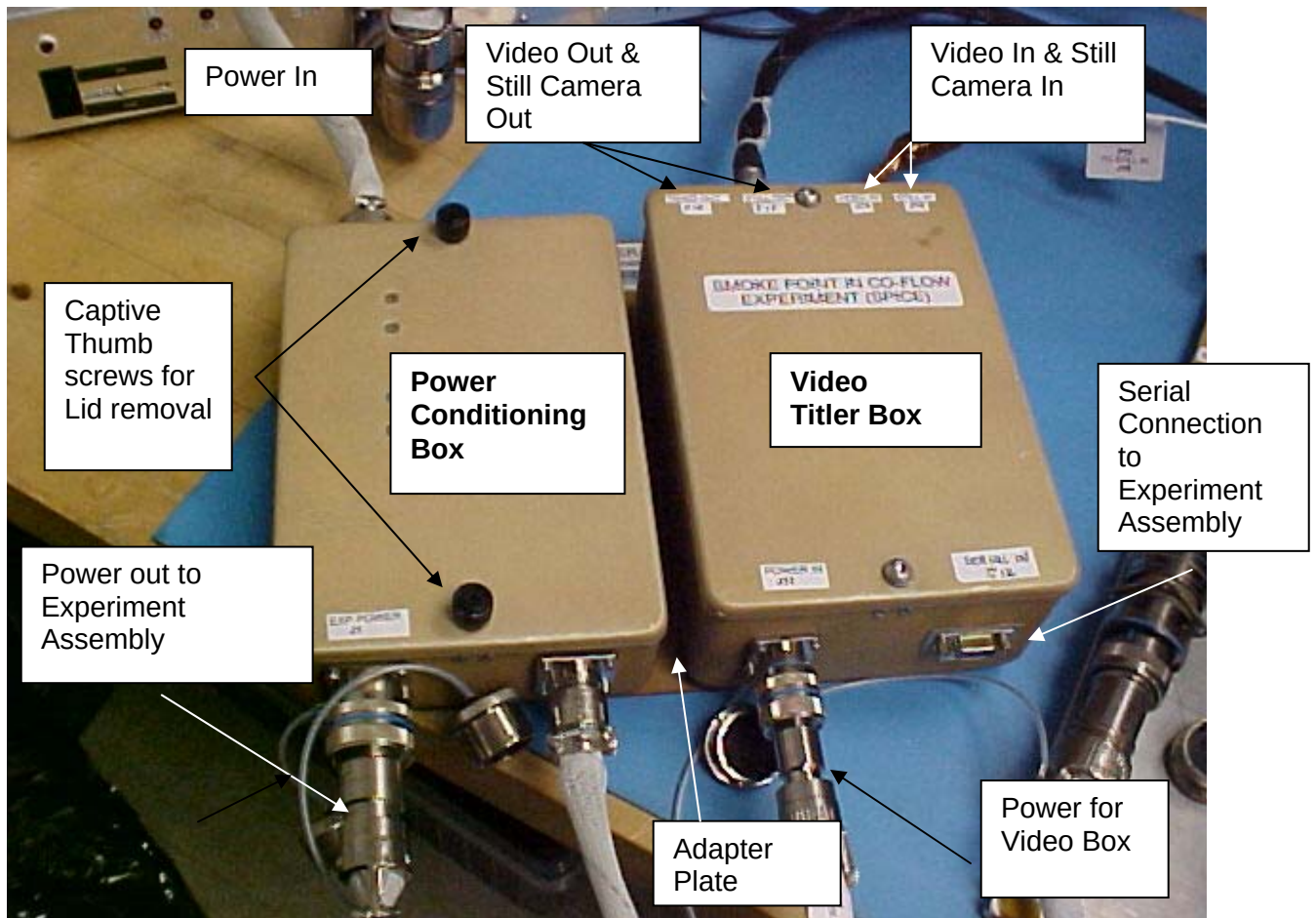
2.3 POWER CONDITIONING BOX

This box is located in the Power/Video Assembly and is attached to the adapter plate. It provides SPICE with power conditioning elements required for proper SPICE operation with the MSG. These include wire gauge/routing sites and additional circuit protection Fuses and spares (see Figure 5). Two captive Thumb Screws secure the lid of the box. Backing off the screws and removing the lid accomplish access to the fuses. Fuses can be replaced within the Power Conditioning Box without having to remove the Box from MSG.

Power conditioning elements are housed in an aluminum enclosure (with lid) very similar to the Video Titler Box. The Power Conditioning Box includes a power input cable that is connected to the MSG Working Volume J303 Power Connector. A terminal board inside the Power Conditioning Box provides power distribution for +/-12, +5, and +28 VDC power provided from MSG. The Power Conditioning Box also contains a 28VDC to 9VDC Converter and Filter that provides 9VDC power to the NIKON D100 Camera. The filter, capacitors, and a resistor are to limit the inrush current when the camera snaps a picture.

Two (2) 5 Amp Cartridge Fuses with one wired to the +5 VDC power and the other wired to +28 VDC power are also located inside the Power Conditioning Box. The fuses provide additional circuit protection to the SPICE Experiment Hardware that is wired to the +5 and +28 VDC power. The SPICE Experiment Hardware that uses +/- 12VDC power does not require additional circuit protection.

Figure 5 Video/Power Assembly



2.4 VIDEO CAMERA ASSEMBLY

The Video Camera Assembly is comprised of a Panasonic WV-CP 654 Video Camera with 8mm lens and Aluminum Mirror assembly that is attached to the Camera Mount. The Camera Mount is attached to the 70mm grid on the MSG Cold Plate using 3 M6 thumbscrews (See Figure 7). The mount itself is made up of anodized aluminum pieces bolted together. The camera is attached to the Video Camera Mount by fasteners. The camera is COTS which has not been modified. The mirror is polished aluminum and is attached to the mount by fasteners.

2.5 STILL CAMERA ASSEMBLY

The Still Camera Assembly is comprised of an off the shelf Nikon D100 35 mm Camera with 35mm Lens attached to the Still Camera Mount. It is clamped in position over the Experiment Assembly (See Figure 1).

The Still Camera Mount (Figure 8) is made up of anodized aluminum pieces bolted together. The Still Camera Mount Base attaches to the 70mm grid on the MSG cold plate using 2 M6 thumbscrews. The Still Camera Mount is also side attached to the Experiment Assembly using 1 M6 thumbscrew. The camera is mounted to the Still Camera Mount by two steel pins with a quick disconnect feature of a spring-loaded pin. The camera is COTS equipment, which has been modified by removing the 9VDC battery. Power is not provided to the camera via the connector normally used for power from an AC power adapter. SPICE provides power to the camera from a 9VDC converter located inside the Power Converter Box. The 9VDC Converter now provides the power to the camera for flash and shutter action.

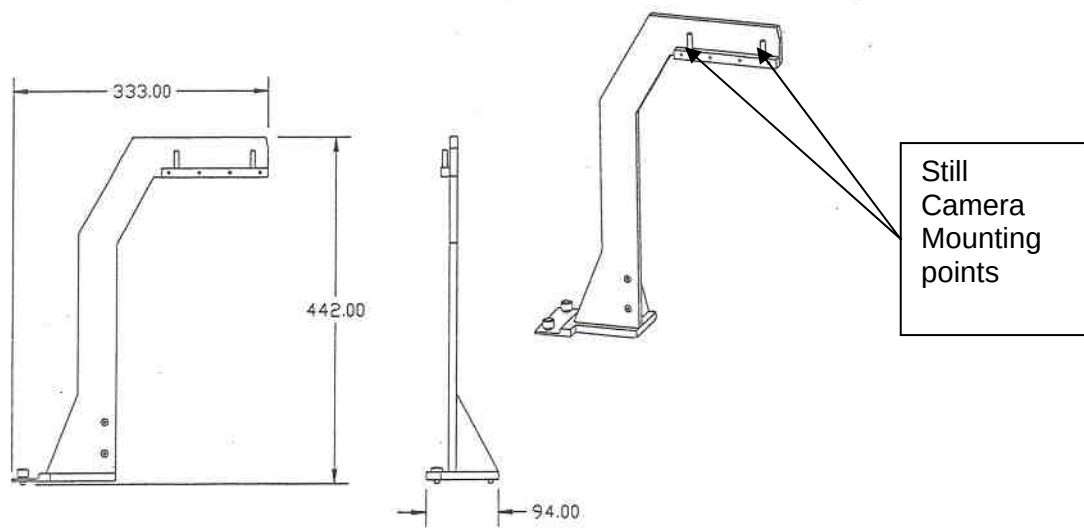
Figure 6 – SPICE Still Camera



Figure 7 – SPICE Video Camera Assembly



Figure 8 – SPICE Still Camera Mount



SPICE Still Camera Mount 60054MAFH400

Note: Dimensions are in millimeters.

2.6 SPICE COMPONENT CASE

The SPICE Component Stowage Case is made out of anodized aluminum and is 36.7 cm X 46.2 cm X 15.7 cm (see Figure 9 & 9a). Inside the Component Stowage Case packed in foam is:

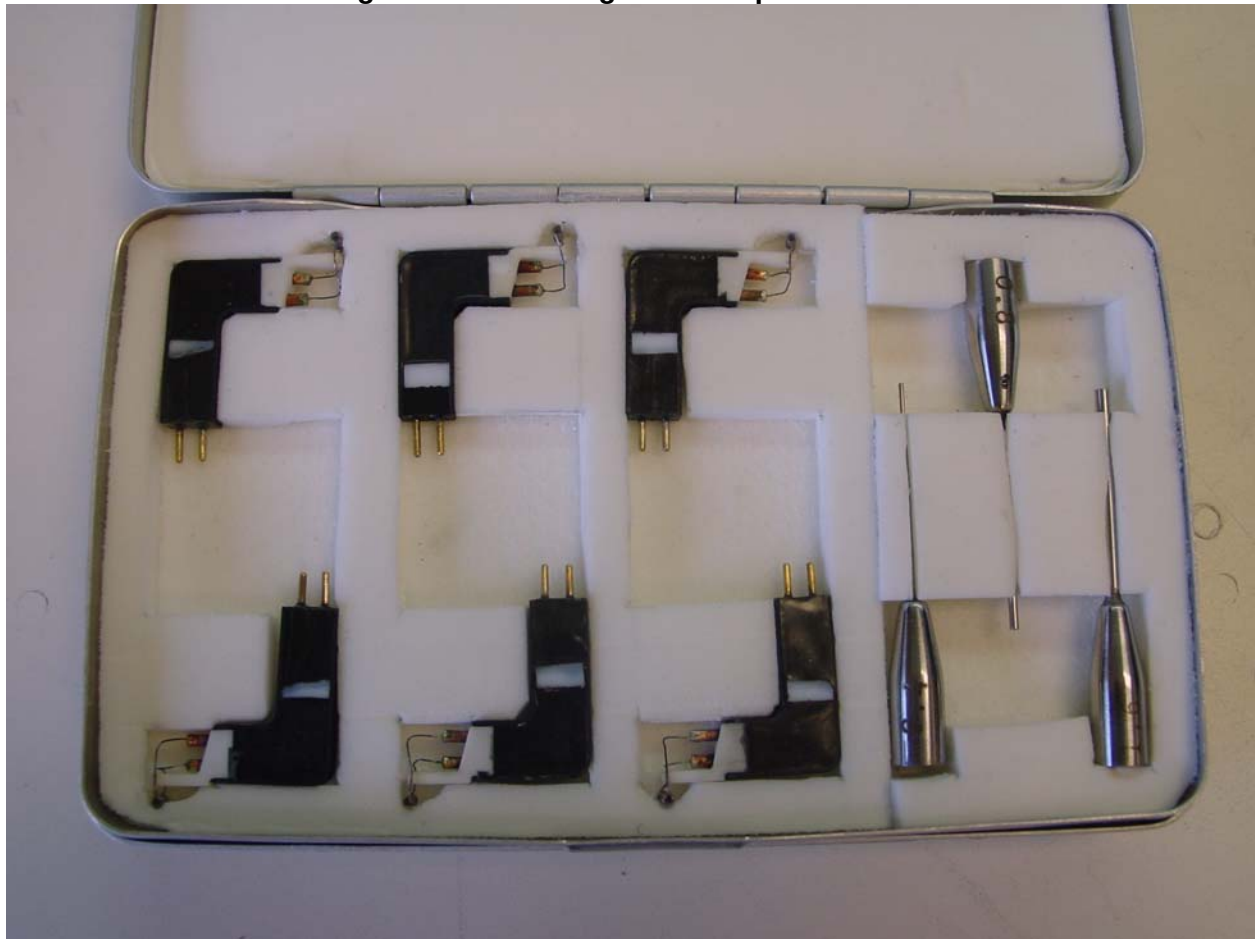
- Control Box Assembly
- Video Camera Assembly
- Still Camera Assembly
- Still Camera Cable Assembly
- Power Cable Assembly
- Video Cable Assembly
- Feed-through Cable Assembly
- 6-Ignitor Assemblies (Stowed in SPICE Ignitor Stowage Case)
- 3-Burner Tube Assemblies, 0.4mm, 0.8mm, 1.6mm (Stowed in SPICE Ignitor Stowage Case)

The Experiment Assembly, Power Box, and Video Box will be stowed in soft stowage.

Figure 9 – SPICE Component Case



Figure 9a – SPICE Ignitor Component Case



2.7 GAS BOTTLE ASSEMBLY STORAGE CASE

Two Fuel Bottle Storage Cases will be used for SPICE. Each storage case holds 6 Gas Bottle Assemblies. Storage Case No. 1 contains Gas Bottle Assemblies 1-6 and Storage Case No. 2 contains Gas Bottle Assemblies 7-12. Each Storage Case is made out of anodized aluminum and is 36.7" X 46.2" X 15.7" (see Figures 9c and 9d). Gas Bottle Assemblies when they are not in use by SPICE will be stored in their applicable Storage Case in closed cell foam.

Figure 9b Gas Bottles Stowage Case. (Aluminum Tape Not Installed)



Figure 9b – SPICE Gas Bottle & Component Cases



2.8 SPICE GAS BOTTLE ASSEMBLIES

Each Gas Bottle Assembly (See Figure 10a) is comprised of a DOT 150cc Stainless Steel Bottle with a Diaphragm Valve, Quick Connect with a Cap Plug that can be connected to the SPICE Experiment Assembly Fuel Delivery System Plumbing once the Cap Plug is removed. There are 12 individual Gas Bottle Assemblies each labeled and containing their applicable fuel mixtures listed in Table 1.

The bottles will have the handles tie wrapped such that the handles can not be opened inadvertently or by launch loads. See Figure 10b. This figure is not the flight configuration. For flight the tie wrap will be installed to prevent sharp edges. The assemblies will also be placed in a case with foam cut outs for the handles which will prevent them from rotating.

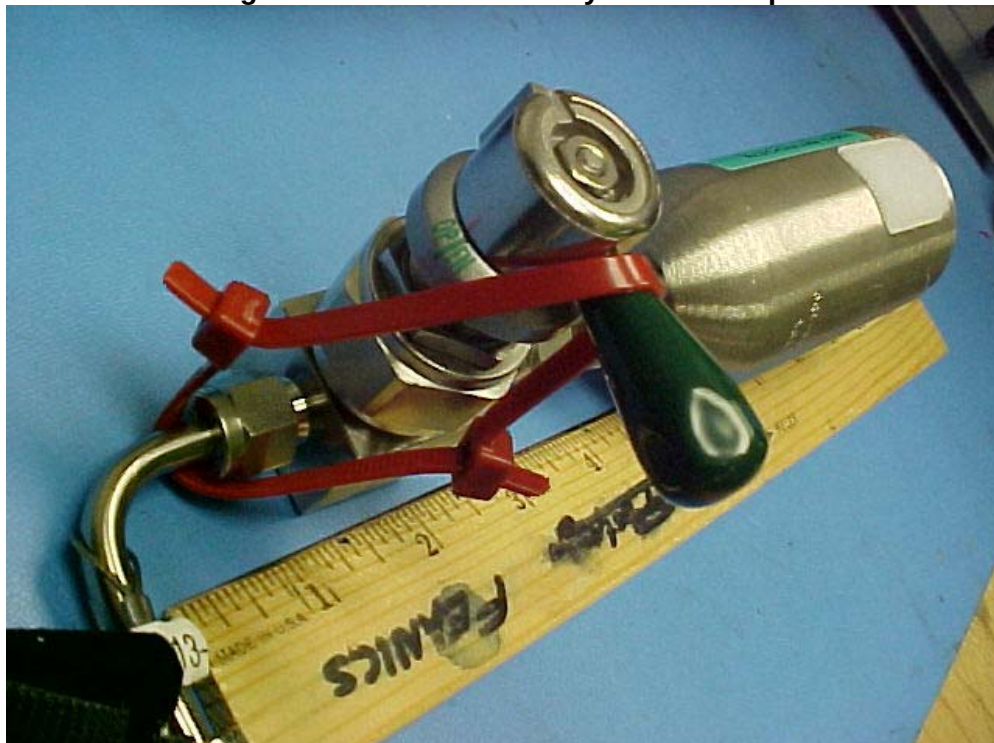
Table 1: Gas Bottle Assemblies

Gas Bottle Assemblies		Part #
Full Name	Fuel Bottle Label/Fuel Type & Percentage	
Smoke Point In Co-flow Experiment Gas Bottle Assembly 1	E1/Ethylene 100% @ 210 psi. (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-1
Smoke Point In Co-flow Experiment Gas Bottle Assembly 2	E2/ Ethylene 100% @ 210 psi. (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-2
Smoke Point In Co-flow Experiment Gas Bottle Assembly 3	M1/Methane 100% @ 210 psi. (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-3
Smoke Point In Co-flow Experiment Gas Bottle Assembly 4	M2/Methane 100% @ 210 psi. (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-4
Smoke Point In Co-flow Experiment Gas Bottle Assembly 5	P1/Propane 100% @ 110 psi. (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-5
Smoke Point In Co-flow Experiment Gas Bottle Assembly 6	P2/Propane 100% @ 110 psi (Note: Stowed inside SPICE Gas Bottle Case 1)	60054MAFH090-6
Smoke Point In Co-flow Experiment Gas Bottle Assembly 7	PR1/Propylene 100% @ 135 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-7
Smoke Point In Co-flow Experiment Gas Bottle Assembly 8	PR2/Propylene 100% @ 135 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-8
Smoke Point In Co-flow Experiment Gas Bottle Assembly 9	PR75-1/Propylene 75%, Diluent GN2 @ 180 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-9
Smoke Point In Co-flow Experiment Gas Bottle Assembly 10	PR75-2/Propylene 75%, Diluent GN2 @ 180 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-10
Smoke Point In Co-flow Experiment Gas Bottle Assembly 11	PR50-1/Propylene 50%, Diluent GN2 @ 180 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-11
Smoke Point In Co-flow Experiment Gas Bottle Assembly 12	PR50-2/Propylene 50%, Diluent GN2 @ 180 psi. (Note: Stowed inside SPICE Gas Bottle Case 2)	60054MAFH090-12

Figure 10a – SPICE Gas Bottle Assembly



Figure 10b Bottle Assembly with tie wrap.



2.9 CABLE ASSEMBLIES

SPICE has the following cable assemblies, Power Cable Assembly, Video Cable assembly, Still Camera Cable Assembly, and Feed-through Cable Assembly. These are all stowed in the SPICE Component Stowage Case. The wires are Teflon wires rated for 200 C and are bundled together with expandable cable mesh. See SPICE Wire Derating Analysis 60054-ANA-017.

The Power Cable Assembly consists of the cables which run from the Power/Video Unit to the Experiment Assembly, to the Video Camera, and to the Still Camera. This cable supplies power to these units.

The Video Cable assembly is a video data cable which connects the Video Camera to the Power/Video Unit for signals.

The Still Camera Cable Assembly. This cable connects the Power/Video unit to the Still Camera for signal.

The Feed-through Cable Assembly has 4 ends and is used for signals except for one 12 volt wire to the experiment assembly. One end connects to the MSG feed through connector to provide communication to the SPICE Control Box. The control box itself has a cable and connector which is permanently attached to the Control Box. The control box consists of two 10 turn pots, three toggle switches, a feed through, a multi-positional switch which switches resistors to set the radiometer gain. Another end of the cable connects to the Still Camera to snap pictures. Another end connects to the Experiment Assembly to start data collection, enable fuel flow, set the radiometer, set the fuel flow, set the fan flow, and enable the ignitor. The fourth end connects to the Power/Video Unit.

2.10 SOFTWARE SUMMARY

Data gathering software resides in the Experiment Assembly's Data Logger. The Control Box Assembly is equipped with switches for controlling the fan speed and fuel flow rate, and activates the Ignitor and data acquisition. Video images are taken with a Video Camera. The video data is recorded on a MSG provided video recorder. Digital pictures are also taken of the top view through the Experiment Assemblies Lexan window cover. The video images, together with the digital data output of the data logger, are sent to a video box, which writes the digital data (air flow velocity, fuel flow, fan voltage, and time) on the video signal. The video signal from this box is then recorded by the MSG video recorder and down linked.

The SPICE experiment will fail-safe in the event of total or partial power loss, i.e., the hardware will not be damaged or pose any hazard in the event of a power disruption. In the event of a total power loss, the SPICE experiment cannot be operated and the normally closed MFC will shut, cutting off fuel flow. Experiment sequencing is controlled by the crew. Software is only used for data recording in the SPICE Experiment.

3 SPICE FLIGHT OPERATION

The SPICE experiment currently has a CEF submitted for launch on ULF-2. This manifest is currently being reviewed. The experiment will be performed inside the Microgravity Science Glove-box (MSG) on the International Space Station. On-orbit operations will consist of SPICE Hardware Set-up and Stowage, SPICE Operations, Fuel Purges, along with Bottle, Burner Tube, and Ignitor Assembly exchanges. Instructions for off-nominal activities are being

developed. Flight Operational Procedures are listed in Table 3 of section 3.1 The SPICE hardware including the Component Stowage Case will be soft stowed in the MPLM for launch and landing. The Gas Bottle Stowage Cases with 6 bottles in each case will be delivered at approximately L-48 hours and launched in the Middeck. The hardware configuration for stowage during launch and landing is still under development. The safety aspects of launch and landing will be covered in the safety section.

3.1 DETAILED OPERATIONS

The SPICE SET UP procedure provides instructions to install SPICE hardware in the MSG WV, configure the SPICE Control Box Assembly outside the MSG Facility, perform an ignitor test, and perform a still camera test.

The SPICE STOW procedure provides instructions to disassemble and stow the SPICE Experiment after SPICE Operations have been completed.

SPICE OPERATIONS procedure provides instructions to conduct combustion experiment tests according to the Test Matrix of Table 4 in section 3.2.

The SPICE GAS BOTTLE EXCHANGE procedure provides instructions to remove and replace Gas Bottle Assemblies from the Experiment Assembly as required by the Test Matrix of Table 4. This includes configuring the MSG video recorders and SPICE Still Camera Assembly, activating the experiment, performing a flame test, powering down, and logging test data.

The SPICE FUEL PURGE procedure provides instructions to conduct a fuel purge of the SPICE Fuel Delivery System so that any unburned gas remaining from the previous combustion experiment test is purged out and replaced with the next combustion test gas.

The SPICE BURNER TUBE EXCHANGE procedure provides instructions to change out the previous Burner Tube diameter size with a new Burner Tube diameter size as required to support the Test Matrix of Table 4.

The SPICE IGNITOR EXCHANGE procedure provides instructions to remove the used ignitor and install a new ignitor in the SPICE Experiment Assembly.

Table 3 lists the flight operation procedures for the SPICE experiment and figure 11 provides the experiment flow.

Table 3- Flight Operations

SPICE Flight Procedures	Summary
MGUEMSGSPCEN001	SPICE SET UP
MGUEMSGSPCEN002	SPICE STOW
MGUEMSGSPCEN003	SPICE OPERATIONS
MGUEMSGSPCEN004	SPICE FUEL PURGE
MGUEMSGSPCEN005	SPICE GAS BOTTLE EXCHANGE
MGUEMSGSPCEN006	SPICE BURNER TUBE EXCHANGE
MGUEMSGSPCEN007	SPICE IGNITOR EXCHANGE

3.2 TEST MATRIX

The SPICE Combustion Experiment Test Matrix is shown below in Table 4.

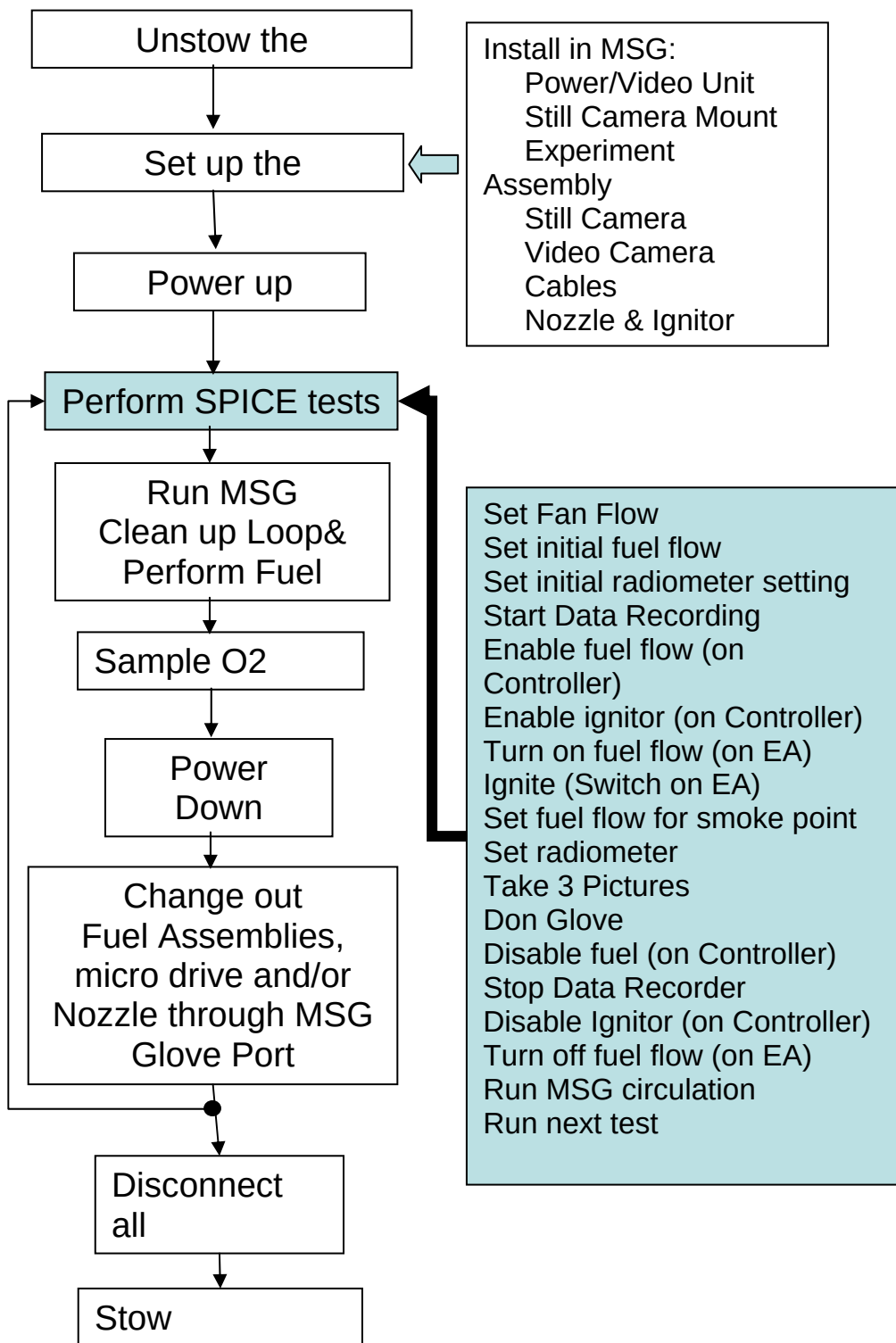
Table 4- SPICE Test Matrix

TEST POINT NUMBER	AIR COFLOW RATE (% MAX COFLOW VELOCITY)	FUEL FLOW RATE (% LAMINAR SMOKE POINT)	BURNER DIAMETER (MM I.D.)
1	50%	100%	0.4
2	50%	50%	0.4
3	100%	100%	0.4
4	50%	100%	0.8
5	50%	50%	0.8
6	100%	100%	0.8
7	50%	100%	1.6
8	50%	50%	1.6
9	100%	100%	1.6
10~14*	50%-100%	50%-100%	0.4, 0.8, 1.6

Test Points 10-14 represent flame tests that will be repeated. They are not pre-determined.

- Fuels 1-6: methane, propane, ethylene, and propylene (0, 75%, and 50% nitrogen dilution)
- The test matrix above is representative of a single fuel.
- There are nine (9) specified test points plus ~ 5 selected repeat tests per fuel. Thus the SPICE test matrix will include 54 predetermined flames plus ~ 30 “repeat” flames.

Figure 11 SPICE Experiment Flow



4 SAFETY ASSESSMENT OF SPICE

This section addresses the hazards investigated for SPICE. If the hazard has not been eliminated by design with its rationale captured in this section, its controls are documented in either the JSC 1230 Form or in unique hazard reports developed for the SPICE experiment. The SPICE Flight Payload Standardized Hazard Control Report (**STD-SPICE-01**) is discussed in Section 4.2 and the JSC 1230 form is provided in Attachment D. Unique Hazards Reports (**SPICE-U1, SPICE-U2, SPICE-U3, SPICE-U4, SPICE-U5, SPICE-U6, and SPICE U8**) are provided in Attachment E.

This section is comprised of three parts, as follows:

- Section 4.1 – Discussion of the safety assessment approach for SPICE.
- Section 4.2 – Discussion of each section of the JSC Form 1230.
- Section 4.3 – Discussion of other safety hazards.
- Section 4.4 – Discussion of hazard report verification classifications.
- Section 4.5 – Discussion of SPICE test failures, anomalies and accidents.

4.1 SAFETY ASSESSMENT APPROACH FOR SPICE

Since SPICE is similar in functional design and operation with its predecessor ELF, this SPICE Phase II Safety Data Package was developed using the hazard reports documented in MSFC Document JA-2378 which was the ELF Phase III SDP for the USMP-4/STS-87 Shuttle Middeck Glovebox (MGBX) mission.

The SPICE experiment also uses new fuels and mixtures and a revised Test Matrix than that of its predecessor ELF and a new hazardous material and combustion By-Product list has been submitted to JSC Toxicology for hazard evaluation and assessment. The new fuel mixtures and combustion By-Products applicable to SPICE experiment are provided in the JSC HMST Attachment A.

4.2 STANDARD HAZARDS (JSC FORM 1230)

This section discusses each Standard Hazard applicable to SPICE as shown in the STD-SPICE-01 Hazard report (JSC Form 1230) Attachment B.

4.2.1 Structural Failure

SPICE hardware is designed to meet the expected launch, on-orbit and return loads specified in MSFC-RQMT-2888, Paragraph 3.1.1.1, and Table 1 which is consistent with SSP 57000 requirements. The stowage loads in MSFC-RQMT-2888 are based on a worst case condition of the Middeck and MPLM stowage loads. SPICE hardware will be stowed in soft transfer bags and transported in the MPLM for transfer to the ISS. All the SPICE hardware will be soft stowed for return. ZIN Tech Engineering has performed an analysis of test levels specified in NASA-STD-7001 (NASA Payload Vibration Test Criteria) and MSFC-RQMT-2888 to define vibration test levels to encompass the launch and landing load levels the hardware will be exposed to and has created a composite vibration loading profile to envelope the levels defined in both MSFC-RQMT-2888 and NASA-STD-7001. The SPICE flight hardware has undergone acceptance vibration testing at these levels. A test report has been prepared to confirm that the test was successfully completed.

The SPICE hardware is also designed to meet the expected on-orbit loads during operation in the MSG. SPICE hardware will be fastened to the MSG working volume using captive, threaded fasteners. This will meet the on orbit load requirements as specified in SSP 57000 and MSFC-RQMT-2888. The hardware inside the MSG will not be exposed to on-orbit kick loads during operation. The control box is subjected to crew induced loads and is capable of withstanding the loads per SSP 57000 and MSFC-RQMT-2888.

The Gas Bottle Assembly Stowage Case and the SPICE component cases have two latches which keep them from opening in stowage. The case itself is packed in foam and placed in the stowage volume which will prevent the case from opening during launch, therefore they are not safety critical.

SPICE structural failure is addressed in Section 1 of STD-SPICE-01.

4.2.2 Structural Failure of Sealed Containers

Structural failure of sealed containers containing hazardous materials are addressed in Unique Hazard report **SPICE-U2** for Rupture/Leakage of the SPICE pressurized fuel delivery system. Contamination Hazard for release of hazardous sample fluids is addressed in Unique Hazard Report **SPICE-U1**. The SPICE Fuel Delivery System is designed to meet the requirements of NASA-STD-5003 for Fracture Control of pressure vessels and pressurized components.

4.2.3 Structural Failure of Vented Containers

The Experiment Assembly is open on each end (fan end and screen end) and has a removable polycarbonate Lexan Window Cover; therefore it will not contain an internal atmosphere and is not hermetically sealed. The design of the Experiment Assembly will be reviewed to verify that it can maintain structural integrity in the event of a rapid depressurization / re-pressurization per SSP 52005B, Para. 4.1.12. (See Table 5). The window on the top of the EA is polycarbonate and is held on by two screws. The window cover is held on the bottom by a metal hinge and on the top by two gun bolt type sliding latches.

The designs of the Nikon Digital Camera, Video Camera, Gas Bottle Assembly and Component Parts Cases will also be reviewed to verify that they can maintain structural integrity in the

event of a rapid depressurization / re-pressurization scenario per SSP 52005B, Par. 4.1.12. (See Table 5) These enclosures are not hermetically sealed and have gaps between the mating edges of the enclosure structure.

The designs of the Control Box, Video Control Box, Power Control Box, will also be reviewed to verify that they can maintain structural integrity in the event of a rapid depressurization / re-pressurization scenario per SSP 52005B, Par. 4.1.12. (See Table 5) These aluminum boxes are not hermetically sealed and have a gap between the box base and window cover.

Structural Failure of Vented Containers is addressed in Section 3 of STD-SPICE-01.

Table 5 – Depressurization Times for SPICE Vented Containers from SPICE-ANA-001

Enclosed Volume Description SPICE	Depressurization (Time in seconds)	Final Pressure Pa (psi)	Depressurization / Repressurization Rate Pa/sec (psi/min)
Experiment Assembly	0.03	160.5 (.0233)	3371315 (29353)
Power/Video Unit (total: 2 enclosures)	0.03	160.5 (.0233)	3371315 (29353)
Igniter/Burner Tube Case	0.01	160.5 (.0233)	10113946 (88060)
Control Box Assembly	0.03	160.5 (.0233)	3371315 (29353)
Still Camera Assembly	0.13	160.5 (.0233)	777996 (6774)
Video Camera Assembly	0.04	160.5 (.0233)	2528486 (22015)
Stowage Boxes (total: 3 enclosures)	0.17	160.5 (.0233)	594938 (5180)
Component Case	0.17	160.5 (.0233)	594938 (5180)

4.2.4 Sharp Edges

The SPICE hardware is designed to meet the intent of SSP 50005 / NASA-STD-3000. The flight drawings will incorporate instructions to break all corners and edges to remove any potential sharp edge. A sharp edge inspection will be performed on all exposed sub-assemblies as well as the total integrated assemblies.

Sharp Edges are addressed in Section 4 of **STD-SPICE-01**.

4.2.5 Shatterable Material Release

Use of frangible material is limited to the fuses, flow straightener, Still Camera lens and Video Camera lens. The SPICE Experiment Assembly Window cover is made out of Polycarbonate Lexan 9034 and is not shatterable.

The camera lenses will be launched with protective covers on since the glass of the Still and Video Camera lens are recessed in their respective housings. During SPICE set-up Crew procedure the cameras will be installed into the MSG Working Volume before the protective covers are removed from the camera lens. The crew will assess/inspect the lens for damage at the time of removal. Once the caps are removed and assessment is acceptable the MSG Working Volume enclosure will then provide containment when the protective covers have been removed from the cameras during experiment operations. These items have also been vibration tested and inspected.

The flow straightener is made of Cordierite which is a ceramic and is contained inside the fan housing on the Experiment Assembly (EA). Similar to acceptable controls for optical glass, the EA is also packaged in a clear plastic bag and inspected by the crew after launch. If it were to break, the pieces could be seen through the window of the EA. The EA is inside MSG when removed from the bag and when operating. If any particles are fine enough to get through the screens they will be ingested in the MSG HEPA filters. Since intake and outlets are screened and inside EA there are no identifiable failure modes on orbit to damage the ceramic. The 3 X 3 X 1 inch piece of Cordierite is also bound along the 3 X 3 perimeter which will prevent damage and break up of this honey comb type material. Similar to control of optical glass such as lenses (form 1230 section 5) the item is vibration tested and inspected for damage afterwards. See figures 13a and 13b.

The fuses in the Power Box are also glass. They are wrapped in Kapton tape, which is never removed, to contain any breakage that might occur. They are contained inside the Power Conditioning Box. The crew does not have access to the interior of the Power Conditioning Box.

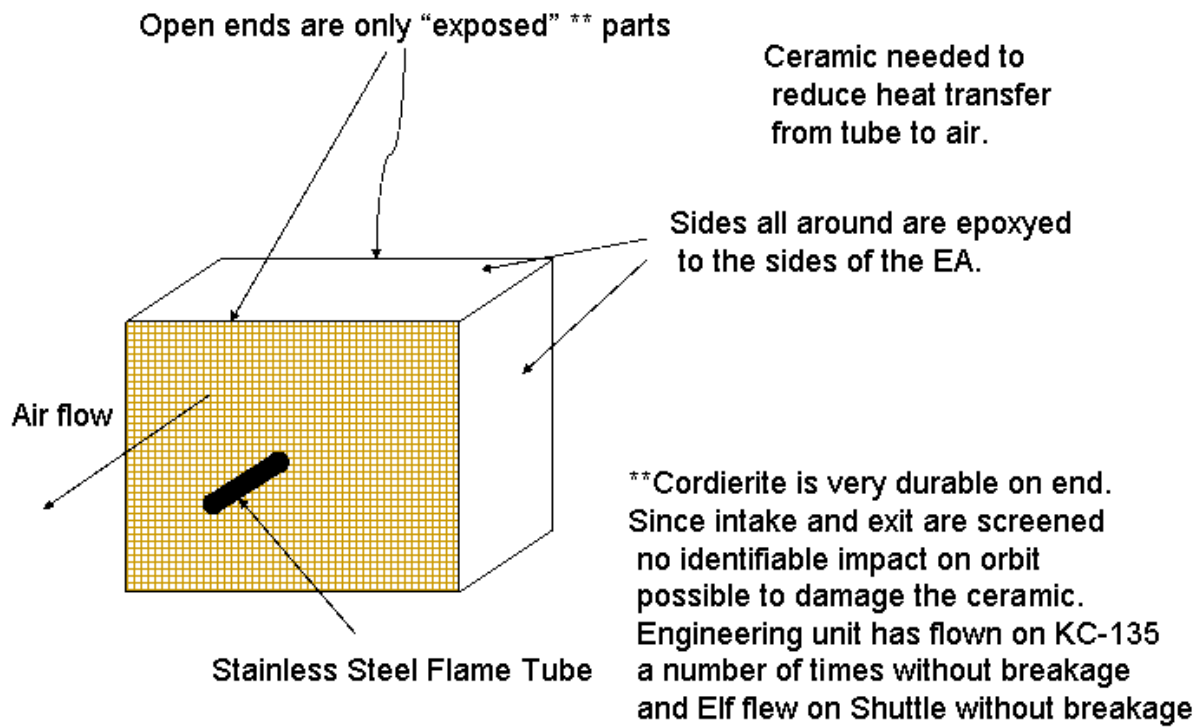


Figure 13a Cordierite Characteristics

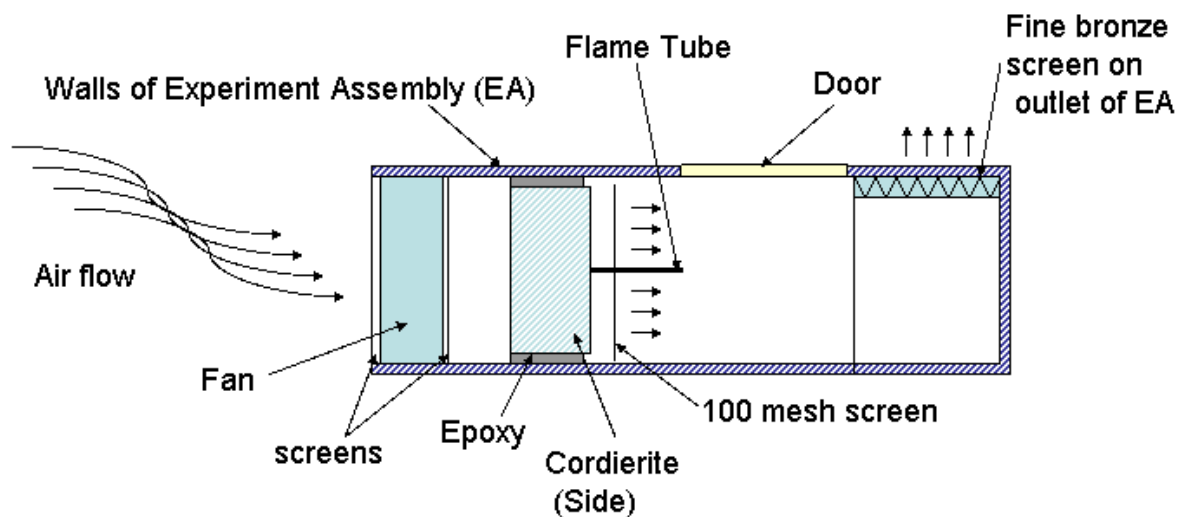


Figure 13b Cordierite Location and Containment.

All frangible components will undergo a vibration test at workmanship levels in a soft stowed configuration followed by a post-test visual inspection and functional test to verify ability to survive launch/landing loads. Shatterable Material Release is addressed in Section 5 of **STD-SPICE-01**.

4.2.6 Flammable Materials

"A" rated materials in accordance with MSFC-SPEC-527/JSC 09604F are selected for experiment hardware to control the hazard associated with the use of flammable materials. Any non-"A" rated materials will be used in a non-flammable configuration in accordance with NSTS 22648, Flammability Configuration Analysis for Spacecraft Applications, or justified for usage via a Materials Usage Agreement (MUA). Materials usage is documented on a Materials Identification and Usage List (MIUL). Materials usage will be analyzed by NASA/GRC/SMAD M&P and certified for flight via the GRC/JSC Materials and Processes Intercenter Agreement. Flammability of fuels is addressed in paragraph 4.3.6.1.

Flammable Materials are addressed in Section 6 of **STD-SPICE-01**.

4.2.7 Materials Offgassing

Representative SPICE hardware, flight-like cables and other miscellaneous items have been toxic off-gas tested in accordance with NASA-STD-6001, Test 7. See letter ED36B(03-060) attached to 1230 form. Offgas test results will be analyzed by NASA/GRC/SMAD and certified for flight via the GRC/JSC Materials and Processes Inter-center Agreement.

Materials Off-gassing is addressed in Section 7 of **STD-SPICE-01**.

4.2.8 Non-Ionizing Radiation (EMI)

4.2.8.1 *Non-Transmitters*

SPICE is designed for compliance with the radiated and conducted emissions requirements specified in MSFC-RQMT-2888 as defined in SSP 30237, Space Station Electromagnetic Emission and Susceptibility Requirements. All electronic assemblies have been tested for conformance to this specification including the determination of the maximum ripple voltage emissions appearing on the SPICE input power leads. The power conversion box contains a filter to meet the requirements of MSFC RQMT-2888. This filter reduces the step from milliamp draw to 3 amp draw by the still camera when taking a picture. See figure 14. SPICE will only be operated inside of the MSG working volume. The test results have been submitted to NASA/MSFC MSG personnel for their review and approval. An ISS Electromagnetic Emissions Panel approved Technical Interface Agreement has been sought for RE02 exceedance.

EMI Radiation is addressed in Section 8.1 of **STD-SPICE-01**.

4.2.8.2 *Lasers*

There are no lasers in the SPICE experiment.

4.2.9 Battery Failure

The Batteries used in the SPICE experiment meet the criteria of small commercial batteries stated in Section 9.2 of the JSC Standardized Hazard Report JSC 1230. The following SPICE hardware contains small commercial batteries that are verified as permanently soldered into their respective electrical circuits and have completed successful verification of in circuit voltage measurements and verification of circuit isolation from external power sources.

The NIKON D100 Digital Camera contains Button Cell 20mmX3.2mm, type Li-Mn-O₂, 3 Volts DC 65mAh, Maxell Model No.: ML2032T6. The battery is fully contained within the Digital Camera and cannot be accessed unless complete Camera disassembly is performed. The purpose of the camera battery is to maintain date, time and camera operational settings. The cell is soldered to board.

The Horita Video Titler Card Cell consists of two cells in series. The cells are 15.5 X6.3 mm Rechargeable Nickel Metal Hydride cells 1.2 VDC each, 60mAh, Model Number GP60BVH, Manufacturer GP Batteries International LTD. The cells will not be charged and are prevented

from doing so by a pull up resistor. The battery is soldered to the board, fully contained with in the Video Control Box and cannot be accessed unless the Video Control Box is disassembled. The purpose of the Video Titler Card Cell is to maintain Video Titler operational settings.

The Dallas Tattle-Tale Card I-Button Data Device, Model Number DS1994, contains a 12.5mmX2.5mm Li-CFx Battery, 3VDC, 50mAh, Model No. BR1225, Manufacturer RAYOVAC. The Lithium Battery is soldered to the boards and fully contained within the I Button Data Device and the I Button Data Device is contained inside the Experiment Assembly. The purpose of the I Button Data Device Lithium (Li-CFx) Battery is to maintain experiment data memory when external power is removed from the Experiment Assembly. The documents for the I-Button Data Device state that it does not use any power to charge the lithium battery. It has a limited life of greater than 10 years based on the low power required to sit idle. Specifically there is a statement on page 46 that the ROM logic can be operated by the parasite power even after the lithium cell is exhausted in 10 years or more. This implies that the other sections of memory may not be operational if the lithium battery dies. When data is being exchanged with the I-Button Data Device, the data line is used to charge a capacitor internal to the I-Button Data Device and that capacitor is used to power the read write operations. Parasite power is an internal I-Button Data Device capacitor that is charged by the data line (converted RS-232).

Battery Failure is addressed in Section 9.2 of **STD-SPICE-01**.

4.2.10 Touch Temperatures

SPICE hardware and electronics have been designed and tested to meet touch temperature limits of 49°C specified in the Interpretation Letter MA2-95-048. Also during on-orbit operation, the SPICE hardware is enclosed inside the MSG working volume with no direct crew contact with the SPICE Experiment Assembly during ignition or combustion processes (experiment runs). Also during post experiment runs the airflow provided by the MSG Air Handling Unit (AHU) during by-product clean up will also aid in the dissipation of any residual heated air within the MSG working volume. SPICE by design does not require external cooling during experiment operations.

A thermal test has been conducted to verify that the touch temperature limits of MA2-95-048 are met. The fuel nozzles were tested and found that the worst case nozzle reached 89 degrees C but cooled to 33 degrees C in less than 60 seconds after flame off. This was without EA air flow which will be present at the end of a burn decreasing cooling time. A wait time will be imposed before EA chamber access after flame off.

Touch Temperature is addressed in hazard report U6.

4.2.11 Electrical Power Distribution

The electrical power distribution system meets station interface circuit protection requirements of SSP 57000 (through the MSG Working Volume secondary power sources) and payload wire sizing and circuit protection requirements of Interpretation Letter TA-92-038. The SPICE hardware will receive circuit protection from the MSG conditioned power sources (28VDC @ 7 Amps), (+/-12VDC @ 2 Amps) and (5VDC @ 4 Amps). SPICE will provide additional circuit protection in the Power Conditioning Box (5-Amp fast-acting cartridge fuses) on the +28 VDC and +5 VDC power lines. The SPICE +/- 12VDC power line does not require additional circuit

protection (fusing) since the wire sizing is appropriately de-rated for the provided MSG conditioned power source.

Besides the circuit protection provided by the MSG circuit breakers and SPICE Fuses, the DC/DC converters also provide additional protection by current limiting their output in the event of an over-current condition. See schematic of the SPICE Power Distribution System attached to hazard report STD-SPICE-01, Section 11. See SPICE Wire De-rating Analysis 60054-ANA-017.

SPICE does not meet the Power Distribution Single Point Grounding requirements of MSFC-RQMT-2888. The Hitachi Video Camera HV-C20U-S4 video camera, (PN)7009.80.AA (specially modified for the MSG Program), the Nikon D100 Digital SLR Camera and other similar commercial off-the-shelf (COTS) cameras (modified or unmodified) violate SSP30240, 3.2.1.2, Secondary and Tertiary Electrical Power (single point grounding) by connecting the incoming 12 Vdc power return to camera structure (the second time the 12 Vdc is grounded to structure). SPICE has submitted a Tailoring Interpretation Agreement for these occurrences. See schematics attached to hazard report STD-SPICE-01, Section 11.

SPICE meets the bonding and grounding requirements of SSP 30245 and SSP 30240. Bonding and grounding of the SPICE hardware is accomplished via several different methods; via the chassis conductor pin provided via Pin L at the P303 Power Connector (i.e., conductor bond path), via the overall shield on the P303 Power Cable (i.e., using a grounded backshell), and via the metal-to-metal bond of SPICE structure to the MSG mounting plate. See schematic attached to hazard report STD-SPICE-01, Section 11.

The SPICE experiment will fail-safe in the event of total or partial power loss, i.e., the hardware will not be damaged or pose any hazard in the event of a power disruption. In the event of a total power loss, the SPICE Experiment cannot be operated. The MFC is normally closed, so power loss will stop fuel flow. See Hazard Report SPICE-U5 for loss of power case effect on fuel flammability control.

Power into SPICE is filtered to prevent surges on the MSG power supply. See Figure 14 for the 28 volt power filter schematic.

A temporary or intermittent loss of power will cause the SPICE control software to "reboot" (the unit will act as if it was being turn "ON" for the first time).

Electrical Power Distribution is addressed in Section 11 of **STD-SPICE-01**.

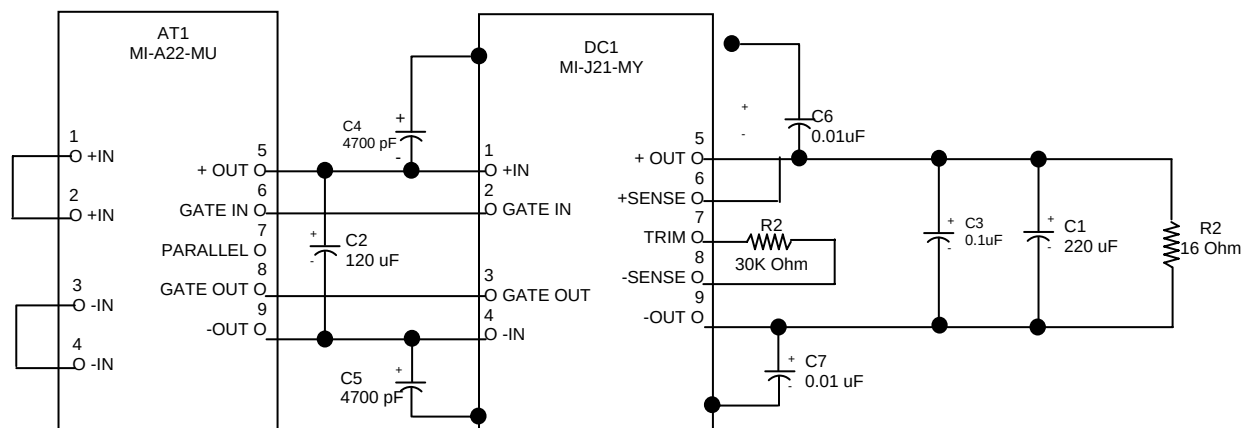


Figure 14 28 volt power filter

4.2.12 Ignition of Flammable Atmospheres in the Payload Bay

Not applicable, as the SPICE hardware is not mounted/transported in the the Shuttle Payload Bay.

4.2.13 Rotating Equipment

The SPICE Experiment Assembly has one internal Fan to (1) study convective flow during combustion and (2) to remove combustion byproducts from the Experiment Assembly in between tests. The fan is DC-powered and is made of fiberglass-reinforced plastic (See Sections 6 and 7 of **STD-SPICE-01** for materials requirements). The fan diameter is 3 inches, with a speed of 2950 RPM. It is a brushless DC motor provided with 12 V. Although the Fan shroud is plastic it is also fully enclosed in aluminum housing with stainless steel mesh (Type 304ss, 71% of open area) covering the fan inlet and outlet openings. Analysis shows that this fan has a low kinetic energy level ($\ll 19,310$ J) and is fully contained with the Experiment Assembly according to NASA-STD-5003 requirements.

Rotating Equipment is addressed in Section 13 of **STD-SPICE-01**.

4.2.14 Mating/Demating of Power Connectors

SPICE meets the requirements of Letter MA2-99-170. The worst-case voltage that would be encountered during mating/demating of SPICE power connector P303 is 29 VDC maximum which is lower than 32 VDC with a corresponding current of 1.0 Amp. The highest current drawn during SPICE ignition is 2.8 Amps on the +5 VDC power line. Crew procedures require that the MSG circuit breakers that provide power to the MSG J303 connector be turned off prior to any crew interface with the SPICE Experiment.

To prevent connector mismating, all SPICE electrical connectors are of different shell sizes and/or keying. In addition, they will all be labeled to help identify the connector interface in

accordance with MSFC-RQMT-2888. The connector and associate harness will also be inspected for proper pin assignment per MSFC-RQMT-2888.

Normal operating power draw of the Experiment is 27 Watts with a maximum power draw, which only occurs during ignition, of approximately 42 Watts for < 10 sec.

All power connectors utilized by SPICE conform to the MIL-C-38999 Series I standard which feature a scoop-proof design that completely encloses the pin/socket contacts during mating and de-mating operations. Additionally, the powered side of the cable connectors on SPICE are terminated in sockets to provide protection against inadvertent shorting that could be caused by external debris.

Mating/Demating of Power Connectors is addressed in Section 14 of **STD-SPICE-01**.

4.2.15 Contingency Return and Rapid Safing

The SPICE experiment is not in operation mode while stowed in Shuttle for launch and landing. While on orbit the SPICE hardware will be either stowed in the ISS stowage, or in operation in the MSG. In either case, the SPICE hardware will not interfere with the IVA envelope or impede emergency IVA egress to the remaining contiguous pressurized volumes.

Design analysis will be performed for the SPICE hardware and planned flight operations for compliance to MA2-96-190.

Contingency Return and Rapid Safing is addressed in Section 15 of **STD-SPICE-01**.

4.2.16 Release of Mercury from Bulbs into Crew Habitable Environment

There is no mercury in any of the SPICE components.

4.3 OTHER SAFETY HAZARDS

The previous sections close discussion on the SPICE Flight Payload Standardized Hazard Control Report (**STD-SPICE-01**). Additional potential hazards and SPICE unique hazards are discussed below.

4.3.1 Pressurized Systems

SPICE pressurized components are the Gas Bottle Assemblies and the Experiment Assembly Fuel Delivery System. These components do not interface with the nitrogen or vacuum sources of the MSG.

Even though all of the components are rated for pressures well above 230 psi, the system MDP is determined by the point at which the fuel can turn to liquid. This pressure also considered the maximum temperature that the bottle could experience. SPICE pressurized lines and fittings will be designed to a factor of safety of 4.0 X System MDP of 230psid. SPICE Pressurized system components will be designed to a factor of safety of greater than or equal to 2.5 X System MDP

of 230 psid. The gas bottles are DOT bottles with a rated pressure of 500 psi which will be pressurized to a maximum of 230 psi. To prevent overpressurization this will be verified after filling that the bottles are the correct mixture and at the correct pressure. The bottles have been proof tested to 1000 psi. The Experiment Assembly is not sealed and will not pressurize in the event of full open flow from the mass flow controller. The maximum flow rate of the fuel system is 20 liters per minute. The fan that is used to cause flow is capable of 1120 liters per minute with a pressure drop of parts of a psi. Therefore the flow rate through the chamber is orders of magnitude greater than the maximum fuel flow into the chamber. Therefore the chamber will have no significant increase in pressure. See hazard Report SPICE U5 for analysis of pressure due to burning of worst case fuel load in the EA. In the event of power loss the Mass Flow Controller is normally closed, therefore fuel flow will stop.

Materials will be selected and evaluated for stress corrosion and/or fluid compatibility in accordance with SSP 30233, Space Station Requirements for Materials and Processes.

Hardware will be built in conformance with approved design drawings and specifications. The gas bottles will be filled to approved procedures to insure that the proper mixture of gas is placed in the correct bottle and each bottle is filled to the correct pressure.

The Gas Bottle Assemblies and the pressurized portions of the Experiment Assembly Fuel Delivery System will be proof pressure tested to 1.5 X MDP and Helium Leak tested for an acceptable leak rate of 1×10^{-4} scc/sec Helium in order to validate structural integrity and containment by design.

During on orbit operations, procedures will be provided to instruct the crew on proper Gas Bottle Exchange, Experiment Set-up and Operation to prevent Gas Bottle and Experiment Assembly Fuel Delivery System leakage.

Reference **Unique Hazard Report SPICE-U2. Flammability issues of the fuels are covered in SPICE-U4 and SPICE U5. Projectile issues are covered in SPICE-U3.**

4.3.2 Collision

Calculations show that for the bottle under full pressure, the propulsive force of the worst case, a break off of a tube, only create 0.6 lbs of force for 0.23 seconds. The force required to break off the tube is 10 lbs at the worst case moment arm with all factors of safety. If the bottle is being held such that 10 lbs of force is on the bottle, the 0.6 lb propulsive force can easily be withstood. Each Gas Bottle Assembly by design incorporates a single fault tolerant design. The Diaphragm Valve and Quick Disconnect are in series on the Gas Bottle Assembly. Should the Diaphragm Valve fail the Quick Disconnect will contain the gas. The failure of the Quick Disconnect is backed up by the containment of the Diaphragm Valve. Both the Diaphragm Valve and Quick Disconnect would need to fail simultaneously in order for the Gas Bottle Assembly to become a projectile. In addition there is a cap on the QC which although is not a radial vent cap it will restrict the flow and disperse it at an angle reducing the force. Even if both failures occurred the Gas Bottle Tubing at the QD end incorporates a 90 degree bend that would prevent the Gas Bottle assembly from gaining any significant speed ($<1.5\text{m/s}$), and force ($<0.6\text{lb}$) or direction that would present a collision hazard.

Reference **Unique Hazard Report SPICE-U3.**

4.3.3 On-Orbit Maintenance

The SPICE experiment hardware does not contain limited life items. No on-orbit maintenance is planned.

4.3.4 Fire Detection and Suppression

SPICE does not have any computer control of any FD& S functions and cannot interface directly with the ISS caution and warning system. SPICE flammability hazards are controlled by material selection, containment, and wire de-rating and circuit protection, (See Section 4.2.6 and 4.2.11). SPICE unburned Fuel contamination hazard is controlled under unique Hazard Report SPICE-U1. Structural Integrity of the SPICE Experiment Fuel Delivery System is controlled under unique Hazard Report SPICE-U2.

SPICE will receive Fire Detection capability from MSG through its Working Volume Rear Wall Temperature Sensor. The crew is required to be present during any burn per MSFC-RQMT-2888. Fire Suppression capability is provided by MSG through the use of a station Portable Fire Extinguisher (PFE) expelling CO₂ into the MSG Working Volume. See Attachment D for the completed Fire Detection and Suppression Reporting Form 1428 for the SPICE Experiment.

4.3.5 Crew Interaction

SPICE is designed such that any required access to SPICE hardware during flight or ground operations can be accomplished with minimum risk of injury to personnel. SPICE hardware components are designed to provide enough clearance for crew interface and usage and do not create a pinch or accessibility hazard.

The SPICE experiment complies with the requirements for Electrostatic Discharge (ESD) compatibility as specified in MSFC-RQMT-2888 document. An analysis will be prepared to demonstrate that that electrical equipment and any parts susceptible to damage by ESD will fail-safe when exposed to an Electrostatic Discharge (ED) equal or less than 4,000 Volts to the case or any pin on the external contacts. These (ED) voltages might be the result of charges accumulated and then discharged from ground personnel or crewmembers during equipment installation or removal. Reference unique hazard SPICE-U8.

Table 5 provides a listing of the SPICE Standard and Unique Hazards that are controlled by operating procedures.

Hazard Number	Hazard Control	Hazard Verification
SPICE STD 01: 5a.3	Shatterable Material Release – All materials are contained.	ZIN Tech Engineering will review crew procedure inputs to ensure that checks of containment and assessment of status prior to utilizing frangible hardware are included, and prepare a report confirming inclusion in the crew procedure inputs.
SPICE STD 01: 14b.1	Mating/demating powered connectors	ZIN Tech Quality will Inspect Crew Procedure inputs to ensure that step is in place to remove power from the MSG using the Control and Monitoring Panel (CMP). Reference PHR MSG-G2 control 6.2. and control 4.3.
SPICE U2: 6.1.1	Experiment Set-up and Operations on orbit are conducted per approved procedures to minimize risk of Gas Bottle Assembly damage. 1) One bottle unstowed at a time; 2) Crew maintains control at all times; 3) QC caps installed during transit and stowage; 4) Gas bottle installed last.	ZIN QA will review crew procedure inputs to ensure adequate instructions are provided for on-orbit Gas Bottle Assembly Experiment set-up and Experiment Operations. Specifically these are assembly sequence, description of mating parts, transfer/handling instructions, removal/replacement of QC cap instructions, and installation instructions.
SPICE U4: 1.3.3	Each Gas Bottle Assembly will have a Main Valve and a QC to prevent fuel flow and approved crew procedures will ensure tie wrap valve restraint is in place before stowage.	ZIN Safety to verify that the crew procedure inputs include valve restraint mechanism engagement before removal from MSG.
SPICE U5: 1.2.1.1	Only approved crew procedures will be used to turn on ignitor before fuel flow which will eliminate fuel built up.	ZIN QA to inspect the crew procedure inputs will be inspected to insure proper sequence.
SPICE U5: 1.2.3.1	Fuel flow is a two step process to enable fuel to flow for which approved crew procedures will be used.	ZIN QA to inspect the crew procedure inputs will be inspected to insure proper sequence.
SPICE U5: 1.3.3.1	Only approved crew procedures specifying turning off flue flow after flame out will be used to monitor burn to adjust flame	ZIN QA to inspect the crew procedure inputs for instructions on turning off fuel flow after a flame out.

Hazard Number	Hazard Control	Hazard Verification
SPICE U5: 2.1.2.1	Only approved crew procedures will be used which state bottle valve will be closed before first power up and before crew "steps" away from MSG (any time unattended).	ZIN QA to inspect crew procedure inputs to stipulate closing Gas Bottle Assembly Valve before leaving set up.
SPICE U5: 2.1.5.1	Only approved crew procedures will be used stating only one bottle will be used in the MSG at a time.	ZIN QA to inspect crew procedure inputs to stipulate only one bottle in a time is in the MSG
SPICE U5: 2.2.2.2	Only approved crew procedures will be used to insure MSG air flow will keep the MSG volume below LFL.	ZIN QA to inspect crew procedure inputs to stipulate that MSG air flow is on during crew access to MSG while bottle is present.
SPICE U6: 2.1.3	The hardware (nozzles) inside the Experiment Assembly will cool from 89 degrees C worst case to a temperature below 49C before crew access which is limited by the use of approved crew procedures.	ZIN Tech QA will inspect the crew procedure inputs for a 1 minute wait time before accessing nozzle.
SPICE U8: 1.1.2	SPICE designed with Fuel Enable Switch on the Controller Box which is turned on by the crew using only approved crew procedures before each burn sequence.	ZIN Tech QA will verify that crew procedure inputs specify Fuel Enable Switch only on during desired combustion.
SPICE U8: 1.2.2	SPICE designed with Fuel Switch on the Experiment Assembly which is turned on by the crew using only approved crew procedures before each burn sequence.	ZIN Tech QA will verify that crew procedure inputs specify Fuel Switch is only on during desired combustion.
SPICE U8: 1.4.1	SPICE Gas Bottle Assembly Valve will be closed by crew using only approved crew procedures when accessing EA chamber.	ZIN Tech QA will verify crew procedures specify Gas Bottle Assembly Valve closed before access to EA chamber.
SPICE U8: 2.1.2	SPICE designed with Ignition Enable Switch on the Controller Box which is turned on by the crew using only approved crew procedures before each burn sequence.	ZIN Tech QA will verify crew procedure inputs specify Ignition Enable Switch on only during time ignition is required.
SPICE U8: 2.2.2	SPICE designed with Ignitor Switch on the EA which is activated by the crew using only approved crew procedures before each burn sequence.	ZIN Tech QA will verify that crew procedure inputs specify Ignitor Switch is only on during time ignition is required.

Hazard Number	Hazard Control	Hazard Verification
SPICE U8: 2.3.2	SPICE designed such that if window access to flame area is open power to ignitor is cut off and crew using only approved crew procedures will ensure it is closed.	ZIN Tech QA will verify crew procedure inputs specify window is closed during the time when ignition is required.
SPICE U8: 3.4.2	MSG air flow verified on by crew using only approved crew procedures will keep the MSG volume below LFL.	ZIN Tech QA will inspect crew procedure inputs to stipulate that MSG air flow is on during crew access to MSG while bottle is present.
SPICE U8: 3.5.1	Crew using only approved crew procedures will ensure only one bottle will be used in the MSG at a time.	ZIN Tech QA will inspect crew procedure inputs to stipulate only one bottle in a time is in the MSG

Table 5- Hazards Controlled by Procedures.

4.3.6 Release of Experiment Materials

4.3.6.1 Unburned Fuel Release

JSC toxicologist assessment of the fuels used in SPICE have a toxicity rating of 0 and a flammability rating of 4, See Attachment A, SPICE Hazardous Material Summary Table (HMST). Also see table 6 for levels of fuel in the lab module.

A concern was raised at the Phase 0/I review concerning effects on the ECLSS due to leaks of these fuels. The action item to determine any issues with these gases in the quantities from SPICE was given. The results of the analysis are that leakage of the total amounts of these fuels is not an issue for ECLSS. Reference the March 25, 2004 email from Tim Smith which is attached to hazard report SPICE-U1. Leakage of fuel from SPICE is being controlled by design and verified by Helium leak testing of the Fuel Delivery System and the bottle assemblies. The gases that have been analyzed for ECLSS will be used to fill the bottle assemblies by using approved procedures, by labeling each bottle assembly for contents, and then by sampling each bottle to test for the correct contents.

Reference Attachment E **Unique Hazard Report SPICE-U1.**

Flammability in the US Lab. Lean Flammability Limit (LFL) Calculations conducted on the HMST listed SPICE Fuel Concentrations that could be released inside the United States Lab (USL) Volume show that all 12 SPICE Gas Bottles could leak their entire contents into the USL Volume and still be below the LFL for the applicable Fuel. The MSG will be in the Columbus module at the time SPICE is run. This module has a smaller volume than the US Lab. A new HMST has been requested and new calculations were performed based on the volume of 64 m3 that is being used by toxicology. See table 6 for the calculations.

Flammability in Stowage. The bottles are launched packed in foam in stowage cases which are in turn packed in foam. The bottles have all been vibration tested in the stowage configuration and leak tested. The fittings are all commercial fittings rated for over 3000 psi. Also each Gas Bottle Assembly incorporates a single fault tolerant design, which would require the Diaphragm Valve and Quick Disconnect to fail simultaneously in order for the Gas Bottle Assembly to leak unburned fuel through the normal flow path. This fault tolerance is accomplished by having the Diaphragm Valve and Quick Disconnect in series on the Gas Bottle Assembly. Should the Diaphragm Valve fail the Quick Disconnect will contain the gas. The failure of the Quick Disconnect is backed up by the containment of the Diaphragm Valve. There will be no ignition sources in the Gas Bottle Assembly stowage cases (no electronics or batteries). The bottles will be electrically connected to the aluminum case by two redundant strips of aluminum tape attached to the foam which will contact each bottle. A calculation will be performed providing the necessary volume in stowage which must be clear of ignition sources assuming two worst case bottles leak. This is based on being below the LFL. See table 6. The current calculation with a 75 % packing factor (there is only 25 % void space) is 156 liters or a 54 cm cube. This size is easily accommodated considering the amount of foam used. Once the fuel leaks into the MPLM or ISS stowage volume it was agreed that it will dissipate rapidly.

Stowage for landing utilizes the same controls, including valve handle restraints. The restraint (a tie-wrap) is capable of withstanding 50 lbs of force, which is sufficient to prevent turning of the valve during launch and to prevent inadvertent turning during handling. See figure 10b for a

picture of the valve with the restraint. This picture is for demonstration of the concept, the flight version will be installed with considerations for the crew (e.g. sharp edges) They will be cut by the crew once the bottle is inside the MSG. The bottles are expected to be used to the point where no gas is available to leak out, therefore for landing there should be less fuel, but the tie wrap will be installed by the crew when the bottle is removed from the MSG. In addition to the tie wrap the foam cut outs for the bottles includes cut outs for the handles which will prevent them from turning.

Reference Attachment E Unique Hazard Report SPICE-U4

Flammability in the MSG. This concern is divided into three concerns, fuel build up in the EA igniting, fuel leak into the MSG igniting and gas leak impinging on a crew member's hand.

For the concern of the EA filling with fuel and igniting during times with out crew access to the inside of the EA, the MFC has been leak tested at MDP to demonstrate that it will not leak into chamber when off, preventing fuel in the chamber from building up or flowing in quantities that could ignite. Air flow from the EA fans keep the EA internal fuel percentage below the LFL. The EA fan is on automatically upon power up of SPICE. Also the air flow rate is verified by the crew from the laptop which has a measured value of air flow rate displayed. The only ignition source in the EA chamber is the ignitor and the ignitor is not turned on until after the EA fan is verified on. The crew will turn on the ignitor before fuel flow which will eliminate fuel build up. In the event that the fan fails, the crew fails to verify the fan speed and there was an ignition source the total energy content of the EA chamber is well below the 14,240 ft-lb limit for a pressure vessel at worst case fuel mixture. See table 6.

A flame out condition has been proposed to cause fuel build up if the fuel is not turned off or fuel is not ignited. For this condition to occur the EA fan would also have to be off. Large amounts of air in the line could cause flame out. An air bubble caused by changing bottles has been hypothesized as causing a flame out. The QC from manufacturer is "drip-less" meaning it has minimal air entrainment (0.3 cc). This air bubble would be pushed out in the first 2 seconds of operation before ignition was confirmed (before steady state operation), therefore the situation of a flame out or failure to ignite in 2-3 seconds would not be a surprise situation for the crew, it would be part of the ignition sequence. Only approved crew procedures will be used to monitor burn to adjust flame. In addition it has been shown that the air bubble would mix in the fuel line filter to a percentage of fuel/air that would not be an issue. See attachments to hazard report SPICE-U5 for analysis.

For the MSG concern, in general Lean Flammability Limit (LFL) Calculations conducted on the HMST listed SPICE Fuel Concentrations that could be released inside the MSG Working Volume show that 2 SPICE Gas Bottles could leak their entire contents into the MSG Working Volume and still be below the LFL for the applicable Fuel, see table 6. Note only one bottle at a time will be permitted in the MSG. This assumes total mixing which is realistic considering that there is a minimum of 180 liters/sec of air flowing in the MSG caused by the MSG fans. The MSG fans are capable of the fuel concentrations below the LFL, and in addition the EA fan will mix the air in the MSG WV. See attachments to hazard report SPICE-U5 for analysis. There are no ignition sources inside MSG to ignite any leak from the gas bottles. The switches used for the SPICE inhibits and ignition will not produce any sparks or electrical ignition sources. SPICE is designed to minimize the potential for leaks, and with the use of approved crew procedures to ensure the MSG fans are running and only one bottle is permitted in MSG at a time there should not be fuel ignition in MSG. See unique hazard SPICE-U8.

In the case of MSG being powered down or power failing there is a concern that a pool of fuel which leaks out of the Bottle Assembly or EA will ignite when the MSG is powered on. This is addressed in the attachment to hazard report SPICE-U5 which shows that in the worst case the fuel will dissipate due to molecular action in less than 28 minutes, but other factors reduce that time. A slow leak would mean less fuel to dissipate or allow a longer dilution time. A fast leak would increase mixing due to gas velocity. Since the crew is necessary for the total operation of SPICE, when the crew leaves the MSG it will be required that the crew turn off the hand valve reducing the leak potential. In the event of a power loss, when power is restored to the lab, racks must be individually turned back on, this includes the MSG. If a wait time of ½ hour were implemented it would insure adequate dilution of the fuel.

In the case of impingement the leak controls mentioned before for the bottles are still valid. In addition, the pressurized area of the EA will be leak tested to insure that it does not leak. The Quick Connects (QC) softgood seal resides on the bottle side of the connection which will be used only once. The EA side which will mate 12 times (once for each bottle) is the metal side, thus reducing the risk of leakage due to seal wear. Electronics are contained inside the EA. The only exposed item is a switch which is a 5 volt switch. 5 volt systems are frequently used in flammable atmospheres (mines) due to their inability to ignite them. In addition this switch is sealed. The ignitor throw is a lever that the crew moves to insert the ignitor into the fuel path and turn on the ignitor. It is a lever on a rod which goes inside the EA to move the ignitor. There is a hall effect sensor (digital) which senses the proximity of the ignitor to the nozzle area and electronically turns on the ignitor. Therefore there are no sparks associated with turning on the ignitor. There are also no sparks in the proximity of the crew's hand to light a fuel impingement. See figure 16.

Each Gas Bottle Assembly will be filled per approved procedure which will ensure all bottles are filled with the proper fuel mixtures to appropriate pressures and labeled to identify filled bottle contents. Structural Integrity of the Gas Bottle Assemblies including the Experiment Assembly Fuel Delivery System is verified by conducting proof pressure to 1.5 X system MDP of 230 psid. The Gas Bottle Assemblies will meet a leak rate of less than 1×10^{-4} scc/sec Helium. The Experiment Assembly Fuel Delivery System will be verified by a decay pressure test

Reference Unique Hazard Report SPICE-U5.

In the case of the crew having access to the inside of the EA and fuel leaking into it and igniting, the fuel is controlled by the MFC which is inhibited by the window switch (open circuit when the EA window is open), fuel enable on control box is off, and the fuel switch is off. All of these inhibit the MFC. In addition the Gas Bottle Assembly valve will be turned off. There are no ignition sources other than the ignitor itself. The inhibit switches are all 5 volt and the hall effect switch for the ignitor is solid state digital. The ignitor is inhibited by a window switch, the control box ignitor inhibit and the fact that the ignitor throw is not thrown. Also when ever there is power to the experiment the EA fan will be on, which keeps the chamber below the LFL. See figure 17. Reference Unique Hazard Report SPICE-U8.

Table 6

Gas	Pressure (psi)	Volume (liters) ***	MW fuel	Mass (g)	MSG Conc, mg/m3 **	Lab Conc mg/m3*	SMAC mg/m3	% in MSG **	LFL %	UFL	Critical Volume # liters	Critical radius ## cm	ft-lbs energy	ft-lb/liter	NRG in EA \$ (1710 cc) 14,240 limit
Methane	210	2.1	16.4	1.45	7250.	23	3800	1.05	5	15	84.00	27.09	58800	28000	4536
Propane	110	1.1	44.1	2.04	10209.	32	900	0.55	2.1	9.5	104.76	29.16	78100	71000	4899
Propylene	135	1.4	42.1	2.48	12401.	38	860	0.7	2.4	11	116.67	30.22	89600	64000	4864
Ethylene	210	2.1	28.1	2.48	12400.	38	340	1.05	2.7	36	155.56	33.26	92820	44200	4950.4
Propylene/ GN2 75/25%	180	1.4	42.1	2.48	12401.	38	860	0.7	2.4	11	116.67	30.22	89600	64000	4864
Propylene/ GN2 50/50%	180	0.92	42.1	1.63	8149.	25	860	0.46	2.4	11	76.67	26.28	58880	64000	4864

* Assumes 64 m3 lab volume

** Assumes 200 liter free volume in MSG

*** Fuel volume only at standard pressure assuming all gas released from bottle (none is left in bottle).

volume of gas in liters to reach LFL diluting 2 bottles

distance from center of leak to outside of sphere volume (sphere radius) required to dilute below LFL

\$ UFL was not used for this calculation, there will not be enough O2 to burn this amount of fuel. Therefore stoichiometric quantities were used.

Figure 16 EA side view.

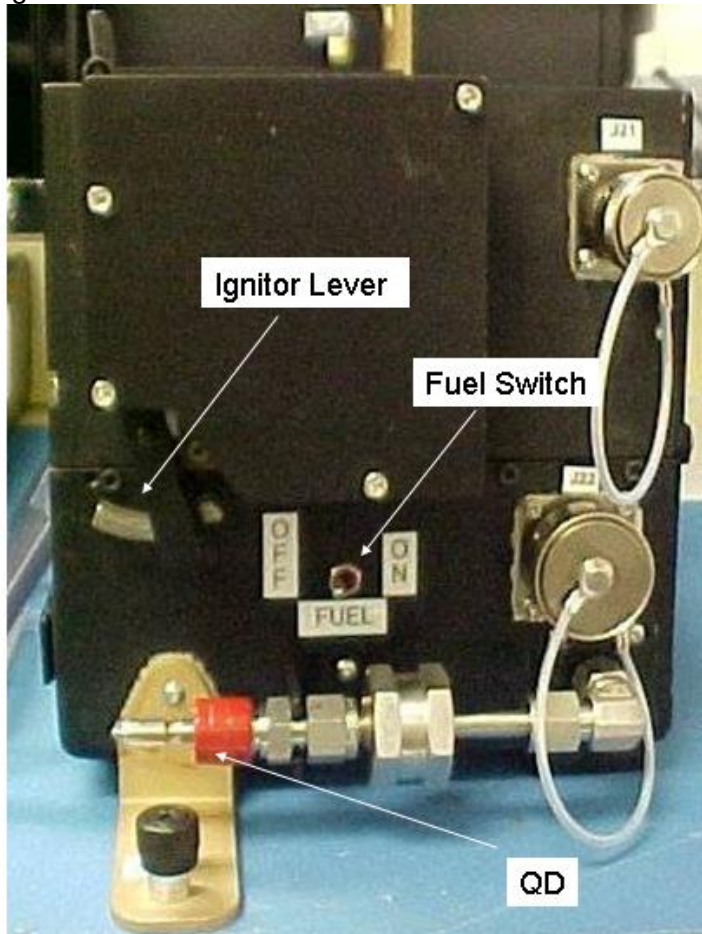


Figure 17

4.3.6.2 Combustion By-Product Release

JSC toxicologist assessments of the combustion by-products of Carbon Monoxide and Carbon Dioxide produced by the SPICE experiment have been given a toxicity rating of 0. See Attachment A, SPICE Hazardous Material Summary Table HMST. To ensure that the combustion by-products are compatible with ECLSS, SPICE will ensure that the correct gases, mixtures, and pressure are used to fill the test bottles.

The SPICE Experiment is operated inside the Working Volume of the Micro-gravity Science Glove Box (MSG) Facility. The MSG Facility provides a level of containment for SPICE Combustion By-Products. The amount of soot which may be created by SPICE is 2-10 mg total. This is based on ELF and ground testing. MSG filtering will contain any soot by-products that are produced.

Reference Attachment E **Unique Hazard Report SPICE-U1.**

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4.3.7 Acoustic Noise

Acoustic noise generated by the small fan in the Experiment Assembly will not be a hazard. The SPICE Experiment will be in compliance to the Acoustic requirements of the MSFC-RQMT-2888.

4.3.8 Stowage

The Gas Bottle assemblies shown in Figure 10 will be pre-packed using two Gas Bottle Storage Cases, shown in figure 9. Each Gas Bottle Storage Case will contain 6 Gas Bottle Assemblies and a third storage Case will be used for SPICE Component storage. See sections 2.6 and 2.7.

The SPICE Experiment Assembly along with the Gas Bottle and Component Cases will be soft stowed in the MPLM for launch and landing. Stowage for launch is planned at L-48 hours. The hardware configuration for stowage during launch and landing is still under development and is in work in conjunction with MSG.

4.4 SPICE TEST FAILURES, ANOMALIES AND ACCIDENTS

SPICE has had no safety related test failures, anomalies and accidents involving Qualification or Flight Hardware. The only test failure is related to EMI susceptibility of the camera.

4.5 AGREEMENTS FROM PHASE 0/1 REVIEW AND SUBSEQUENT REVIEWS

The following agreements/action items were taken at phase 0/1. After each action item is an answer or reference to the SDP text or hazard report which addresses the comment.

Phase 0/1 Agreements

3.1 The PO agreed to perform a maintenance safety assessment to encompass fuse replacement in the Power Conditioning Box, and other nominal equipment changeouts. **A maintenance safety assessment has been completed and is included as an attachment to unique Hazard Report U-7.**

3.2 The PO agreed to note that the hardware transfer case latches are not safety-critical, since the case will be soft stowed and packed in foam. **This information is now included in section 4.2.1.**

3.3 The PO agreed to provide more information and pictures, along with a safety assessment, for the valve handle tie-wrap. **Figure 10 b shows the valve handle tie-wrap constraint as well as figures attached to the Hazard Report SPICE U-3. An assessment of the restraint is covered in paragraph 4.3.6.1.**

3.4 The PO agreed to confirm that there would still be airflow if the Control Box Assembly fan setting were set to zero. **The airflow is wired to be on whenever the experiment is powered. Also the procedures require the crew to confirm the air flow by reading the measured air flow value off the laptop. See hazard report U-5.**

3.5 The PO agreed to attach a shatterable materials matrix to the 1230 hazard reporting form. **This has been attached to the 1230 form.**

3.6 The PSEs agreed to provide a shatterable materials matrix example to the PO. **This was provided by Daniel Porter and was used to generate the matrix for SPICE.**

3.7 The PO agreed to determine if they could use a flow straightener made of non-frangible materials. **This was investigated. Substitute materials did not have the same properties and replacement of this item would be difficult. It would also cause the acceptance vibration test to be voided. See controls provided in the 1230 form section 5.**

3.8 The PO agreed to send correspondence from Nikon to the PSEs and coordinate with the PSEs to set up a working group meeting to discuss COTS battery systems and schematics with EP5. **This has been initiated (there have been a few interactions with Nikon) Removal of the battery from the camera before flight will be done. The SPICE Project has determined that the removal of the battery reduces the reliability of the Camera and is seeking to keep the battery in place for flight.**

3.9 The PO agreed to provide the EMI waiver and all bonding, grounding, power, and wire sizing schematics as attachments to the 1230 hazard reporting form at Phase II. **These items have been added to the end of the 1230 form.**

3.10 MSFC/FD31/T. Smith agreed to provide to the PO input from ECLSS concerning the effects of unburned fuels and combustion byproducts on the air scrubbers of Shuttle and ISS. **Tim Smith provided this information and its conclusion has been included in 4.3.6.1 of the SDP.**

3.11 The PSEs agreed to compare SPICE to its predecessor, ELF, and other similar payloads to see how similar hazards were handled previously and how it may apply to SPICE. **This was provided by John Fonner.**

3.12 The PO agreed to provide better rationale on the projectile assessment for the pressurized bottle assemblies. **This analysis has been updated and attached to the hazard report U-3.**

3.13 The PO agreed to develop a unique hazard report to address fuel leakage in the bottle stowage cases and the MPLM: SPICE-U4, Fire and Detonation in MPLM and ISS Stowage. **The hazard report has been updated to include controls reflecting the concerns raised from the 0/1 review on fuel leakage in the bottle stowage cases stowed both in MPLM and ISS.**

3.14 The PO agreed to develop a unique hazard report to address premature/inadvertent fuel ignition, as well as failure to ignite fuel: SPICE-U5, Fire and Detonation in MSG. **The hazard report has been updated to include controls reflecting the concerns raised from the 0/1 review.**

3.15 The PO agreed to develop a unique hazard report for Touch Temperature, providing a wait time for cool-down of the burner tube prior to changeout: SPICE-U6, Touch Temperature. **This hazard report has been created and added to the unique hazard reports. A wait time has been added.**

Post Phase 0/1 Comments

Create operational hazard controls matrix for Phase II per the 13830 requirements. Many operational controls were identified in the hazard reports, but they will need to be listed individually in a hazard control matrix as well. **Table 5 has been updated to include this information.**

Sequentially number the pages of your HR attachments for Phase II, especially the 1230 attachments. Hand-written numbers are fine. **Page numbering has been done using Adobe on the finished converted file.**

At Phase II please provide the following:

1. Updated information on the progress/success of removing the battery from the Nikon D100 camera. **The battery was removed from an engineering camera and as a result the camera does not function properly. The battery is now planned to remain in the camera.**
2. Value of Vcc depicted in the schematic for the Horita Titler. **Power to Vcc is supplied by a regulator U6 (lower middle of schematic) which is an LM7805. This is a 5Volt regulator.**
3. If the I-Button reader is not to be used on-orbit, there are no other power sources that can charge the battery. If the reader is to be used, information on the "Parasite Power" voltage and current protection for the cell will be required. **Information was added to the STD SPICE 01 hazard report information with the rest of the battery information.**

As noted at the Phase 0/1 review, please do not state "adequate instructions." At Phase II add the appropriate details that are required to be in the procedures to Hazard Control 6.1. This will make it clear to the PSRP what you will verify and easier on you to complete your verifications. **Have added details to SPICE-U2 hazard control 6.1 and hazard control.**

Phase II Agreements

MINUTES FOR SMOKE POINT IN CO-FLOW EXPERIMENT PHASE II FLIGHT SAFETY REVIEW ON JANUARY 27,

**NASA JSC
Payload Safety Review Panel
Smoke Point in Co-flow Experiment
Phase II Flight Safety Review**

**Minutes of Meeting
January 27, 2005**

1.0 INTRODUCTION

1.1 General: The Payload Safety Review Panel (PSRP), chaired by JSC/NA3/S. Wolf, met on January, 2005, with representatives of GRC, the Payload Organization (PO), and MSFC at the Regents Park III Conference Facility for a Smoke Point in Co-flow Experiment (SPICE) Phase II Flight Safety Review (FSR). JSC/NC44/J. Fonner and M. Highsmith, the supporting Payload Safety Engineers (PSEs), introduced the meeting and attendees (see Attachment 1).

1.2 Background: The Phase 0/I FSR was held March 3, 2004.

1.3 Scope: This meeting focused on the Phase II FSR. The PSRP assigned no previous Action Items (AIs) to this payload. The PO also summarized the Phase 0/I agreement disposition.

1.4 Conclusion: One agreement and one action item resulted from this meeting (see Attachment 2). All nine Hazard Reports (HRs) were signed and approved for Phase II. This meeting completed the Phase II FSR.

2.0 SIGNIFICANT SAFETY DISCUSSION

2.1 SPICE Science and Hardware Overview: SPICE continues to study fundamental phenomena to understand the mechanisms controlling the stability and extinction of jet diffusion flames. SPICE will stabilize an enclosed laminar flame, measure the overall flame shape to validate the theoretical and numerical predictions, measure the flame stabilization heights, and measure the temperature field to verify flame structure predictions. The objective of the SPICE experiment is to find the laminar smokepoint properties of nonbuoyant jet diffusion flames (i.e., the properties of the largest laminar jet diffusion flames that do not emit soot) for several fuels under different nozzle diameter/ co-flow velocity configurations. Luminous flame shape measurements would also be made to verify models of the flame shapes under co-flow conditions. The soot point is a simple measurement that has been found useful to study the influence of flow and fuel properties on the sooting propensity of flames. This information would help support current understanding of soot processes in laminar flames and by analogy in turbulent flames of practical interest. Unfortunately, while the laminar smoke-point properties of buoyant jet diffusion flames at normal gravity (ng) are well known, corresponding information for nonbuoyant flames is limited and questionable. Thus, the measurements of the laminar smoke-point properties proposed here would also reduce the risk of future experiments involving hydrocarbon-fueled laminar jet diffusion flames in spacecraft.

2.2 SPICE Hardware Overview: Hardware stowed in during launch and landing includes:

- Experiment Assembly: Chamber for combustion, Hot-Wire Igniter Arm, Window, Flow Straightener, Heat Sink/Filter, Fuel Delivery System, Nozzle, Fan, Data Logger, Radiometer.
- Fuel Bottle Assembly: Commercial DOT bottle with Diaphragm Valve, SS tubing, QC fitting.
- Controller Box: External to MSG, controls: radiometer, Data Recording, Fuel Enable, Ignitor Enable, Fuel Flow set point, Fan Flow set point and camera shutter.
- Image Acquisition Nikon Digital still camera and a Panasonic WV-CP 654 Video Camera System
- Power-Video Unit: Video Titler and Power Conditioning Assemblies.
- Cables: Power Cable Assembly, Video Cable Assembly, Still Camera Cable Assembly, and Feed Through Cable Assembly.
- Video Tapes: 30 Digital Video tapes.
- Data Storage: 6 IBM micro drives 1 GB.

The PO indicated that the radiometer measures the temperatures of the flames.

2.3 Integration and Operations: The PO indicated that the fan will operate for chamber cooling on-orbit. The crew can verify that the igniter is on prior to turning on the gas. The PO anticipates 10-14 hours total operational time for the experiment with a set-up time of 45-60 minutes. Each experiment run lasts just a few minutes. The PO is hoping for use of the hardware and the Microgravity Science Glovebox (MSG) for several hours at a time. The PO clarified that the fuel nozzle changeout is manual. The crew procedure hazards are listed in a table.

2.4 Safety Assessment:

2.4.1 Form 1428, Fire Detection and Suppression Reporting Form: *Approved as discussed in the meeting.*

2.4.2 Form 622 - Reflown and Series Payload Hardware Reflight Assessment Reporting Sheet: *Not applicable to this payload.*

2.5 Hazard Report Discussion: The PO provides crew procedure inputs to the Payload Operations Data File (PODF) at MSFC.

2.5.1 Form 1230, Flight Payload Standardized Hazard Control Report: *Approved as modified in this meeting.* The MSG integrated/series assessment will appear at Phase III. The PO indicated that the controller box is velcroed in place inside the MSG and that there are no crew entanglement concerns with the cables associated with the hardware package. Associated with Shatterable Material Release, the PO indicated that the flow straightener is made of Cordierite (a ceramic) and is contained inside the fan housing on the Experiment Assembly (EA) with a 100 mesh screen over it. The EA will be packaged in a clear plastic bag and will be inspected for particles before use. Any particles that escape the EA mesh should be captured by the MSG filter and the MSG HEPA filter. The PO clarified that the minor electromagnetic exceedences affect the cameras, but this is a performance issue, not a safety issue. The PSRP recommended that the PO match the voltage rating of the fan with the power supply to the fan (12V supply).

2.5.2 SPICE-U1, Contamination of Orbiter/ISS Habitable Areas by SPICE Unburned Fuels or Combustion By-Products Results in Injury/Illness: *Approved as modified in this meeting.* The PO stated that the by-products consist of carbon dioxide and water, both scrubbed by the MSG filters. The Hazardous Materials Summary Table (HMST) is attached and under review. The PO analyzed the amounts, kinds, and release rates of the fuel bottle contents and determined that the contamination

hazards are reasonably benign in this payload. The PO also stated that each bottle is individually proof tested.

2.5.3 SPICE-U2, Pressure System Rupture: Approved as modified in this meeting. The PO indicated that each gas bottle has been leak tested and filled to an appropriate pressure corresponding to the fuel mixtures. The PO keeps a pressure vessel log on each bottle. The QC caps will be installed during stowage and transit to MSG.

2.5.4 SPICE-U3, Projectile Hazard Gas Bottle: Approved as modified in this meeting. The PO deleted hazard cause 2 and controls 1.2, 1.3, and 1.4 as a result of the discussion.

2.5.5 SPICE-U4, Fire and Detonation in MPLM and ISS Stowage: Approved as modified in this meeting. Crew procedures have the crew marking spent bottles to prevent the crew from grabbing an empty gas bottle for installation. The PO described the aluminum strip inside the bottle case that is the bonding strap to preclude electrostatic discharge inside the bottle case: the aluminum strip is stuck to the interior of the top of the case in such a way that it touches the metal of the bottles when the case is closed. The PO indicated that any labels on the bottles will be positioned to assure the metal-to-metal contact between the bottles and the grounding strip. Discussion of the temporary aisle stowage of the bottle case led the PSRP to assign AI 1 to the PO to determine if the Station-supplied fuel stowage location complies with the SPICE flammability assessment restrictions. The PO will provide more details at Phase III.

2.5.6 SPICE-U5, Fire in MSG: Approved as modified in this meeting. The PSRP recommended that the PO delete the 14,242 foot pounds reference in favor of the estimates in the table as the limits of released energy from the gas bottles. The pressure release exceeds the MSG interface control document limit, which makes this a catastrophic hazard rather than the usual critical designation.

2.5.7 SPICE-U6, Touch Temperature Injury: Approved as modified in this meeting.

2.5.8 SPICE-U7, Maintenance Assessment Hazard Controls: Approved as modified in this meeting.

2.5.9 SPICE-U8, Flame Impingement During Crew Access: Approved as modified in this meeting. The PO clarified that the igniter cools down within less than 30 seconds. The PO has not seen the nozzles work loose due to gas flow—a matter of running torque versus seating torque—and indicated that gas flow pressure is not high enough to cause a problem. The PO has four inhibits on fuel flow and three inhibits on ignition. The PO agreed to provide a logic diagram depicting all of the inhibits that prevent fuel ignition at Phase III (Agreement 3.1).

3.0 AGREEMENTS

3.1 The PO agreed to provide a logic diagram depicting all of the inhibits that prevent fuel ignition at Phase III.

Original Signed by M. Highsmith for:

JSC/NC44/J. Fonner
Payload Safety Engineer

Original Signed by:

JSC/NC44/W. Stauffer
Technical Writer

Original Signed by:

May 2008

JSC/NC44/M. Highsmith
Payload Safety Engineer

Status of Hazard Reports Presented
(Note: See the text of the minutes for more details.)

Number	TITLE	Status	Comments
Form 1230	Form 1230, Flight Payload Standardized Hazard Control Report	Approved/mods	2.5.1
Form 1428	Form 1428, Fire Detection and Suppression Reporting Form	Approved	2.4.1
SPICE-U1	Contamination of Orbiter/ISS Habitable Areas by SPICE Unburned Fuels or Combustion By-Products Results in Injury/Illness	Approved/mods	2.5.2
SPICE-U2	Pressure System Rupture	Approved/mods	2.5.3
SPICE-U3	Projectile Hazard Gas Bottle	Approved/mods	2.5.4
SPICE-U4	Fire and Detonation in MPLM and ISS Stowage	Approved/mods	2.5.5
SPICE-U5	Fire in MSG	Approved/mods	2.5.6
SPICE-U6	Touch Temperature Injury	Approved/mods	2.5.7
SPICE-U7	Maintenance Assessment Hazard Controls	Approved/mods	2.5.8
SPICE-U8	Flame Impingement During Crew Access	Approved/mods	2.5.9

Previous Action Item Status
The PSRP assigned no previous AIs for this payload.

ATTACHMENT 1

Payload Safety Review Attendance Log

Payload: SPICE Phase II Flight Safety Review
Meeting Date: January 27, 2005

Mail Code	Name	Phone 281	X
CHAIRMAN			
NA3	Wolf, S.	483-4096	X
SUPPORT PERSONNEL			
CB	Rickard, J.	483-3760	X
DA8/USA	Knutson, D.	483-4405	X
EA441	Henning, G.N.	483-5502	X
MS3/USA	Kunkel, S.	280-6844	X
NA3	Moreland, D. W.	483-5549	X
MSFC/JS-20	Johnson, P.T.	256-961-4646	X
SM	Spann, R.	483-3807	X
EP4/LMES	Manha, W.	483-6439	X
LMES/C20	Russell, D.	333-6892	X
NC44/SAIC	Fonner, J.	483-3790	X
NC44/SAIC	Highsmith, H.	244-1822	X
NC44/SAIC	Rehm, R.	244-1819	X
NC44/SAIC	Sobota, A.	244-1839	X
NC44/SAIC	Stauffer, P. W.	483-6066	X

Name	Mail Code	Employee	Phone Number	Technical Discipline	Internet Address
W. Schoren		GRC	216.433.2356		William.r.schoren@grc.nasa.gov
S. Burchell		USA	256.961.4393	POIC Safety	Steven.D.Burchell@msfc.nasa.gov
S. James		MSFC			
J. Wetherholt		ZIN	216.977.0365	Safety	Jon.wetherholt@zin-tech.com
E. Selent		ZIN	216.977.0415	Task Lead	Edward.selent@zin-tech.com
D. Truong		GRC	216.433.8394	Project Lead	Duc.truong@grc.nasa.gov
C. Azzarita	CB	NASA	281.244.8609	Crew Office	Cynthia.j.azzarita1@jsc.nasa.gov

ATTACHMENT 2

NASA JSC
Payload Safety Review Panel
Smoke Point in Co-flow Experiment
Phase II Flight Safety Review

Summary of Action Items

January 27, 2005

AI	Action	Date Due
1 Assigned to: GRC/D. Truong HR: SPICE-U4	Determine if the Station-supplied fuel stowage location complies with the SPICE flammability assessment restrictions.	Date: Phase III Mandatory Reviewer(s): PSRP

Original Signed by:

JSC/NA3/S. Wolf
PSRP Chairman

January 27, 2005
Date

Original Signed by:

GRC/D. Truong
Payload Organization

January 27, 2005
Date

Response to the agreement from the SPICE Phase II safety review

Agreement: The PO agreed to provide a logic diagram depicting all of the inhibits that prevent fuel ignition at Phase III.

Two diagrams are attached to SPICE unique hazard SPICE-U8 that show all of the ignition inhibits that prevent fuel ignition.

Response to Action Item from the SPICE Phase II safety review

Action Item: Determine if the Station-supplied fuel stowage location complies with the SPICE flammability assessment restrictions.

ATTACHMENT A : SPICE HMST
SPICE Hazardous Material Summary Table
HMST

STS-118/ ISS-13A.1
HAZARDOUS MATERIALS SUMMARY TABLE
SpaceHab Up To ISS US Lab

Verification Status
V-1: Preliminary Data
V-2:

SPICE: Smoke Point in Co-flow Experiment

Max Tox Lvl	HMST Record # Expt Subsystem IMS Numbers Label of ID	Chemicals or Biological Materials	Maximum Concentration	Maximum Volume or Amount	Principal Toxic Hazards	Toxic Haz'd Level	Medical Protocol	Flam- mability	Record update history/ Contact Person/ Comments
Part #s									
	HMST Record #: 11876	Fuel gas: Ethylene (C2H4)	100%	2 x 150 cc @ 210 psia	None	0	None	4	Last modified: 7/26/04 Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part # 60054M AFH090- 1, 60054M AFH090- -2	Subsystem: DOT Pressure bottle Label: E1, E2	Combustion products: Carbon monoxide Carbon dioxide Water	41.4 ppm 41.4 ppm N/A	100 m3 100 m3 100 m3		0 0 0			
	HMST Record #: 11877	Fuel gas: Methane (CH4)	100%	2 x 150 cc @ 210 psia	None	0	None	4	Last modified: 7/26/04 Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part # 60054M AFH090- 3, 60054M AFH090- 4	Subsystem: DOT Pressure bottle Label: M1, M2	Combustion products: Carbon monoxide Carbon dioxide Water	20.7 ppm 20.7 ppm N/A	100 m3 100 m3 100 m3		0 0 0			
	HMST Record #: 11878	Fuel gas: Propane (C3H8)	100%	2 x 150 cc @ 110 psia	None	0	None	4	Last modified: 7/26/04 Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part # 60054M AFH090- 5, 60054M AFH090- 6	Subsystem: DOT Pressure bottle Label: P1, P2	Combustion products: Carbon monoxide Carbon dioxide Water	30.4 ppm 30.4 ppm N/A	100 m3 100 m3 100 m3		0 0 0			

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STS-118/ ISS-13A.1
HAZARDOUS MATERIALS SUMMARY TABLE
SpaceHab Up To ISS US Lab

Verification Status
V-1: Preliminary Data
V-2:

SPICE: Smoke Point in Co-flow Experiment

Max Tox Lvl	HMST Record # Expt Subsystem IMS Numbers Label of ID	Chemicals or Biological Materials	Maximum Concentration	Maximum Volume or Amount	Principal Toxic Hazards	Toxic Haz'd Level	Medical Protocol	Flam- mability	Record update history/ Contact Person/ Comments
Part #s	HMST Record #	Fuel gas:			None				
	11879	Propylene (C3H6)	100%	2 x 150 cc @ 135 psia		0	None	4	<i>Last modified: 7/26/04</i> Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part #	Subsystem:	Combustion products:							
60054M	DOT Pressure	Carbon monoxide	38.3 ppm	100 m3		0			
AFH090-	bottle	Carbon dioxide	38.3 ppm	100 m3		0			
7,		Water	N/A	100 m3		0			
60054M	Label:								
AFH090-	PR1, PR2								
8									
	HMST Record #	Fuel gas:			None				
	11880	Propylene (C3H6)	75%	2 x 150 cc @ 180 psia		0	None	4	<i>Last modified: 7/26/04</i> Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part #	Subsystem:	Nitrogen	25%						
60054M	DOT Pressure	Combustion products:							
AFH090-	bottle	Carbon monoxide	39.4 ppm	100 m3		0			
9,		Carbon dioxide	39.4 ppm	100 m3		0			
60054M	Label:	Water	N/A	100 m3		0			
AFH090-	PR75-1, PR75-2								
10									
	HMST Record #	Fuel gas:			None				
	11881	Propylene (C3H6)	50%	2 x 150 cc @ 180 psia		0	None	4	<i>Last modified: 7/26/04</i> Contact: brian.loucks@zin-tech.co m; 216-977-0374
Part #	Subsystem:	Nitrogen	50%						
60054M	DOT Pressure	Combustion products:							
AFH090-	bottle	Carbon monoxide	31.0 ppm	100 m3		0			
11,		Carbon dioxide	31.0 ppm	100 m3		0			
60054M	Label:	Water	N/A	100 m3		0			
AFH090-	PR50-1, PR50-2								
12									

ATTACHMENT B : SPICE STANDARD HAZARD REPORT
SPICE STANDARD HAZARD
REPORTING FORM 1230

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
1. Structural Failure (payloads must comply with the listed requirements for all phases of flight) <i>Note: Locker and Soft Stowage items only.</i>	Designed to meet the standard modular locker stowage requirements of NSTS 21000-IDD-MDK or equivalent IDD _____.	☒	1a.1 ZIN Tech Engineering will review pre-launch stowage list and re-stow procedure to confirm that SPICE experiment and associated hardware complies with NSTS 2100-IDD-MDK and ISS-MPLM-IDD-006 for launch and landing. Status – Closed to VTL.	
2. Structural Failure of Sealed Containers Note 1: Only sealed containers made of conventional metal (metal alloy) can use the 1230 form. Sealed containers made of unconventional metals or non-metallic materials shall be documented on Unique Hazard Reports. Note 2: The 1230 form is not applicable for “sealed container”(s) employed in levels of containment control of hazardous fluid.	Sealed containers must meet the criteria of NASA-STD-5003, Para. 4.2.2.4.2.3a, contain a substance which is not a hazard if released, be made of conventional metals (e.g. Al, inconel, monel, steel or titanium), contain less than 14,240 foot-pounds (19,130 Joules) of stored energy due to pressure, and have a maximum delta pressure of 1.5 atm.	☐	N/A- SPICE Gas Bottle Assemblies and pressurized section of the Experiment Assembly contain hazardous materials and are documented under Unique Hazard Reports SPICE-U1 for Contamination Hazard and SPICE-U2 for Rupture/Leak hazard of the Pressurized Fuel Delivery System. All other containers are vented.	
APPROVAL	PAYLOAD ORGANIZATION		SSP/ISS	
PHASE I				
PHASE II				
PHASE III				

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
3. Structural Failure of Vented Containers	For intentionally vented containers, vents are sized to maintain a 1.4 factor of safety for Shuttle or a 1.5 factor of safety for Station with respect to pressure loads. Meets all of the applicable pressure rates defined for one or more of the following. a) Shuttle payload bay – ICD 2-19001, Para. 10.6.1 b) Station environment – SSP 52005, paragraph 4.3 or equivalent payload specific ICD ____. c) Station PFE discharge – SSP 57000, Para. 3.1.1.4K, or equivalent payload specific ICD ____. d) Shuttle Middeck – NSTS 21000-IDD-MDK, Section 6.1.	☐ ☒ ☐ ☒	3b.1 Analysis. ZIN Engineering will perform an analysis to verify that the SPICE vented containers will maintain structural integrity in the event of rapid depressurization/repressurization with a factor of safety of 1.5 per SSP 52005. Status - Closed to VTL. See Vented Container Table (table B-1).	
4. Sharp Edges, Corners, and/or Protrusions.	Meets the <u>intent</u> of one or more of the following: a) NASA-STD-3000 / SSP 50005 b) SLP 2104 c) NSTS 07700 Vol. XIV App. 7 (EVA hardware) d) NSTS 07700 Vol. XIV App. 9 (IVA hardware) / SSP 57000 e) SSP 41163, Para. 3.3.6.12.3 (EVA), Para. 3.3.6.12.4 (IVA).	☒ ☐ ☐ ☐ ☐	4a.1 Analysis. ZIN Tech Engineering will conduct drawing review to ensure the requirements to remove sharp edges, corners, and protrusions have been met and prepare a report confirming design requirements. Status – Closed 12/10/03. 60054-ANA-021. Sharp Edge Analysis 4a.2 Inspection. ZIN Tech Quality will perform a sharp edge inspection of the SPICE hardware and record results in an inspection report. Status – Closed to VTL.	

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
5. Shatterable Material Release	a) All materials are contained. b) Optical glass (i.e. lenses, filters, etc.) components of crew cabin experiment hardware that are non-stressed (no delta pressure) and have passed both a vibration test at flight levels and a post-test visual inspection. c) Payload bay hardware shatterable material components that weigh less than 0.25 lb and are non-stressed (no delta pressure) or non-structural.	☒ ☒ ☐	5a.1 Analysis. ZIN Tech Engineering will ensure that SPICE design provides for containment of all shatterable materials (see list below) and prepare a report confirming design requirements have been met. Status – Closed 08/07/03. 60054-ANA-026, Containment of Shatterable Materials Analysis. 5a.2 Inspection. ZIN Tech Quality will verify that the as-built flight hardware conforms to the approved drawings and record results in an inspection report. Status – Closed 03/01/07. SPICE-INS-020-R1, As-Built Inspection Report 5a.3 Inspection. ZIN Tech Engineering will review crew procedure inputs to ensure that checks of containment and assessment of status prior to utilizing frangible hardware are included, and prepare a report confirming inclusion in the crew procedure inputs. Status - Closed to VTL. 5b.1 Test. ZIN Tech Engineering will arrange for SPICE flight hardware to undergo acceptance vibration testing and prepare a test report confirming test was successfully completed and passed inspection for no breakage. Note even though this control is for Optics, the flow straightener has been tested, see list of all items in table B-2. Status - Closed to 02/07/07. SPICE-TST-007-R1. Stowed Vibration Test Report See shatterable materials table, table B-2.	
6. Flammable Materials	a) A-rated materials selected from MAPTIS, or b) Flammability assessment per NSTS 22648	☒ ☒	6ab.1 Analysis. ZIN Tech Quality will compile a MIUL that records materials specified in approved design documents, evaluate the materials identified per NSTS 1700.7B / ISS Addendum requirements and current MAPTIS material ratings, and generate MUAs to substantiate usage of all non-A-rated materials. Any required flammability assessment per NSTS 22648 will be included in usage rationale of the MUAs. Status – Closed to VTL. 6ab.2 Analysis. NASA/GRC SMAD reviews the MIUL and MUAs and when approved provides a material certification memorandum in accordance with intercenter agreements. Status - Closed to VTL.	
7. Materials Offgassing	Offgassing tests of assembled article per NASA-STD-6001 (previously published as NHB 8060.1C).	☒	7a.1 Test. ZIN Tech Engineering will arrange offgas testing in accordance with NASA-STD-6001, Test 7 and prepares a report. Status – Closed 2/09/07. SPICE-RPT-001. Toxic Off-gas Test Report 7a.2 Analysis. NASA/GRC SMAD reviews offgas test report and when approved provides material certification memorandum in accordance with intercenter agreements. Status – Closed to VTL.	

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
8. Nonionizing Radiation	Meets all that apply:		8.1.a.1 Test. ZIN Tech Engineering performs EMI/EMC test per SSP 30238 using the requirement of MSFC-REQ-2888 and prepares a report confirming requirements have been met. Status – Closed 09/30/03. GRC-EMI-RPT-201	
8.1 Non-transmitters	a) Pass ICD-2-19001, 10.7.3.2.2 / SSP 30238 EMI compatibility testing, or	☒	8.1c.1 Analysis. NASA ISS/EMEP will review and approve a TIA based on EMI test results. Waivers and deviations have been submitted and a TIA has been requested. Status – Closed to VTL.	
	b) NSTS/USA approved analysis ICD Section 20, or	☐		
	c) ISS/EMEP approved TIA	☒		
	d) Meets SSP 41163, Para. 3.3.6.6 and SSP 50094, Para. 3.4.	☐	The power box and the video titler (together know as the power/video unit), the two cameras, the control box and the experiment assembly are all powered and were tested for EMI emissions.	
8.2 Lasers	Meet ANSI Z136.1-2000 for class 1, 2, or 3a Lasers (as measured at the source).	☐	N/A- No Lasers.	
NOTE: Lasers operating at class levels 3b and 4, meeting ANSI Z136.1, shall be documented on Unique Hazard Reports.				
9. Battery Failure	a)			
9.1 Alkaline cells and batteries made of alkaline cells, connected in series or in parallel, up to 12 V and with up to 60 Watt-hours capacity, no potential charging source and cells are not in a gas-tight compartment.	Cells and batteries pass acceptance tests that include loaded and open circuit voltage measurements, visual examination, and leakage check under vacuum (e.g. 6 hours at 0.1 psia).	☐	N/A Submission of EP-Form-03(including the form's approved reference number) for EP5 approval Information to be included on the EP5 reviewed/approved Form-03: application and protective circuit schematics	

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
9.2 Primary button cells such as Li-(CF)x, Li-iodine, LiV ₂ O ₅ , LiMnO ₂ , Ag-Zn and rechargeable button cells such as LiV ₂ O ₅ , Li-ion, Ni-Cd, Ni-MH, Ag-Zn cells or batteries, which have a capacity of 300 mAh or less and no more than 3 cells per common circuit, and cells are not in a gas-tight compartment.	Cells and batteries pass acceptance tests that include loaded and open circuit voltage measurements, visual examination, and leakage check under vacuum (e.g. 6 hours at 0.1 psia). Note: Above acceptance testing for button cells in Section 9.2 which are soldered to a circuit board in commercial equipment (not applicable to those button cells in a spring-loaded clip) is limited to a functional check of the equipment utilizing the subject battery.	☒ ☐ ☐	9.2.a.1 Test. Since all batteries are soldered to the boards, ZIN Tech Engineering will conduct an acceptance functional test on each item and prepare reports confirming requirements have been met. Status – Closed to VTL. NIKON D100 Digital Camera Button Cell 20mmX3.2mm, type Li-Mn-O2, 3 Volts DC 65mAh, Maxell Model No.: ML2032T6. (This battery will be removed before flight.) Dallas I-Button data device Cell, 12.5mm X 2.5mm, type Li-CFx, 3 Volt DC, 50mAh, Model No. BR1225, Manufacturer: RAYOVAC. The Horita Video Titler card contains two cells in series. The cells are 15.5 X6.3 mm Rechargeable Nickle Metal Hydride cells, 1.2 VDC each, 60mAh, Model Number GP60BNK, Manufacturer GP Batteries International LTD EP-Form-03 submitted for each battery. As submitted forms attached. Submission of EP-Form-03(including the form's approved reference number) for EP5 approval Information to be included on the EP5 reviewed/approved Form-03: application and protective circuitry.	
9.3 COTS NiMH, NiCd and Ag/Zn cells and batteries for IVA use up to 20 V and 60 Wh	Cells and batteries purchased in one lot; pass acceptance tests that include loaded and open circuit voltage measurements, visual examination, leakage check under vacuum (e.g 6 hours at 0.1psia) and vibration to workmanship levels with functional checks which include charge/discharge cycles for rechargeable batteries.	☐	N/A Submission of EP5 Form-03(including the form's approved reference number) for approval Information to be included on the EP5 reviewed/approved Form-03: manufacturer's specification, battery protective features and charger schematics	
9.4 COTS Li-ion batteries up to 10 V and 60 Wh for IVA use	Batteries and charger shall be from a single lot; shall show one-fault tolerance at battery level and shall pass acceptance tests that include loaded and open circuit voltage measurements, visual examination, leakage check under vacuum (e.g 6 hours at 0.1psia) and vibration to environment or double the workmanship level, whichever is higher; and functional checks which include charge/discharge cycles.	☐	.N/A	

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
10. Touch Temperature	a) Within IVA touch temperature range of –18 Degrees C. (0 Degrees F.) and 49 Degrees C. (120 Degrees F.) and satisfies the intentional contact constraints of letter MA2-95-048 (if applicable). b) Meets EVA touch temperature criteria of NSTS 07700 Vol. XIV App. 7.	☐ ☐	See unique HR SPICE-U6	
11. Electrical Power Distribution	a) Shuttle-powered payloads – Meets all circuit protection requirements of Letter TA-92-038. b) Station-powered payloads – Meets station interface circuit protection requirements of SSP 57000 and payload circuit protection requirements of Letter TA-92-038.	☐ ☒	11b.1 Analysis . ZIN Tech Engineering will perform an analysis of the SPICE design to verify proper wire sizing for circuit protection provided by MSG and prepare a report confirming design requirements have been met. Status – Closed 6/3/03. Report 60054-ANA-017, SPICE Wire Derating Analysis demonstrates that wire size is adequate for the fusing. 11b.2 Analysis . ZIN Tech Engineering will perform an analysis to verify that the SPICE design incorporates proper circuit protective devices in accordance with SSP 57000 and TA-92-038 and prepare a report confirming design requirements have been met. Status – Closed 6/3/03. 60054-ANA-017, SPICE Wire Derating Analysis 11b.3 Analysis. JSC/EP5 review and approval of integrated schematic indicating bonding and grounding methodology for compliance with SSP 30245 and SSP 30240. Status – Closed 6/3/03 Schematic is provided as an attachment to the 1230 Form. 60054-ANA-015, Grounding & Cable Analysis. SPICE-TST-016-R1, As-Run Grounding Test Report. See SPICE-NCR-003 and SPICE-DEV-001. 11b.4 Inspection. ZIN Tech Quality will verify that the as-built electrical circuitry is in accordance with approved drawings and parts lists and record results in an inspection report. Status – Closed 3/01/07. SPICE-INS-027-R1, As-Built Electrical Circuitry Inspection Report Note: See attached figures on the electrical system: B-3, B-4 and B-5.	
12. Ignition of Flammable Atmospheres in Payload Bay	All ignition sources in the Payload bay, for launch and landing, are controlled as required in Letter NS2/81-MO82, and MLI grounded per ICD 2-19001.	☐	N/A- SPICE will not be transported to ISS in Payload Bay.	

FLIGHT PAYLOAD STANDARDIZED HAZARD CONTROL REPORT		A. NUMBER	B. PHASE	C. DATE
		STD- SPICE-01	III	May 2008
D. PAYLOAD, DTO, DSO or RME (Include Part Number(s), if applicable)		HAZARD TITLE		E. VEHICLE
Smoke Points In Co-flow Experiment (SPICE)		STANDARD HAZARDS		STS/ISS
F. DESCRIPTION OF HAZARD:	G. HAZARD CONTROLS: (complies with)	H. APP.	I. VERIFICATION METHOD, REFERENCE, AND STATUS:	
13. Rotating Equipment	Rotating equipment meets criteria of NASA-STD-5003 for obvious containment.	☒	13.1 Analysis. ZIN Tech Engineering will review SPICE design to ensure that the rotating component which is the 3 inch, 2950 rpm Fan located in the Experiment Assembly meets the criteria of NASA-STD-5003 for containment and prepare a report confirming design requirements have been met. Status – Closed 8/1/03. 60054-ANA-028, Rotating Components Compliance Analysis.	
14. Mating/demating powered connectors	a) Meets the low power criteria of letter MA2-99-170 or, b) Meets the paragraph 1 criteria of letter MA2-99-170 (e.g., IVA and open circuit voltage no greater than 32 volts).	☐ ☒	14b.1 Inspection. ZIN Tech Quality will Inspect Crew Procedure inputs to ensure that step is in place to remove power from the MSG experiment power connector prior to mate/demate of connectors. Status – Closed to VTL. 14b.2 Inspection. ZIN Tech Quality will Inspect electrical connectors to comply with MSFC-RQMT-2888 connection requirements for 28 volt connections. Status – Closed 2/13/07. SPICE-INS-011-R1, Electrical Connectors and Pin Assignment Inspection Report. 14b.3 Inspection. ZIN Tech Quality will Inspect the SPICE hardware for design features that completely enclose or shroud the pins and sockets. Status – Closed 4/19/07. SPICE-INS-003 Connector Protection and Shape Design Inspection. 14b.4 Inspection. ZIN Tech Quality will Inspect the SPICE connectors for powered side terminating in sockets rather than pins. Status – Closed to 4/19/07. SPICE-INS-003 Connector Protection and Shape Design Inspection.	
15. Contingency Return and Rapid Safing	a) Shuttle Environment: If middeck payload – can be stowed within 50 min. (see paragraph 3 of letter MA2-96-190). b) If SPACEHAB transfer item – can establish a safe for return configuration within 3 min. (see paragraph 5 of letter MA2-96-190). Station Environment: c) Payload design does not impede emergency IVA egress to the remaining adjacent pressurized volumes.	☒ ☐ ☒	15a.1 Analysis. ZIN Tech Engineering will perform an analysis to show that hardware can be stowed in allowed time. Status – Closed 12/10/03. 60054-ANA-030, ISS Emergency Egress Compliance Analysis 15c.1 Analysis. ZIN Tech Engineering will perform design analysis of SPICE hardware to show the design does not impede emergency IVA egress to adjacent pressurized volumes and prepare a report confirming design requirements have been met. Status – Closed 12/10/03. 60054-ANA-030, ISS Emergency Egress Compliance Analysis	

Vented Container Table B-1 (for STD SPICE-01 Description of Hazard 3)

Container	Housing Material	Size	Energy contained (ft-lbs) \$	Container Volume (in³) No packing factor	Vent Size (in²)
Control Box	Aluminum	4.8X2.6X1.4 inches	20	16.8	0.3
Digital Camera	Plastic	*8X 5X8	370	320	0.32
Video Camera	Aluminum	* 5.9X2.6X2.1	37	32.3	0.31
Power Control Box	Aluminum	8.1X5.7X2.4 inches	128	111	0.86
Video Control Box	Aluminum	8.1X5.7X 3 inches	160	139	0.86
Gas Bottle Assembly Cases	Aluminum	18.2X14.5X6.2 inches	1891	1636	1.20
Component Parts Case	Aluminum	18.2X14.5X6.2 inches	1891	1636	1.20
Experiment Assembly	Aluminum and Lexan	13.6X10.4X6.7	1095	947	19.5
Ignitor/Burner case	Aluminum	6.1X3.5X1.1	24.9	21.5	0.30

Note none of these contain any chemicals other than air except the Gas Bottle Assembly Cases which will contain the Fuel Bottles which are tox level 0 if they are all released.

*The camera is not a box, these are the extreme outside dimensions and includes the lens, they are used for conservatism.

@ This is based on depressurization. For the gas bottle assembly case, although any individual bottle if dumped instantly would increase the pressure in the box by 4.4 psi if the box were sealed, this occurring at the same time as a depressurization does not seem credible.

\$ This is based on the container being sealed and at 14.7 psi differential pressure due to depressurization.

Shatterable Materials/Components Table B-2 (for STD SPICE-01 Description of Hazard 5)

System (or Subsystem)	Material name/type	Location/use	Protection & containment	# of levels of containment	Rationale for acceptance
Video Camera Lens	Glass lens	In MSG	Non stressed lens	1	The glass lens parts are small and not stressed. The camera was tested in vibration in stowage configuration. Item is contained during launch (stowed item) and is in MSG during use. Camera will be packed inside transparent bag and will be examined for glass splinters by the crew before unpacking. Verification shall be done by review and approval of inputs to crew procedures.
Still Camera Lens	Glass lens	In MSG	Non stressed lens	1	The glass lens parts are small and not stressed. The camera was tested in vibration in stowage configuration. Item is contained during launch (stowed item) and is in MSG during use. Camera will be packed inside transparent bag and will be examined for glass splinters by the crew before unpacking. Verification shall be done by review and approval of inputs to crew procedures.
Flow Straightener	Cordierite (ceramic)	In MSG	Non stressed, bound 4 sides, and contained	1	The part is small and not stressed. The EA was tested in vibration in stowage configuration. Item is stowed during launch. Item is contained between 100 mesh screens at all times. The item is 1 inch thick honey comb which is bound on 4 sides (3 inch sides) exposing only the ends of the square tubes formed by the ceramic.

Reference letter for STD SPICE-01 hazard 7

National Aeronautics and
Space Administration

George C. Marshall Space Flight Center
Marshall Space Flight Center, AL 35812



September 24, 2003

Reply to Attn of: ED36B (03-060)

TO: SD43/TBE/Neal Scott

FROM: ED36/Dennis E. Griffin

SUBJECT: Toxic Offgassing of The Smoke Point In Co-flow Experiment (SPICE)
Flight Assembly

The subject assembly has been tested for toxic offgassed products by the procedures outlined in NASA-STD-6001, Test 7. The assembly met the acceptance criteria with a rating of A. The test parameters and results are provided in Enclosure 1. The test analysis is provided in Enclosure 2.

Cleaning can adversely affect test results. The MSFC Materials, Processes, and Manufacturing Department must approve any post-test cleaning. Any unapproved cleaning or modifications will invalidate these test results and require retesting.

Enclosed also is a copy of the MCRF Quality and Satisfaction Survey. Please help us by completing the enclosed survey and return it to us as indicated on the form.

Direct any questions regarding this test to Steve Whitfield at 256-961-0358.

A handwritten signature in black ink, appearing to read "Dennis E. Griffin".

Dennis E. Griffin
Group Leader
Chemistry Group

3 Enclosures

cc: (w/o Enclosure 3)
ED36B/File copy
ED36B/S. Whitfield
ED36B/ICRC/C. Cooper

Mission Success Starts with Safety

EP-FORM-03 forms as submitted for STD SPICE 01 Hazard 9.2

EP5 BATTERY DESIGN EVALUATION FORM		Page 1 of 7
Section 1 to be completed by EP/Division Secretary:		
1a. EP Tracking Number:	1b. Assigned EP5 Reviewer:	
Section 2 to be completed by the Hardware Point-of-Contact:		
2a. Hardware Point-of-Contact: (Name/Phone/Company/Mail Code) Tibor Lorik/ZIN Technologies Inc	2b. Hardware Name: Smoke Point in Co-Flow Experiment Hardware Part Number: 60054MAFH400 Hardware Acronym: SPICE	
2c. Hardware Managing Group, Company, or Agency: NASA GRC/ZIN Technologies		
2d. Hardware Flight Information (i.e. flight application, target flight number and/or date, number of flights anticipated, etc.): The SPICE hardware is scheduled for Flight ULF-2 - 10/2008. The hardware contains 3 different batteries. The first battery is located in the Nikon Camera. It is rechargeable although it will not be configured for charging. A second is on the Video Tiler Board, and a third in the Dallas Semiconductor I-Button. The batteries are used to maintain configuration settings and clock. The hardware will be flown aboard the Shuttle and transferred to ISS where it will be stowed awaiting its scheduled use in the MSG Facility.		
2e. Hardware environmental Requirements: Thermal (max, min, operational and non-operational ranges): 10 – 46 C both operating and non-operating. The hardware was tested to these levels. Pressure (EVA, IVA): Cabin Pressure. Hardware has been analyzed to be safe when exposed to pressures of 0 to 15.2 PSIA. Life (calendar/shelf, cycle): 10 years		
Section 3 to be completed by Hardware Point-of-Contact:		
3a. Battery Hardware Description: Are the cells/ the battery pack Commercial-off-the-shelf (COTS)? YES Function/Operating mode (pulse? intermittent? clock backup? memory?): Clock Backup and Configuration Settings in Memory Battery interfaces (connectors? equipment?): Batteries are installed on Printed Circuit boards and are soldered. Stowage location (installation and use locations): Hardware will be launched in Middeck or MPLM. On-orbit hardware is placed in on-orbit stowage during operations hardware is located in the MSG Facility. Battery/cell access during flight: None Short circuit/safe-touch temperature protection: Unsure if this rechargeable battery has short circuit protection. Battery is embedded within the Nikon Still camera with no crew contact of the battery possible. Packaging (battery box material, coating, terminal protection): Battery is housed within the Nikon camera and soldered into the circuit board. Wicking material: N/A Vent design and operating characteristics (opening pressure? redundancy scheme?): N/A Test data for the cells, battery, or complete hardware package: Soft Stowage Vibration Test to 6.8 gRMS of each SPICE subassembly. Thermal Test for 10 cycles from -10 to 45 C of the complete SPICE Assembly.		

EP5 BATTERY DESIGN EVALUATION FORM	Page 2 of 7
3b. Chemical and Electrical Description: <i>If more than one battery is contained in the hardware, provide information for <u>all</u> the batteries.</i>	
<u>Cell Description:</u> Chemistry: Li-Mn-O2 Size: 20mmX3.2mm Manufacturer: Maxell Model number: ML2032T6 Nominal open circuit voltage: 3.0 V Expected load voltage: 3.0 V Rated capacity: 65mAh <u>Battery Description: (9 V COTS batteries shall be considered as single units)</u> Quantity of cells per battery: 1 Connectivity of cells (serial? parallel?): N/A Operating voltage range: Operating temperature range: -20 to +60 C Average Load and duration: Unknown/Continuous Peak Load and duration: Unknown Capacity Required: Unknown Circuit description and electrical schematic (attach electronically or indicate that a hard copy is provided): See attached. Circuit protection (i.e. fuses, diodes, MOSFETs, resistors, source isolation, etc.): Diode	
3c. Is the battery to be charged on-orbit? Yes ☐ No ☒ <i>If yes, describe charging scenario, hardware, and protective device. Attach charger schematics.</i> For operation on-orbit the camera will be powered via its DC input connector. The camera can also be powered by a battery pack although this is not part of the flight configuration. When the camera is powered via the DC input connector the battery is NOT being charged. This information was provided to the SPICE team during discussions with Nikon via email dated January 20, 2004.	

EP5 BATTERY DESIGN EVALUATION FORM

Page 3 of 7

3d. Summarize the pre-flight processing plan for the hardware. (See Section 4.2 of EA-CWI-033 for an overview of the processes to be considered for battery pre-flight processing.) If the preflight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation.

There battery has no specific pre-flight processing, however, the SPICE Assembly including all hardware containing a battery will or has undergone workmanship vibration testing and thermal testing as described previously.

The Phase II review presented that the battery in the Nikon Camera would be removed. The battery in the qualification unit was removed, however, as a result the camera hasd become non-operational. The SPICE project was cancelled following the Phase II Safety Review in January 2004. The project was resurrected in 2006. The new project management believes removing the battery from the camera is a technical risk and thus this form submittal is to obtain approval

3e. Summarize the on-orbit processes and operational constraints for the hardware. (See Section 4.3 of EA-CWI-033 for an overview of the processes to be considered for on-orbit battery usage.) If the on-orbit processes and operational constraints are documented elsewhere, indicate where they are documented and provide a copy of the documentation.

The hardware will be assembled into the MSG facility. The MSG facility will be sealed. The crew will operate the hardware via the MSG glove ports. The MSG facility will be opened and the SPICE hardware removed and returned to stowage.

3f. Summarize the post-flight processing plan for the hardware. (See Section 4.4 of EA-CWI-033 for an overview of the steps to be completed as part of post-flight processing.) If the post-flight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation.

There is currently no post-flight processing plan for the hardware other than the hardware is transferred to shuttle and the hardware returned to the ground.

EP5 BATTERY DESIGN EVALUATION FORM

Page 3 of 7

3d. Summarize the pre-flight processing plan for the hardware. (See Section 4.2 of EA-CWI-033 for an overview of the processes to be considered for battery pre-flight processing.) If the preflight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation.

There battery has no specific pre-flight processing, however, the SPICE Assembly including all hardware containing a battery will or has undergone workmanship vibration testing and thermal testing as described previously.

The Phase II review presented that the battery in the Nikon Camera would be removed. The battery in the qualification unit was removed, however, as a result the camera hasd become non-operational. The SPICE project was cancelled following the Phase II Safety Review in January 2004. The project was resurrected in 2006. The new project management believes removing the battery from the camera is a technical risk and thus this form submittal is to obtain approval

3e. Summarize the on-orbit processes and operational constraints for the hardware. (See Section 4.3 of EA-CWI-033 for an overview of the processes to be considered for on-orbit battery usage.) If the on-orbit processes and operational constraints are documented elsewhere, indicate where they are documented and provide a copy of the documentation.

The hardware will be assembled into the MSG facility. The MSG facility will be sealed. The crew will operate the hardware via the MSG glove ports. The MSG facility will be opened and the SPICE hardware removed and returned to stowage.

3f. Summarize the post-flight processing plan for the hardware. (See Section 4.4 of EA-CWI-033 for an overview of the steps to be completed as part of post-flight processing.) If the post-flight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation.

There is currently no post-flight processing plan for the hardware other than the hardware is transferred to shuttle and the hardware returned to the ground.

EP5 BATTERY DESIGN EVALUATION FORM

Page 4 of 7

Section 4 to be completed by EP5 Reviewer:

4. Evaluation Assessment:

4a. Does the design address:

Two-fault tolerance to catastrophic failure?	Yes ☐	No ☐	N/A ☐
Cell/battery gas generation mitigation or containment?	Yes ☐	No ☐	N/A ☐
Cell/battery electrolyte leakage mitigation or containment?	Yes ☐	No ☐	N/A ☐
Electrolyte toxicity memo obtained?	Yes ☐	No ☐	N/A ☐
Cell/battery electrical circuit protection?	Yes ☐	No ☐	N/A ☐
Cell/battery high temperature protection?	Yes ☐	No ☐	N/A ☐
Cell/battery materials compatibility and offgassing?	Yes ☐	No ☐	N/A ☐
Materials certification memo obtained?	Yes ☐	No ☐	N/A ☐

Are the battery requirements and hazard controls adequately addressed per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review: January 2004

If no, describe additional hazard controls required for adequate control. If N/A, provide rationale.

4b. Is the cell screening or battery pack screening adequate per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review:

If no, describe additional screening required for cells or packs.

4c. Does the pre-flight processing plan address:

Flight cell/battery pack qualification/certification testing?	Yes ☐	No ☐	N/A ☐
Flight cell/ battery pack verification acceptance testing?	Yes ☐	No ☐	N/A ☐
Installation of fresh, flight-acceptance tested cells/battery packs?	Yes ☐	No ☐	N/A ☐
Maintaining storage conditions?	Yes ☐	No ☐	N/A ☐

Is the pre-flight processing plan adequate per JSC-20793 guidelines and per Section 4.2 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the pre-flight processing plan.

4d. Do the on-orbit processing plan and operational constraints address:

Crew procedures to inspect and checkout hardware prior to usage?	Yes ☐	No ☐	N/A ☐
Crew procedures for removing, discarding, and replacing non-rechargeable, depleted cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedure for removing, recharging, and storing rechargeable cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedures for stowage of equipment?	Yes ☐	No ☐	N/A ☐
Crew procedures for documentation of anomalies?	Yes ☐	No ☐	N/A ☐

Are the plans for on-orbit processing and operational constraints adequate per Section 4.3 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the on-orbit processing plan and operational constraints.

4e. Does the post-flight processing plan address:

Post-flight performance evaluation of hardware?	Yes ☐	No ☐	N/A ☐
Removal of cells and battery packs?	Yes ☐	No ☐	N/A ☐
Discarding or downgrading removed cells for non-flight use?	Yes ☐	No ☐	N/A ☐
Inspection of battery compartment and contacts?	Yes ☐	No ☐	N/A ☐
Hardware storage (minus cells and battery packs)?	Yes ☐	No ☐	N/A ☐

Is the post-flight processing plan adequate per Section 4.4 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the post-flight processing plan.

EP-Form-03 Rev A (4/15/2005)

Verify this is the correct version before use.

Energy Systems Division

Original Signed and approved by Patricia A. Petete

EP5 BATTERY DESIGN EVALUATION FORM		Page 5 of 7
4f. Subsequent battery review held to verify:		
4.f.1 Adequate hazard controls implemented? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.2 Cell-screening or battery pack screening plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.3 Pre-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 On-orbit processing plan and operational constraints adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 Post-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:

EP5 BATTERY DESIGN EVALUATION FORM	Page 6 of 7
Section 5 to be completed by EP5 Reviewer:	
5a. Evaluation Results: The results of the battery evaluation conducted for the hardware outlined in Section 2 are as follows: ☐ Battery design for use in the stated application is approved for the following purposes: ☐ Ground Testing ☐ Shuttle ☐ Space Station ☐ Russian Vehicles (specify) ☐ Other (specify) ☐ Battery design is not approved for the following reason(s): ☐ Evaluation request withdrawn by Hardware Point-of-Contact ☐ Hardware Point-of-Contact states requested modifications will not be incorporated ☐ Requested modifications could not be incorporated; new request for alternate hardware will be initiated by Hardware Point-of-Contact ☐ Other (specify)	
5b. Approval of EP5 Battery Reviewer: <i>Typed Name</i> <i>Signature</i> <i>Date</i>	
Section 6 to be completed by Hardware Point-of-Contact:	
6. Concurrence of Hardware Point-of-Contact: I concur with the results of the battery evaluation conducted for the hardware outlined in Section 2. <i>Typed Name</i> <i>Signature</i> <i>Date</i>	

Data Sheet

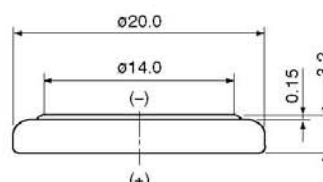
Lithium Manganese Dioxide Rechargeable Battery

ML2032

maxell

1/2

Model	ML2032	
System	(Li-Al) alloy-Manganese Dioxide/Organic Electrolyte	
Nominal Voltage (V)	3	
Nominal Capacity (mAh)*	65	
Nominal Discharge Current (μA)	200	
Charge/Discharge Cycle Lifetime		
Depth of Discharge = 10%	1,000 (6.5 mAh discharge) (total capacity 6,500 mAh)	
Depth of Discharge = 20%	300 (13 mAh discharge) (total capacity 3,900 mAh)	
Temperature Ranges (deg. C)	min.	max.
Operating	-20	+60
Weight (g)**	3.0	
Dimensions (mm)**		
Diameter	20	
Height	3.2	
UL Recognition	MH12568	

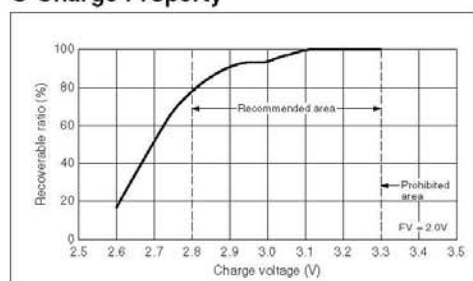


Available Terminals and Wire Connectors

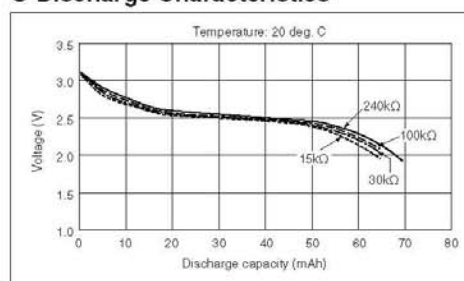
Check http://www.maxell.co.jp/e/products/industrial/battery/pdf/ml2032tw_e.pdf for diagrams of batteries with terminals.

Characteristics

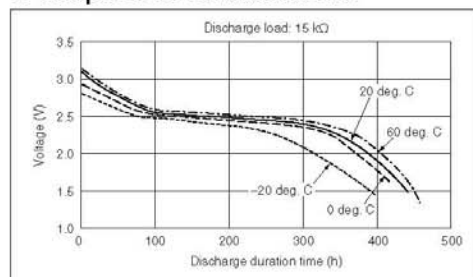
● Charge Property



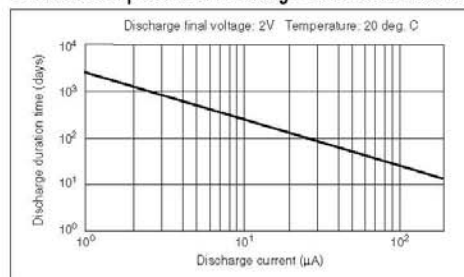
● Discharge Characteristics



● Temperature Characteristics



● Relationship between Discharge Current and Duration Time



Data Sheet

Lithium Manganese Dioxide Rechargeable Battery

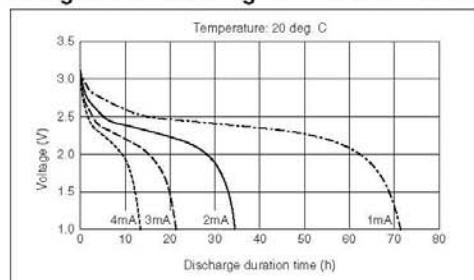
ML2032

maxell

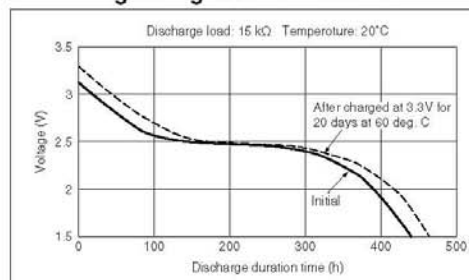
2/2

Characteristics

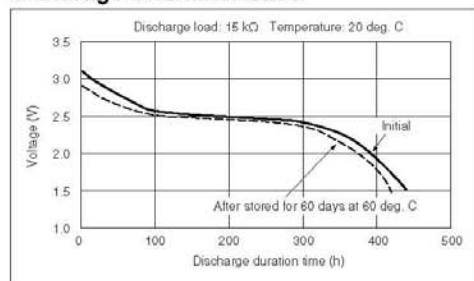
● High Rate Discharge Characteristics



● Floating Charge Characteristics



● Storage Characteristics



* Nominal capacity indicates duration until the voltage drops down to 2.0V when discharged at a nominal discharge current at 20 deg. C.

** Dimensions and weight are for the battery itself, but may vary depending on the shape of terminals or other factors.

Hitachi Maxell, Ltd.
2-18-2, Iidabashi, Chiyoda-ku, Tokyo 102-8521, Japan
<http://www.maxell.com>

• Data and dimensions are just reference values. For further details, please contact us at your nearest Maxell office.
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Date of issue: April, 2007

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Button

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LR

ML

Lithium Manganese Dioxide Rechargeable Batteries

Outline of Products

Warnings

Maxell ML (lithium manganese dioxide rechargeable) batteries are available only for equipment manufacturers as a built-in part. Therefore, Maxell does not supply these batteries for replacement directly to users of equipment with these batteries. When built in ML batteries need to be replaced, please contact your equipment manufacturer. If you are planning to use Maxell ML batteries in your equipment as a built-in part, please contact Maxell.

Products

Model	Nominal Voltage (V)	Nominal Capacity (mAh) ^{nom}	Nominal Discharge Current (μA)	Charge/Discharge Cycle Lifetime		Dimensions*		Weight (g) [†]	Data Sheets	With terminals	Wa
				Depth of Discharge = 10%	Depth of Discharge = 20%	Diameter (mm)	Height (mm)				
ML 2032	3	65	200	1,000	300	20	3.2	3.0	46kb	81kb	
ML 2016	3	25	200	1,500	500	20	1.6	1.8	132kb	52kb	6
ML 1220	3	18	100	1,500	500	12.5	2.0	0.7	70kb	77kb	

* Dimensions and weight are for the battery itself, but may vary depending on the shape of terminals or other factors.

** Nominal capacity indicates duration until the voltage drops down to 2.0V when discharged at a nominal discharge current at 20 deg. C.

▪ Data and dimensions are not guaranteed but reference values. For further details, please contact us at your nearest Maxell office.

<http://www.maxell.co.jp/e/products/industrial/battery/ml/index.html>

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Product Safety Data Sheet

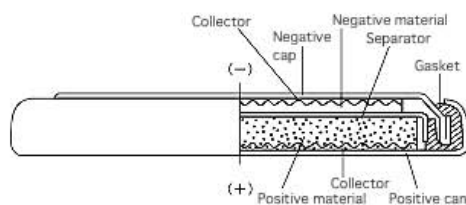


[Download\(44kb\)](#)

Overview

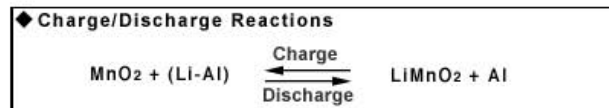
This battery employs specially treated manganese dioxide for the positive material and a lithium-aluminum compound for the negative material. A specially formulated organic electrolyte is also used, yielding excellent discharge characteristics with low self-discharge.

Construction

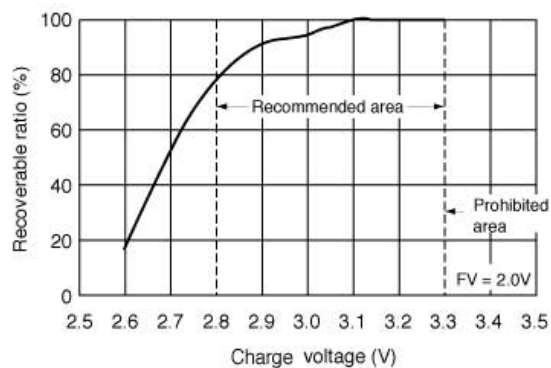


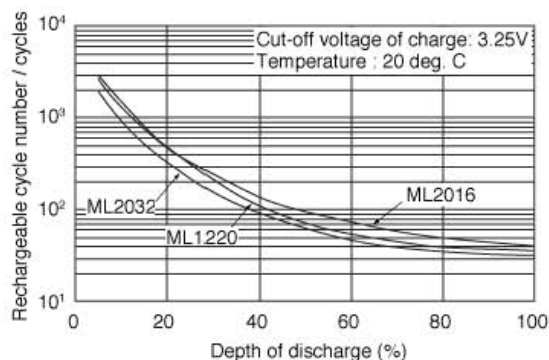
Principle and Reactions

The coin-type lithium manganese dioxide rechargeable battery is a 3V battery using specially treated manganese dioxide for the positive material, a lithium-aluminum compound for the negative material, and a specially formulated organic electrolyte.

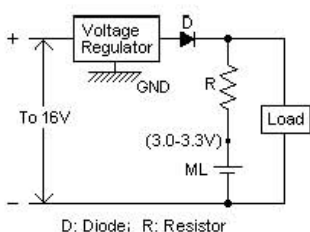


Charge Property and Charge/Discharge Cycle Performance





Recommended circuits



Model	Output Voltage of Voltage Regulator	
	3.1V	3.2V
ML2032	> 550 ohm	> 600 ohm
ML2016	> 550 ohm	> 600 ohm
ML1220	> 1.1K ohm	> 1.2K ohm

D: Diode; R: Resistor

Nominal Charge Current by Model

	ML2032	ML2016	ML1220
Charge current	less than 2mA	less than 2mA	less than 1mA

For details, refer to [Warning - On use of Maxell Coin-Type Lithium Manganese Dioxide Rechargeable Battery.](#)

Features

Approx. 2.5V operating voltage

The operating voltage is about twice that of nickel cadmium rechargeable batteries. Displays a high discharge voltage of 2.8 V when at 10% of nominal capacity (depth of discharge is 10% or less), when charged at 3.0 to 3.3 V.

Superior discharge characteristics

Achieves 1,000 cycles of discharging to 10% of nominal capacity (Depth of discharge = 10%). The total discharge capacity is quite high at 100 times nominal capacity.

Wide -20 deg. C to 60 deg. C usable temperature range

Demonstrates stable operating voltage in temperatures as low as -20 deg. C and as high as 60 deg. C.

Low self-discharge and superior leakage resistance

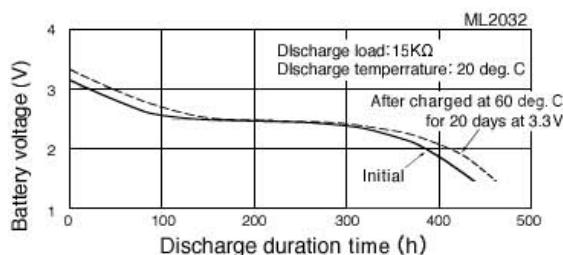
Self-discharge at 20 deg. C is no more than 2% per year. Supplies a nominal capacity of about 95% even when stored at 20 deg. C for roughly five years (according to accelerated test conducted by Maxell). And since organic electrolyte is used, the battery has superior leakage resistance (shipped fully charged).

Excellent overcharge characteristics

A specially formulated organic electrolyte is employed to provide stable discharge characteristics even if charged for a year at 3.3 V at 20 deg. C (according to accelerated test conducted by Maxell).

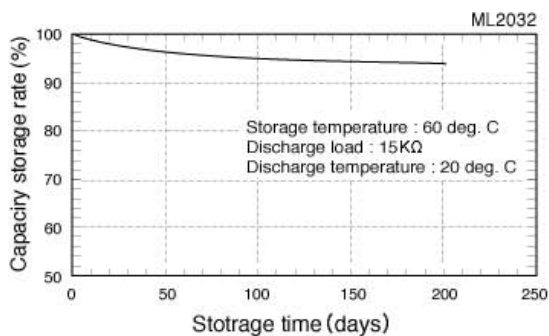
UL (Underwriters Laboratories Inc.) Recognized Components

Superior Floating Charge Characteristics



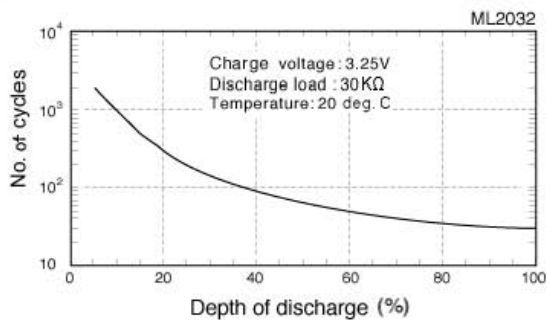
Battery discharge duration time accelerated to 20 days at 60 deg. C is equivalent to one year at 20 deg. C

Low Self-discharge

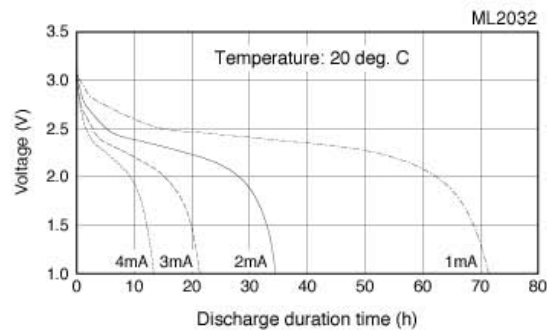


(when accelerating 20 days at 60 deg. C to 1 year at 20 deg. C)

Superior Charge and Discharge Characteristics



Superior High Rate Discharge Characteristics



Applications

PCs / note PCs / PDA / digital cameras / mobile phones / electronic calculators / video-camera combos / faxes / timepieces / VTRs / memory cards / timer-equipped home electronics / other info-communications devices

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EP5 BATTERY DESIGN EVALUATION FORM		Page 1 of 7
<i>Section 1 to be completed by EP/Division Secretary:</i>		
1a. EP Tracking Number:	1b. Assigned EP5 Reviewer:	
<i>Section 2 to be completed by the Hardware Point-of-Contact:</i>		
2a. Hardware Point-of-Contact: (Name/Phone/Company/Mail Code) Tibor LorikZIN Technologies Inc	2b. Hardware Name: Smoke Point in Co-Flow Experiment Hardware Part Number: 60054MAFH400 Hardware Acronym: SPICE	
2c. Hardware Managing Group, Company, or Agency: NASA GRC/ZIN Technologies		
2d. Hardware Flight Information (i.e. flight application, target flight number and/or date, number of flights anticipated, etc.): The SPICE hardware is scheduled for Flight ULF-2 - 10/2008. The hardware contains 3 different batteries. The first battery is located in the Nikon Camera. It is rechargeable although it will not be configured for charging. A second is on the Video Titler Board, and a third in the Dallas Semiconductor I-Button. The batteries are used to maintain configuration settings and clock. The hardware will be flown aboard the Shuttle and transferred to ISS where it will be stowed awaiting its scheduled use in the MSG Facility.		
2e. Hardware environmental Requirements: Thermal (max, min, operational and non-operational ranges): 10 – 46 C both operating and non-operating. The hardware was tested to these levels. Pressure (EVA, IVA): Cabin Pressure. Hardware has been analyzed to be safe when exposed to pressures of 0 to 15.2 PSIA. Life (calendar/shelf, cycle): 10 years		
<i>Section 3 to be completed by Hardware Point-of-Contact:</i>		
3a. Battery Hardware Description: Are the cells/ the battery pack Commercial-off-the-shelf (COTS)? YES Function/Operating mode (pulse? intermittent? clock backup? memory?): Configuration Settings in Memory Battery interfaces (connectors? equipment?): Batteries are installed on Printed Circuit boards and are soldered. Stowage location (installation and use locations): Hardware will be launched in Middeck or MPLM. On-orbit hardware is placed in on-orbit stowage during operations hardware is located in the MSG Facility. Battery/cell access during flight: None Short circuit/safe-touch temperature protection: Unsure if this rechargeable battery has short circuit protection. Battery is embedded within the Video Titler Assembly with no crew contact of the battery possible. Packaging (battery box material, coating, terminal protection): Battery is housed within the Video Titler Assembly and soldered into the circuit board. Wicking material: N/A Vent design and operating characteristics (opening pressure? redundancy scheme?): N/A Test data for the cells, battery, or complete hardware package: Soft Stowage Vibration Test to 6.8 gRMS of each SPICE subassembly. Thermal Test for 10 cycles from -10 to 45 C of the complete SPICE Assembly.		

EP5 BATTERY DESIGN EVALUATION FORM	Page 2 of 7
3b. Chemical and Electrical Description: <i>If more than one battery is contained in the hardware, provide information for <u>all</u> the batteries.</i>	
<u>Cell Description:</u> Chemistry: Nickel Metal Hydride Size: 15.5 mmX 6.3 mm Manufacturer: GP Batteries Intl Model number: GP60BVH Nominal open circuit voltage: 1.2 V Expected load voltage: 1.2 V Rated capacity: 60mAh <u>Battery Description: (9 V COTS batteries shall be considered as single units)</u> Quantity of cells per battery: 1 Connectivity of cells (serial? parallel?): N/A Operating voltage range: Operating temperature range: 0 - 35 C Average Load and duration: Unknown/Continuous Peak Load and duration: Unknown Capacity Required: Unknown Circuit description and electrical schematic (attach electronically or indicate that a hard copy is provided): See attached. Circuit protection (i.e. fuses, diodes, MOSFETs, resistors, source isolation, etc.): Diode	
3c. Is the battery to be charged on-orbit? Yes ☐ No ☒ <i>If yes, describe charging scenario, hardware, and protective device. Attach charger schematics.</i>	

EP5 BATTERY DESIGN EVALUATION FORM	Page 3 of 7
3d. Summarize the pre-flight processing plan for the hardware. (See Section 4.2 of EA-CWI-033 for an overview of the processes to be considered for battery pre-flight processing.) If the preflight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation. The battery has no specific pre-flight processing, however, the SPICE Assembly including all hardware containing a battery will or has undergone workmanship vibration testing and thermal testing as described previously.	
3e. Summarize the on-orbit processes and operational constraints for the hardware. (See Section 4.3 of EA-CWI-033 for an overview of the processes to be considered for on-orbit battery usage.) If the on-orbit processes and operational constraints are documented elsewhere, indicate where they are documented and provide a copy of the documentation. The hardware will be assembled into the MSG facility. The MSG facility will be sealed. The crew will operate the hardware via the MSG glove ports. The MSG facility will be opened and the SPICE hardware removed and returned to stowage.	
3f. Summarize the post-flight processing plan for the hardware. (See Section 4.4 of EA-CWI-033 for an overview of the steps to be completed as part of post-flight processing.) If the post-flight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation. There is currently no post-flight processing plan for the hardware other than the hardware is transferred to shuttle and the hardware returned to the ground.	

EP5 BATTERY DESIGN EVALUATION FORM

Page 4 of 7

Section 4 to be completed by EP5 Reviewer:

4. Evaluation Assessment:

4a. Does the design address:

Two-fault tolerance to catastrophic failure?	Yes ☐	No ☐	N/A ☐
Cell/battery gas generation mitigation or containment?	Yes ☐	No ☐	N/A ☐
Cell/battery electrolyte leakage mitigation or containment?	Yes ☐	No ☐	N/A ☐
Electrolyte toxicity memo obtained?	Yes ☐	No ☐	N/A ☐
Cell/battery electrical circuit protection?	Yes ☐	No ☐	N/A ☐
Cell/battery high temperature protection?	Yes ☐	No ☐	N/A ☐
Cell/battery materials compatibility and offgassing?	Yes ☐	No ☐	N/A ☐
Materials certification memo obtained?	Yes ☐	No ☐	N/A ☐

Are the battery requirements and hazard controls adequately addressed per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review: January 2004

If no, describe additional hazard controls required for adequate control. If N/A, provide rationale.

4b. Is the cell screening or battery pack screening adequate per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review:

If no, describe additional screening required for cells or packs.

4c. Does the pre-flight processing plan address:

Flight cell/battery pack qualification/certification testing?	Yes ☐	No ☐	N/A ☐
Flight cell/ battery pack verification acceptance testing?	Yes ☐	No ☐	N/A ☐
Installation of fresh, flight-acceptance tested cells/battery packs?	Yes ☐	No ☐	N/A ☐
Maintaining storage conditions?	Yes ☐	No ☐	N/A ☐

Is the pre-flight processing plan adequate per JSC-20793 guidelines and per Section 4.2 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the pre-flight processing plan.

4d. Do the on-orbit processing plan and operational constraints address:

Crew procedures to inspect and checkout hardware prior to usage?	Yes ☐	No ☐	N/A ☐
Crew procedures for removing, discarding, and replacing non-rechargeable, depleted cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedure for removing, recharging, and storing rechargeable cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedures for stowage of equipment?	Yes ☐	No ☐	N/A ☐
Crew procedures for documentation of anomalies?	Yes ☐	No ☐	N/A ☐

Are the plans for on-orbit processing and operational constraints adequate per Section 4.3 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the on-orbit processing plan and operational constraints.

4e. Does the post-flight processing plan address:

Post-flight performance evaluation of hardware?	Yes ☐	No ☐	N/A ☐
Removal of cells and battery packs?	Yes ☐	No ☐	N/A ☐
Discarding or downgrading removed cells for non-flight use?	Yes ☐	No ☐	N/A ☐
Inspection of battery compartment and contacts?	Yes ☐	No ☐	N/A ☐
Hardware storage (minus cells and battery packs)?	Yes ☐	No ☐	N/A ☐

Is the post-flight processing plan adequate per Section 4.4 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the post-flight processing plan.

EP-Form-03 Rev A (4/15/2005)

Verify this is the correct version before use.

Energy Systems Division

Original Signed and approved by Patricia A. Petete

EP5 BATTERY DESIGN EVALUATION FORM		Page 5 of 7
4f. Subsequent battery review held to verify:		
4.f.1 Adequate hazard controls implemented? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.2 Cell-screening or battery pack screening plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.3 Pre-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 On-orbit processing plan and operational constraints adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 Post-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:

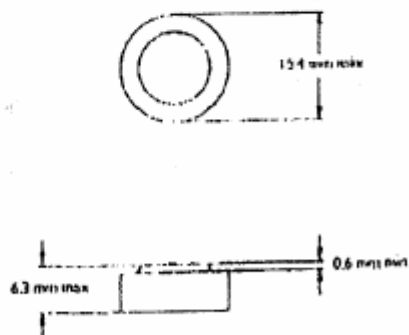
EP5 BATTERY DESIGN EVALUATION FORM	Page 6 of 7
Section 5 to be completed by EP5 Reviewer:	
5a. Evaluation Results: The results of the battery evaluation conducted for the hardware outlined in Section 2 are as follows: ☐ Battery design for use in the stated application is approved for the following purposes: ☐ Ground Testing ☐ Shuttle ☐ Space Station ☐ Russian Vehicles (specify) ☐ Other (specify) ☐ Battery design is not approved for the following reason(s): ☐ Evaluation request withdrawn by Hardware Point-of-Contact ☐ Hardware Point-of-Contact states requested modifications will not be incorporated ☐ Requested modifications could not be incorporated; new request for alternate hardware will be initiated by Hardware Point-of-Contact ☐ Other (specify)	
5b. Approval of EP5 Battery Reviewer: <i>Typed Name</i> <i>Signature</i> <i>Date</i>	
Section 6 to be completed by Hardware Point-of-Contact:	
6. Concurrence of Hardware Point-of-Contact: I concur with the results of the battery evaluation conducted for the hardware outlined in Section 2. <i>Typed Name</i> <i>Signature</i> <i>Date</i>	

GP. SYLVA • CHARGE

Data Sheet for GP60BNK

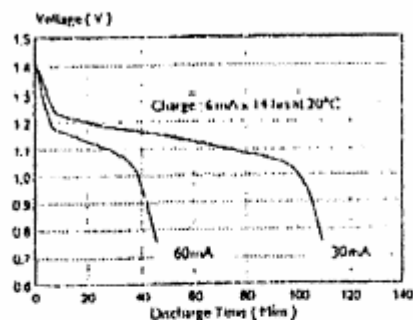
Specifications :

Type	: Rechargeable Nickel Cadmium Button Cell
Model	: GP60BNK
Nominal Dimension	: $\Phi = 15.4$ mm max
For Bare Cell	: H = 6.3 mm max
Applications	: Recommended discharge current 0.6 to 60 mA
Nominal Voltage	: 1.2 V
Nominal Capacity	: 60 mAh when discharge at 12 mA to 1.0 V at 20°C
Charging Condition	: 6 mA for 14 hrs at 20°C
Service Life	: 500 - 1000 Cycles
Continuous Overcharge	: 6 mA maximum permissible current. No conspicuous deformation and/or leakage.
Weight	: 3.7 g
Internal Resistance	: Under 1300 m Ω upon fully charged
Max. Charging Voltage	: 1.5 V at 6 mA charging
Temperature Range	: Charging : 0°C to 35°C Discharging : 0°C to 45°C Storage : -20°C to 35°C

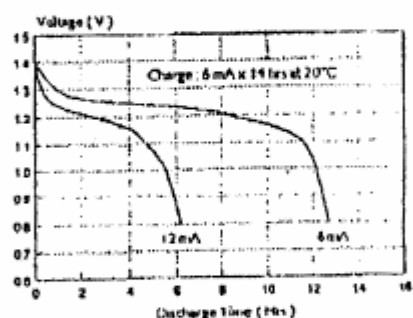


Characteristics :

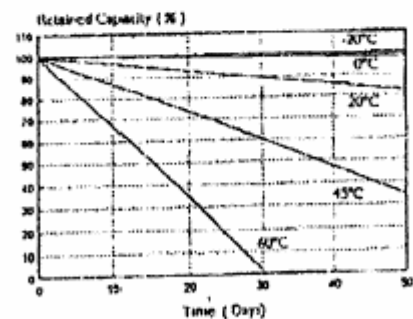
Discharge (At High Rate)

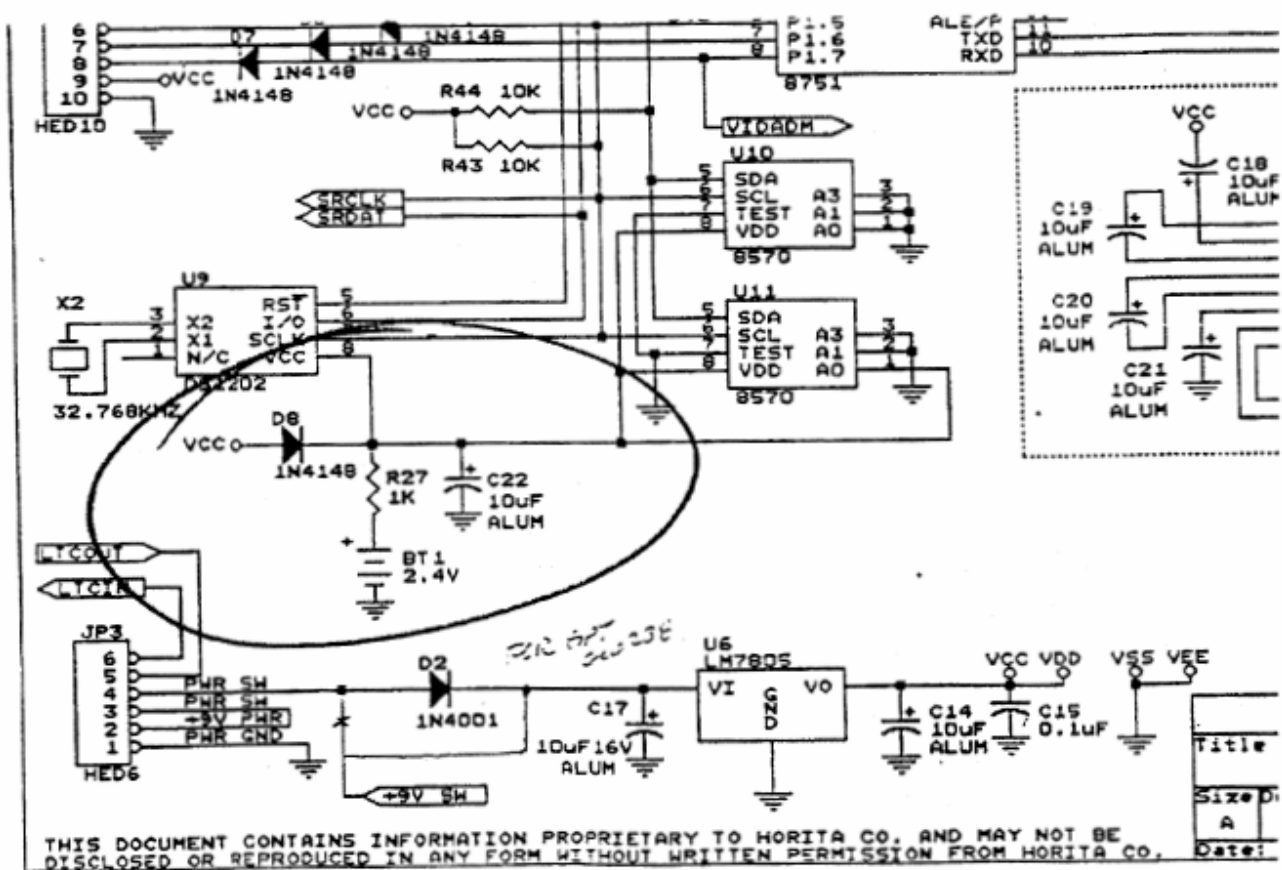


Discharge (At Low Rate)



Charge Retention Characteristics





EP5 BATTERY DESIGN EVALUATION FORM		Page 1 of 7
<i>Section 1 to be completed by EP/Division Secretary:</i>		
1a. EP Tracking Number:	1b. Assigned EP5 Reviewer:	
<i>Section 2 to be completed by the Hardware Point-of-Contact:</i>		
2a. Hardware Point-of-Contact: (Name/Phone/Company/Mail Code) Tibor LorikZIN Technologies Inc	2b. Hardware Name: Smoke Point in Co-Flow Experiment Hardware Part Number: 60054MAFH400 Hardware Acronym: SPICE	
2c. Hardware Managing Group, Company, or Agency: NASA GRC/ZIN Technologies		
2d. Hardware Flight Information (i.e. flight application, target flight number and/or date, number of flights anticipated, etc.): The SPICE hardware is scheduled for Flight ULF-2 - 10/2008. The hardware contains 3 different batteries. The first battery is located in the Nikon Camera. It is rechargeable although it will not be configured for charging. A second is on the Video Tiler Board, and a third in the Dallas Semiconductor I-Button. The batteries are used to maintain configuration settings and clock. The hardware will be flown aboard the Shuttle and transferred to ISS where it will be stowed awaiting its scheduled use in the MSG Facility.		
2e. Hardware environmental Requirements: Thermal (max, min, operational and non-operational ranges): 10 – 46 C both operating and non-operating. The hardware was tested to these levels. Pressure (EVA, IVA): Cabin Pressure. Hardware has been analyzed to be safe when exposed to pressures of 0 to 15.2 PSIA. Life (calendar/shelf, cycle): 10 years		
<i>Section 3 to be completed by Hardware Point-of-Contact:</i>		
3a. Battery Hardware Description: Are the cells/ the battery pack Commercial-off-the-shelf (COTS)? YES Function/Operating mode (pulse? intermittent? clock backup? memory?): Configuration Settings in Memory Battery interfaces (connectors? equipment?): Batteries are installed on Printed Circuit boards and are soldered. Stowage location (installation and use locations): Hardware will be launched in Middeck or MPLM. On-orbit hardware is placed in on-orbit stowage during operations hardware is located in the MSG Facility. Battery/cell access during flight: None Short circuit/safe-touch temperature protection: Battery is embedded within the Experiment Assembly with no crew contact of the battery possible. Packaging (battery box material, coating, terminal protection): Battery is housed within the Experiment Assembly and soldered into the circuit board. Wicking material: N/A Vent design and operating characteristics (opening pressure? redundancy scheme?): N/A Test data for the cells, battery, or complete hardware package: Soft Stowage Vibration Test to 6.8 gRMS of each SPICE subassembly. Thermal Test for 10 cycles from -10 to 45 C of the complete SPICE Assembly.		

EP5 BATTERY DESIGN EVALUATION FORM	Page 2 of 7
3b. Chemical and Electrical Description: <i>If more than one battery is contained in the hardware, provide information for <u>all</u> the batteries.</i>	
<u>Cell Description:</u> Chemistry: Li-CFx Size: 12.5 mm X 2.5 mm Manufacturer: Rayovac Model number: BR1225 Nominal open circuit voltage: 3 V Expected load voltage: 3 V Rated capacity: 50mAh <u>Battery Description: (9 V COTS batteries shall be considered as single units)</u> Quantity of cells per battery: 1 Connectivity of cells (serial? parallel?): N/A Operating voltage range: Operating temperature range: -40 - 70 C Average Load and duration: Unknown/Continuous Peak Load and duration: Unknown Capacity Required: Unknown Circuit description and electrical schematic (attach electronically or indicate that a hard copy is provided): See attached. Circuit protection (i.e. fuses, diodes, MOSFETs, resistors, source isolation, etc.): Diode	
3c. Is the battery to be charged on-orbit? Yes ☐ No ☒ <i>If yes, describe charging scenario, hardware, and protective device. Attach charger schematics.</i>	

EP5 BATTERY DESIGN EVALUATION FORM	Page 3 of 7
3d. Summarize the pre-flight processing plan for the hardware. (See Section 4.2 of EA-CWI-033 for an overview of the processes to be considered for battery pre-flight processing.) If the preflight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation. The battery has no specific pre-flight processing, however, the SPICE Assembly including all hardware containing a battery will or has undergone workmanship vibration testing and thermal testing as described previously.	
3e. Summarize the on-orbit processes and operational constraints for the hardware. (See Section 4.3 of EA-CWI-033 for an overview of the processes to be considered for on-orbit battery usage.) If the on-orbit processes and operational constraints are documented elsewhere, indicate where they are documented and provide a copy of the documentation. The hardware will be assembled into the MSG facility. The MSG facility will be sealed. The crew will operate the hardware via the MSG glove ports. The MSG facility will be opened and the SPICE hardware removed and returned to stowage.	
3f. Summarize the post-flight processing plan for the hardware. (See Section 4.4 of EA-CWI-033 for an overview of the steps to be completed as part of post-flight processing.) If the post-flight processing plan is documented elsewhere, indicate where it is documented and provide a copy of the documentation. There is currently no post-flight processing plan for the hardware other than the hardware is transferred to shuttle and the hardware returned to the ground.	

EP5 BATTERY DESIGN EVALUATION FORM

Page 4 of 7

Section 4 to be completed by EP5 Reviewer:

4. Evaluation Assessment:

4a. Does the design address:

Two-fault tolerance to catastrophic failure?	Yes ☐	No ☐	N/A ☐
Cell/battery gas generation mitigation or containment?	Yes ☐	No ☐	N/A ☐
Cell/battery electrolyte leakage mitigation or containment?	Yes ☐	No ☐	N/A ☐
Electrolyte toxicity memo obtained?	Yes ☐	No ☐	N/A ☐
Cell/battery electrical circuit protection?	Yes ☐	No ☐	N/A ☐
Cell/battery high temperature protection?	Yes ☐	No ☐	N/A ☐
Cell/battery materials compatibility and offgassing?	Yes ☐	No ☐	N/A ☐
Materials certification memo obtained?	Yes ☐	No ☐	N/A ☐

Are the battery requirements and hazard controls adequately addressed per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review: January 2004

If no, describe additional hazard controls required for adequate control. If N/A, provide rationale.

4b. Is the cell screening or battery pack screening adequate per JSC-20793 guidelines?

Yes ☐ No ☐ Date of review:

If no, describe additional screening required for cells or packs.

4c. Does the pre-flight processing plan address:

Flight cell/battery pack qualification/certification testing?	Yes ☐	No ☐	N/A ☐
Flight cell/ battery pack verification acceptance testing?	Yes ☐	No ☐	N/A ☐
Installation of fresh, flight-acceptance tested cells/battery packs?	Yes ☐	No ☐	N/A ☐
Maintaining storage conditions?	Yes ☐	No ☐	N/A ☐

Is the pre-flight processing plan adequate per JSC-20793 guidelines and per Section 4.2 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the pre-flight processing plan.

4d. Do the on-orbit processing plan and operational constraints address:

Crew procedures to inspect and checkout hardware prior to usage?	Yes ☐	No ☐	N/A ☐
Crew procedures for removing, discarding, and replacing non-rechargeable, depleted cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedure for removing, recharging, and storing rechargeable cells or batteries?	Yes ☐	No ☐	N/A ☐
Crew procedures for stowage of equipment?	Yes ☐	No ☐	N/A ☐
Crew procedures for documentation of anomalies?	Yes ☐	No ☐	N/A ☐

Are the plans for on-orbit processing and operational constraints adequate per Section 4.3 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the on-orbit processing plan and operational constraints.

4e. Does the post-flight processing plan address:

Post-flight performance evaluation of hardware?	Yes ☐	No ☐	N/A ☐
Removal of cells and battery packs?	Yes ☐	No ☐	N/A ☐
Discarding or downgrading removed cells for non-flight use?	Yes ☐	No ☐	N/A ☐
Inspection of battery compartment and contacts?	Yes ☐	No ☐	N/A ☐
Hardware storage (minus cells and battery packs)?	Yes ☐	No ☐	N/A ☐

Is the post-flight processing plan adequate per Section 4.4 of EA-CWI-033?

Yes ☐ No ☐ Date of review:

If no, describe additional items to be added to the post-flight processing plan.

EP-Form-03 Rev A (4/15/2005)

Verify this is the correct version before use.

Energy Systems Division

Original Signed and approved by Patricia A. Petete

EP5 BATTERY DESIGN EVALUATION FORM		Page 5 of 7
4f. Subsequent battery review held to verify:		
4.f.1 Adequate hazard controls implemented? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.2 Cell-screening or battery pack screening plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.3 Pre-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 On-orbit processing plan and operational constraints adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:
4.f.4 Post-flight processing plan adequate? Comments from additional review:	N/A ☐ Yes ☐	Date of review:

EP5 BATTERY DESIGN EVALUATION FORM	Page 6 of 7
Section 5 to be completed by EP5 Reviewer:	
5a. Evaluation Results: The results of the battery evaluation conducted for the hardware outlined in Section 2 are as follows: ☐ Battery design for use in the stated application is approved for the following purposes: ☐ Ground Testing ☐ Shuttle ☐ Space Station ☐ Russian Vehicles (specify) ☐ Other (specify) ☐ Battery design is not approved for the following reason(s): ☐ Evaluation request withdrawn by Hardware Point-of-Contact ☐ Hardware Point-of-Contact states requested modifications will not be incorporated ☐ Requested modifications could not be incorporated; new request for alternate hardware will be initiated by Hardware Point-of-Contact ☐ Other (specify)	
5b. Approval of EP5 Battery Reviewer: <i>Typed Name</i> <i>Signature</i> <i>Date</i>	
Section 6 to be completed by Hardware Point-of-Contact:	
6. Concurrence of Hardware Point-of-Contact: I concur with the results of the battery evaluation conducted for the hardware outlined in Section 2. <i>Typed Name</i> <i>Signature</i> <i>Date</i>	



DS1994 4kb Plus Time Memory iButton

www.iButton.com

SPECIAL FEATURES

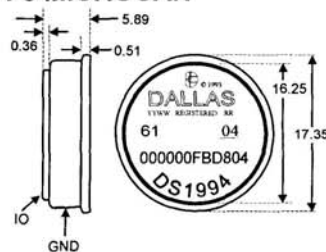
- 4096 bits of Read/Write Nonvolatile Memory
- 256-bit Scratchpad Ensures Integrity of Data Transfer
- Memory Partitioned into 256-bit Pages for Packetizing Data
- Data Integrity Assured with Strict Read/Write Protocols
- Contains Real-Time Clock/Calendar in Binary Format
- Interval Timer Can Automatically Accumulate Time When Power is Applied
- Programmable Cycle Counter can Accumulate the Number of System Power-On/Off Cycles
- Programmable Alarms Can Be Set to Generate Interrupts for Interval Timer, Real-Time Clock, and/or Cycle Counter
- Write-Protect Feature Provides Tamperproof Time Data
- Programmable Expiration Date That Limits Access to SRAM and Timekeeping
- Clock Accuracy is Better Than ± 2 Minutes/Month at 25°C
- Operating Temperature Range from -40°C to +70°C
- Over 10 Years of Data Retention

COMMON iButton® FEATURES

- Unique, Factory-Lasered, and Tested 64-bit Registration Number (8-bit Family Code + 48-bit Serial Number + 8-bit CRC Tester) Assures Absolute Traceability Because No Two Parts Are Alike
- Multidrop Controller for MicroLAN
- Digital Identification and Information by Momentary Contact
- Chip-Based Data Carrier Compactly Stores Information

- Data Can Be Accessed While Affixed to Object
- Economically Communicates to Bus Master with a Single Digital Signal at 16.3kbps
- Standard 16mm Diameter and 1-Wire® Protocol Ensure Compatibility with iButton Family
- Button Shape is Self-Aligning with Cup-Shaped Probes
- Durable Stainless Steel Case Engraved with Registration Number Withstands Harsh Environments
- Easily Affixed with Self-Stick Adhesive Backing, Latched by its Flange, or Locked with a Ring Pressed onto its Rim
- Presence Detector Acknowledges when Reader First Applies Voltage
- Meets UL#913 (4th Edit.); Intrinsically Safe Apparatus, Approved under Entity Concept for Use in Class I, Division 1, Group A, B, C and D Locations

F5 MICROCAN



All dimensions shown in millimeters.

ORDERING INFORMATION

DS1994L-F5 F5 MicroCan

iButton and 1-Wire are registered trademarks of Dallas Semiconductor.

EXAMPLES OF ACCESSORIES

DS9096P Self-Stick Adhesive Pad
DS9101 Multi-Purpose Clip
DS9093RA Mounting Lock Ring
DS9093F Snap-In Fob
DS9092 iButton Probe

iButton DESCRIPTION

The DS1994 Memory iButton is a rugged read/write data carrier that acts as a localized database, easily accessible with minimal hardware. The nonvolatile memory and optional timekeeping capability offer a simple solution to storing and retrieving vital information pertaining to the object to which the iButton is attached. Data is transferred serially through the 1-Wire protocol that requires only a single data lead and a ground return.

The scratchpad is an additional page that acts as a buffer when writing to memory. Data is first written to the scratchpad where it can be read back. After the data has been verified, a copy scratchpad command transfers the data to memory. This process ensures data integrity when modifying the memory. A 48-bit serial number is factory lasered into each DS1994 to provide a guaranteed unique identity that allows for absolute traceability. The durable MicroCan package is highly resistant to environmental hazards such as dirt, moisture, and shock. Its compact, coin-shaped profile is self-aligning with mating receptacles, allowing the DS1994 to be easily used by human operators. Accessories permit the DS1994 to be mounted on almost any surface including plastic key fobs, photo-ID badges, and PC boards.

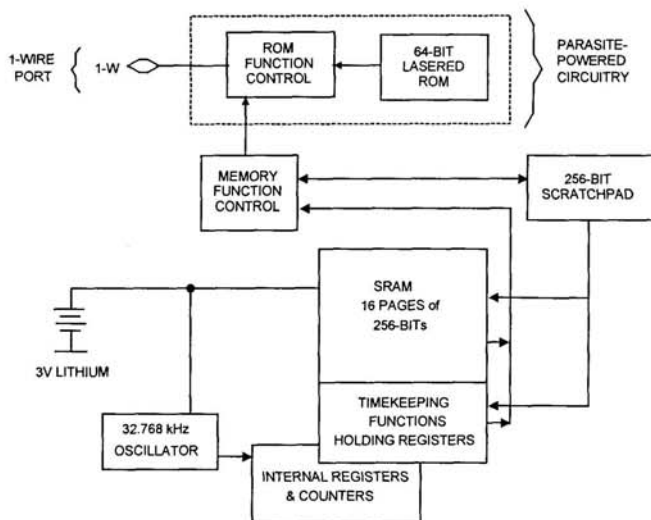
The DS1994 also includes time-keeping functions, a real-time clock/calendar, interval timer, cycle counter, and programmable interrupts, in addition to the nonvolatile memory. The internal clock can be programmed to deny memory access based on absolute time/date, total elapsed time, or the number of accesses. These features allow the DS1994 to be used to create a stopwatch, alarm clock, time and date stamp, logbook, hour meter, calendar, system power cycle timer, interval timer, and event scheduler.

OPERATION

The DS1994 has four main data components: 1) 64-bit lasered ROM, 2) 256-bit scratchpad, 3) 4096-bit SRAM, and 4) timekeeping registers. The timekeeping section utilizes an on-chip oscillator that is connected to a 32.768kHz crystal. The SRAM and time-keeping registers reside in one contiguous address space referred to hereafter as memory. All data is read and written least significant bit first.

The memory functions are not available until the ROM function protocol has been established. This protocol is described in the ROM functions flowchart (Figure 9). The master must first provide one of four ROM function commands: 1) read ROM, 2) match ROM, 3) search ROM, or 4) skip ROM. After a ROM function sequence has been successfully executed, the memory functions are accessible and the master can then provide any one of the four memory function commands (Figure 6).

Figure 1. DS1994 BLOCK DIAGRAM



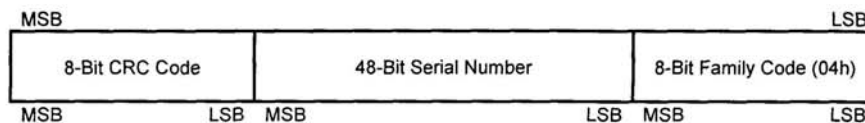
PARASITE POWER

The block diagram (Figure 1) shows the parasite-powered circuitry. This circuitry steals power whenever the data input is high. The data line provides sufficient power as long as the specified timing and voltage requirements are met. The advantages of parasite power are two-fold: 1) by parasiting off this input, lithium is conserved, and 2) if the lithium is exhausted for any reason, the ROM can still be read normally.

64-BIT LASERED ROM

Each DS1994 contains a unique ROM code that is 64 bits long. The first 8 bits are a 1-Wire family code. The next 48 bits are a unique serial number. The last 8 bits are a CRC of the first 56 bits. (See Figure 2.) The 1-Wire CRC is generated using a polynomial generator consisting of a shift register and XOR gates, as shown in Figure 3. The polynomial is $X^8 + X^5 + X^4 + 1$. Additional information about the Dallas 1-Wire Cyclic Redundancy Check is available in the Book of DS19xx iButton Standards. The shift register bits are initialized to zero. Then starting with the least significant bit of the family code, 1 bit at a time is shifted in. After the 8th bit of the family code has been entered, then the serial number is entered. After the 48th bit of the serial number has been entered, the shift register contains the CRC value. Shifting in the 8 bits of CRC should return the shift register to all zeros.

Figure 2. 64-BIT LASERED ROM



PHYSICAL SPECIFICATIONS

Size	See mechanical drawing
Weight	3.3 grams (F5 package)
Humidity	90% RH at 50°C
Altitude	10,000 feet
Expected Service Life	10 years at 25°C
Safety	Meets UL#913 (4th Edit.); Intrinsically Safe Apparatus, Approved under Entity Concept for use in Class I, Division 1, Group A, B, C, and D Locations

ABSOLUTE MAXIMUM RATINGS*

Voltage on any Pin Relative to Ground	-0.5V to +7.0V
Operating Temperature	-40°C to +70°C
Storage Temperature	-40°C to +70°C

* This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operation sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods of time may affect reliability.

DC ELECTRICAL CHARACTERISTICS

($V_{PUP} = 2.8V$ to $6.0V$; $-40^{\circ}C$ to $+70^{\circ}C$)

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
Logic 1 (Note 1)	V_{IH}	2.2			V
Logic 0 (Note 1)	V_{IL}	-0.3		+0.8	V
Output Logic Low at 4mA (Note 1)	V_{OL}			0.4	V
Output Logic High (Notes 1, 2)	V_{OH}		V_{PUP}		V
Input Load Current (Note 3)	I_L		5		μA

CAPACITANCE

($t_A = 25^{\circ}C$)

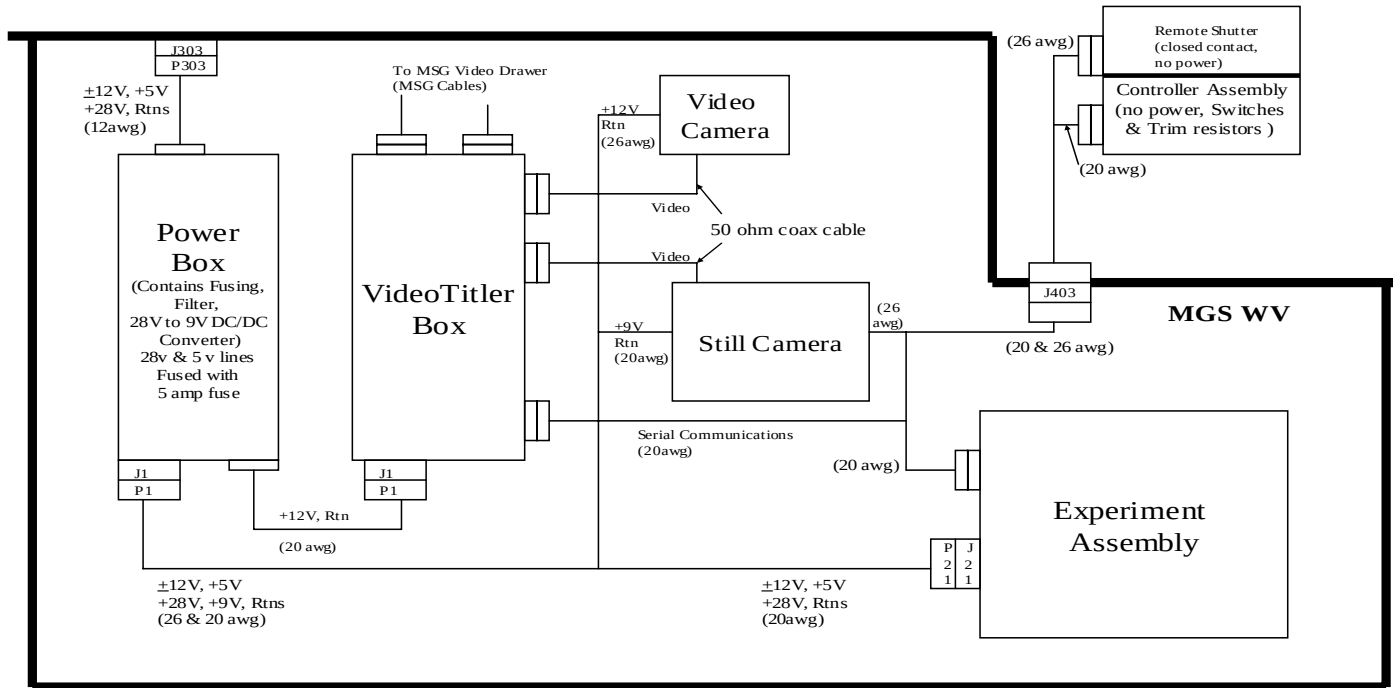
PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
I/O (1-Wire) (Notes 6, 8)	$C_{IN/OUT}$		100	800	pF

AC ELECTRICAL CHARACTERISTICS

($V_{PUP} = 2.8V$ to $6.0V$; $-40^{\circ}C$ to $+70^{\circ}C$)

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
Time Slot	t_{SLOT}	60		120	μs
Write 1 Low Time	t_{LOW1}	1		15	μs
Write 0 Low Time	t_{LOW0}	60		120	μs
Read Data Valid	t_{RDV}	exactly 15			μs
Release Time	$t_{RELEASE}$	0	15	45	μs
Read Data Setup (Note 5)	t_{SU}			1	μs
Interrupt	t_{INT}	960		4800	μs
Recovery Time	t_{REC}	1			μs
Reset Time High (Note 4)	t_{RSTH}	480			μs

Power Distrubution Schematic **Figure B-3 (for STD SPICE-01 Description of Hazard 11)**



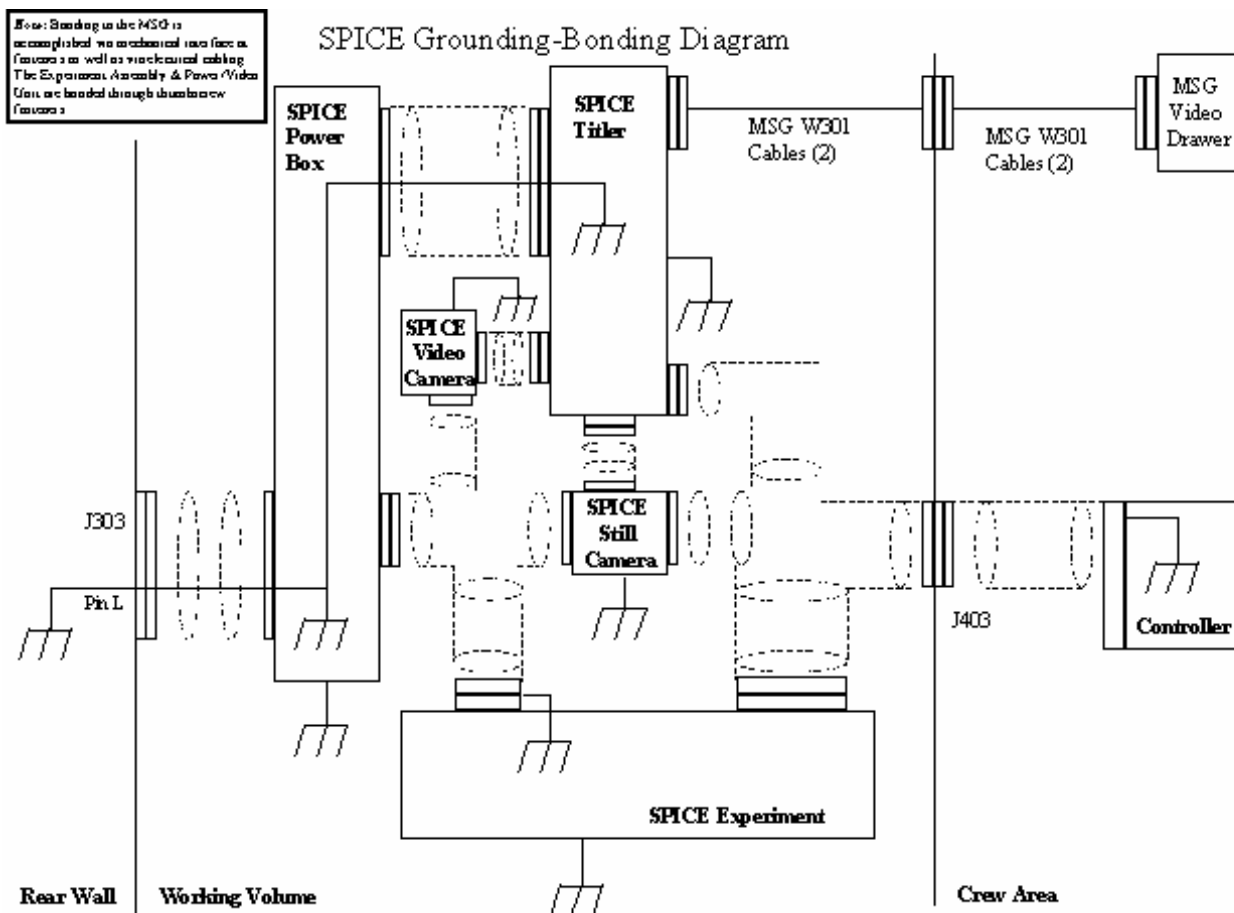
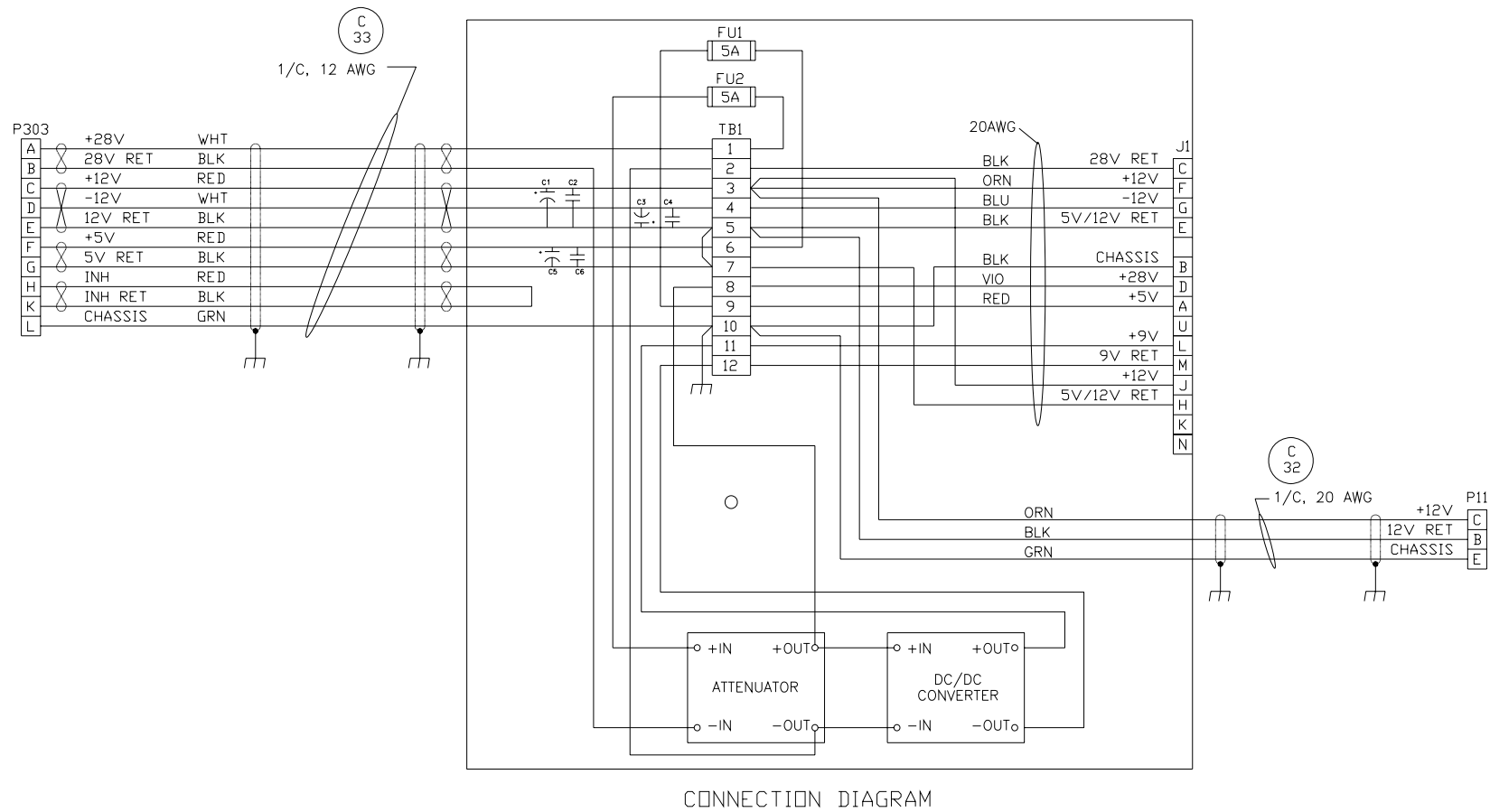


Figure B-4 Bonding and Grounding (For STD SPICE-01 Hazard 11)

Figure B-5 Connection Diagram (for STD SPICE-01 Description of Hazard 11)



ATTACHMENT C: SPICE UNIQUE HAZARD REPORTS
SPICE UNIQUE HAZARDS
SPICE-U1, SPICE-U2, SPICE-U3, SPICE-U4, SPICE-U5, SPICE-
U6, SPICE-U8

Submitted under separate cover.

ATTACHMENT D: SPICE FIRE DETECTION & SUPPRESSION REPORTING FORM 1428

Fire Detection and Suppression Reporting Form				
a. Number: SPICE			b. Mission Phase: On-Orbit Operations LF-1	
c. Payload Name: Smoke Points In Coflow Experiment				
d. Fire Prevention: Hazard Report STD-SPICE-01 6(ab).1 and 6(ab).2 for Materials.				
e. Payload Volume	f. Fire Detection Method	g. Verification Method, Reference and Status	h. Suppression Method	i. Verification Method, Reference and Status
MSG Working Volume	Air Temperature Monitoring 49C Caution to ISS, 70C Warning to ISS.	MSG Test Verification Report	PFE Access to MSG Working Volume.	CO2 Distribution analysis
SPICE Assembly Storage	No ignition sources are present.	Inspection/assessment no ignition sources are within the stowage assembly. (See Hazard Report SPICE U4)	PFE with the area nozzle.	Analysis demonstrating PFE will displace O2 to below 10.5% with in 60 seconds in stowage assembly.
j. SIGNATURES:		PAYLOAD ORGANIZATION		SSP/ISS
PHASE I				
PHASE II				
PHASE III				

JSC Form 1428 (August 14, 2002) (MS Excel)

ATTACHMENT E: SPICE SVTL SPICE SAFETY VERIFICATION TRACKING LOG

Submitted under separate cover.

ATTACHMENT F: SPICE CERTIFICATE OF SAFETY COMPLIANCE

CERTIFICATE OF NSTS/ISS PAYLOAD SAFETY COMPLIANCE FOR

a. PAYLOAD/MISSION/CARRIER: SPICE /STS-126/MPLM

b. Applicable to:

☒ Payload Design and Flight Operations

☐ Ground Support Equipment Design and Ground Operations

c. The Payload Organization Hereby Certifies that:

☐ For STS, the Payload Complies with all Applicable Requirements of the NSTS 1700.7 (current issue), "Safety Policy and Requirements for Payloads Using the National Space Transportation System," and/or KHB 1700.7, "Space Shuttle Payload Ground Safety Handbook."

☒ For ISS, the Payload Complies with all Applicable Requirements of the NSTS 1700.7 (current issue), "Safety Policy and Requirements for Payloads Using the National Space Transportation System," NSTS 1700.7 ISS Addendum (current issue), "Safety Policy and Requirements for Payloads Using the International Space Station," and/or KHB 1700.7, "Space Shuttle Payload Ground Safety Handbook"; and,

1) The Payload is Safe with a Design Life of: 5 years.

2) The Payload is Safe for an Operational Life of: 5 years

d. Approved Waivers/Deviations: Not Applicable

e. Approved (Payload Organization Manager):

Robert J. Hovensack

f. Date: 4/11/08

JSC Form 1114A (Rev February 20, 2001) (MS Word September 1997)